Simulation of Squeezed Light Generation in an Optical Resonator

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Abstract

We investigate a potential method for converting beams of optically encoded quantum information from conventional operational wavelengths to telecom wavelengths which, if viable, could lead to a scalable and economically viable method for rapidly deploying quantum networks. The system of interest is the four-level, diamond configuration, energy substructure of rubidium-87 composed of states $5S_{1/2} - 5P_{1/2} - 5P_{3/2} - 4D_{1/2}$. To achieve efficient frequency conversion with an economically scalable device, the phenomenon of electromagnetically induce transparency enhanced four wave mixing in a Doppler broadened hot atomic medium would need to be produced. The use of a resonant cavity can enhance electromagnetically induced transparency, and four wave mixing can generate squeezed light such that quantum uncertainty can be reduced in one quadrature which gives a method to reduce transmission signal noise. Towards the goal of realizing this system, the theory for all the relevant physical processes is presented, a set of possible system Hamiltonians are derived and presented, Lindbladian decay operators are given to allow for steady-state density matrices to be found via computation, and the Maxwell-Bloch equations for the system are derived and presented to give a method for calculating theoretical field propagation throughout the medium. An algorithmic method for solving Doppler broadened systems with the combined equations is presented. Known example systems are solved to give intuition as to how the computational methods should be applied, and a set of problem specific computational tools are developed and presented. Steps for future work are outlined.

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I Project Motivation

The past century since the formulation of quantum mechanics has led to unrivaled technological progress, especially in recent years. Methods in material sciences, such as density functional theory for computational modelling of solid state quantum mechanical systems [2], have brought commercial transistors down to the scale of nanometers. With greater understanding of and ability to control quantum phenomena, further possibilities become available. Modern avenues for advancing quantum technologies include quantum key distribution, quantum sensing, and quantum computation, to name a few.

Any emerging technology tends to follow the rule ascribed to increasing system complexity sometimes stated as 'more is different' with reference to the 1972 paper by Physicist Philip W. Anderson [3]. This especially applies to quantum networks. As greater quantum control is achieved, greater quantum networking capabilities will be required to reap maximal benefits, thereby necessitating the development of improved methods for transmitting quantum information.

One near-term threat which looms over modern encryption methods is the threat of the 'harvest now, decrypt later' information gathering strategy by bad actors [4]. Since quantum algorithms like Shor's algorithm [5] (or even brute force GPU techniques at massive scale) have the potential to compromise 'securely' transmitted information, finding alternative quantum-secure methods is considered an issue of national defence. Quantum key distribution in which photonic quantum bits encode information has been proposed as one such method [6] and could be applied to critical information channels in the near-term given a scalable method for building quantum networks.

Currently, there are two main hurdles in developing large scale quantum networks: creating a continent-spanning network compatible with current methods of producing quantum information; and transmitting quantum information with sufficiently low loss/noise such that the information sent may actually be used. There is potential that the first problem has already been solved, however. Fibre optic networks already span the world, but have a limited transparency window to which information can be transmitted with minimal losses, and current quantum techniques do not operate produce photonic quantum information in this range. So if a method of efficiently producing quantum information with minimal losses at fibre optic wavelengths is found, then progress towards both problems would be obtained.

After this, the problem becomes advancing techniques for minimizing losses in quantum coherence during fibre optic transmission [7] and all other forms of efficiency such that amount of information transmitted through the channel may be maximized [8].

Progress has been made on forming quantum-optical networks, such as in America where an advanced quantum network architecture has been developed on Long Island [9], or in China where a 4600 kilometer land-based quantum network [10].

This thesis covers a theoretically viable method for frequency conversion of quantum information encoded in an electromagnetic field (continuous variable quantum information). The technique involves interfering three electromagnetic fields within a sample of Rubidium-87 gas to induce transitions such that one additional field is produced. Rubidium-87 is a widely studied candidate for applications in short-timescale quantum memory [11][12], quantum repeaters [13], and quantum computation [14]. We will focus on a 4-level transition substructure with one transition frequency compatible with modern quantum methods, one transition frequency compatible with telecom frequencies in the conventional band (C-band), and two frequencies which will be leveraged for interference effects in a process known as four wave mixing (FWM) [15].

In Sec. II, the theory of electromagnetically induced transparency (EIT) and FWM, the Lindblad master equation and Maxwell-Bloch equation solution methods for quantum systems, and an algorithmic procedure for pairing the Lindblad and Maxwell-Bloch strategies are presented. Sec. III explicitly outlines the problem definition by defining the theoretical system, outlining the effects and mathematical treatment of Doppler broadening, presenting multiple forms for the system Hamiltonian and discussing how each should be applied, and presenting system-specific Maxwell-Bloch equations to calculate field propagation through the medium. Sec. IV covers how to apply the QuTiP python library by reproducing known results, then outlines a set of computational tools designed to solve the system. Sec. V covers the broad ideas encompassed in this work, defines what was accomplished, explains shortcomings, and sets up the next steps for future research. Sec. VI summarizes the work.

II Background Information

This project involves understanding how hot atomic vapours interact with external electromagnetic fields to induce desirable physical phenomena. To convert frequencies, FWM can be applied with the added bonus that it can generate squeezed light [16], which is useful in continuous variable quantum information applications since optical squeezing causes uncertainty in one quadrature (amplitude or phase) to be reduced while being increased in the other. FWM may be enhanced by EIT [17], a process in which an atomic medium becomes effectively transparent to a frequency of light near-resonance with one of its transition frequencies under certain conditions. We will tackle these topics in reverse order to build up to the idea of optical squeezing.

In order to solve for signal propagation, we must find the steady state system dynamics via the Lindblad master equation, then use the result to solve for the field propagation through the medium step by step using the Maxwell-Bloch equations. The master equation is a function of field strength, so we can produce a combined solution algorithm to obtain the field strength of the converted frequency field at the end of the sample. This result tells us how much of the signal has been converted via FWM.

A. Electromagnetically Induced Transparency

In order to gain an intuition on EIT, we will start with a classical toy model of the atom. As is a common theme in physics, a hydrogenic atom in a time dependent electromagnetic field can be approximately modelled as a driven harmonic oscillator. So rather than dealing with quantized photons, we will consider a classical oscillating electromagnetic fields with frequencies nearly resonant with those of the atomic transitions.

To do this, a few assumptions will be made [18]: first, the wavelengths of the light sources interacting with the atom have wavelengths much greater than the Bohr radius of the atom ($\lambda \gg a_0$); second, the detunings of incoming beams are small in proportion to the transition frequencies considered. These two assumption happen to be consistent with one another.

The first assumption allows for us to approximate the electromagnetic field experienced by the atom as constant in space but varying in time since the change in electromagnetic field strength between the two ends the atom in the transverse wave direction is relatively small.

There exists the relation between the electric and magnetic field strengths of light which arises from Maxwell's equations

$$B = \frac{E}{c}. \tag{1}$$

The Lorentz force is

$$\vec{F} = q\left(\vec{E} + \vec{v} \times \vec{B}\right) \tag{2}$$

where q is charge, and $\vec{v}$ is particle velocity. Using the Bohr model for approximation, we have that quantized electron velocities for a hydrogenic atom have magnitude

$$v_n = \alpha \frac{Z}{n} c \tag{3}$$

where α is the fine structure constant, Z is the atomic number, and n is the principle quantum number [19]. We can therefore see by rough approximation that we have

$$|\vec{v} \times \vec{B}| \lessapprox \alpha Z c \frac{E}{c} = \alpha Z E. \tag{4}$$

We are considering Rubidium with atomic number $Z = 37$, but this neglects the effects of screening caused by the electrons in inner shells of the atom. Electron position is distributed in a cloud about the atom for any orbital, though we can expect the 36 electrons of inner shells between the valence electron (singular in its shell) and the nuclear charge of $37e$ to roughly cancel $36e$ of charge such that the valence electron experiences about $1e$. By this line of reasoning we have

$$|\vec{v} \times \vec{B}| \lessapprox \alpha E. \tag{5}$$

Thus, the force due to the external magnetic field is about $1/137$ that of the external electric field. So we may neglect the external magnetic field for simplification.

$$\vec{F} \approx q\vec{E} \tag{6}$$

All of these approximations can be justified from the perspective that this model is a first estimate for system behaviour which will be refined and iterated upon.

This brings us to a physicist's favorite abstraction: the harmonic oscillator. The potential of which the valence electron experiences can be expanded in a Taylor series and approximated as quadratic. The mass of the nucleus is much greater than the electron mass, so we may approximate the nucleus to be fixed. So we have a mass with some equilibrium distance from a fixed point under a quadratic potential, i.e. a simple harmonic oscillator. We may then add in our external time-dependent driving force to obtain a driven harmonic oscillator.

When a charge accelerates, it emits photons as its movement produces a changing electric and magnetic field. So when our system is driven on resonance, then an electromagnetic field of that same frequency will be produced. Here lies the idea behind electromagnetically induced transparency: by driving a cloud of these hydrogenic atoms on resonance, the field essentially 'passes through' the medium as the dipole response of the medium matches the driving frequency. This abstraction is known as the Lorentz oscillator [20].

It is helpful to imagine the dipole response of the medium as a vector field coupled to the external driving field of frequency ω. When we quantize the system, this is encapsulated by the expectation value of the dipole moment of the atomic medium $\langle\hat{\vec{d}}\rangle = -e\langle\hat{\vec{r}}\rangle$ since the polarization is given by $\vec{P} = N\langle\hat{\vec{d}}\rangle$, where $\vec{r}$ is the relative position vector and N is the number of atoms in the medium [21].

So far we have considered the atomic system to behave with a linear response to the external field, however, this is not the case. The true polarization of the system actually behaves in a nonlinear fashion with respect to the driving field. This response is medium dependent and encapsulates all the interactions of the system, so it can be quite complicated. For this reason, it is often abstractly written in the expanded polynomial form [22]

$$\vec{P} = \epsilon_0 \left(\chi^{(1)}\vec{E} + \chi^{(2)}\vec{E}\vec{E} + \chi^{(3)}\vec{E}\vec{E}\vec{E} + \dots\right) \tag{7}$$

where $\chi^{(i)}$ is the i^{th} order electric susceptibility rank $i+1$ tensor. For weak electric fields, the linear approximation may be made to give

$$\vec{P} \approx \epsilon_0\chi^{(1)}\vec{E} \tag{8}$$

then by $N\langle\hat{\vec{d}}\rangle = \epsilon_0\chi^{(1)}\vec{E}$ we have

$$\chi^{(1)} = \frac{N\langle\hat{d}\rangle}{\epsilon_0 E}. \tag{9}$$

What frequencies are allowed to pass through the atomic medium is decided by the linear electric susceptibility of the material $\chi^{(1)}$. For this reason, the above form of Eq. (9) will not actually be applied as the relevant equation is frequency dependent, but in order to obtain any frequency dependent equation for $\chi^{(1)}$ the system must first be defined. Thus, Eq. (9) is merely an illustrative equation to depict the physics in broad detail.

When light travels through some material with complex electric susceptibility (i.e. complex index of refraction), then the light will receive some phase shift due to the real-valued component of susceptibility $Re\left(\chi^{(1)}\right)$ and some decay due to the imaginary-valued component $Im\left(\chi^{(1)}\right)$, such that the oscillating electric field will propagate as

$$E = E_0 e^{i(kz-\omega t)} e^{iRe\left(\chi^{(1)}\right)kz/2} e^{-Im\left(\chi^{(1)}\right)kz/2} \tag{10}$$

where the rate of decay then grows exponentially with distance traveled z through the medium. The decay term $e^{-Im\left(\chi^{(1)}\right)kz/2}$ is known as Beer-Lambert law [23].

So, we can say that when the imaginary component of linear susceptibility goes to zero, the medium becomes 'transparent' to the driving field. This gives us a mathematical way to determine if EIT is present in a system.

One additional phenomenon which happens in tandem with EIT is slowed light. The index of refraction of a material is

$$n = \sqrt{1+\chi} \tag{11}$$

and may be approximated by $n = \sqrt{1+\chi^{(1)}}$ for sufficiently weak electromagnetic fields. $\chi^{(1)}$ for the Lorentz oscillator may be expressed as [20]

$$\chi^{(1)}(\omega) = \frac{Ne^2}{\epsilon_0 m_e}\frac{1}{\omega_0^2 - \omega^2 - i\gamma\omega} \tag{12}$$

where ω_0 is the resonant frequency of the system, ω is the driving frequency, γ is the decay rate (takes on the role of drag in the oscillator approximation), e is the electron charge, and m_e is electron mass. The group velocity of a wave can be written [24]

$$v_g = \frac{\partial \omega}{\partial k} = \frac{c}{n + \omega \frac{\partial n}{\partial \omega}}. \tag{13}$$

Combining the information from these 3 equations tells us that $\chi^{(1)}$ tends to be very sensitive with respect to the driving frequency near $\omega \approx \omega_0$, so if the driving laser is tuned near resonance such that EIT occurs, then the derivative $\partial_\omega n$ may become sufficiently large such that the group velocity of the propagating electromagnetic field is significantly reduced. Thereby, the dipole response of the medium may result in the propagation of slowed light when EIT is present.

Unfortunately, an atom is not a mass on a spring, but this model gives us a good idea of how EIT works as a first step. In reality, we must consider the structure of atomic states and the interference of transition pathways.

1. *Structure of atomic spectra*

Atomic states for the electron of a hydrogenic atom are enumerated by the quantum numbers n (principle quantum number), l (angular momentum quantum number), m_l (magnetic quantum number), and m_s (spin quantum number) [25].

$$n \in \mathbb{N} \tag{14}$$
$$l \in \{0, 1, \ldots, n-1\} \tag{15}$$
$$m \in \{-l, \ldots, l\} \tag{16}$$

Notably, there is also the electron spin s quantum number which may take on values $s = \pm 1/2$ for hydrogenic atoms.

The energy levels for an unperturbed hydrogenic atom modelled with just the coulomb potential are quite simple as they just depend on the principal quantum number.

$$E_n = -\frac{(Z\alpha)^2 mc^2}{2n^2} = -\frac{Z^2 Ry}{n^2} \tag{17}$$

where Z is the atomic number, α is the fine structure constant, m is the electron mass, and c is the speed of light [18]. These unperturbed energy levels each have degeneracy of n^2. The real picture, is more complicated, however. There are fine structure and hyperfine structure perturbations on these energy levels, splitting energy levels and decreasing their degeneracies.

Fine structure corrections arise due to relativistic corrections to the non-relativistic Schrödinger's equation. In particular, we have a pure relativistic correction from considering electron momentum (which is a non-negligible fraction of the speed of light), a spin-orbit coupling correction, and the Darwin correction [18]. To first order perturbation theory, the combined energy shift of these contributions is

$$\Delta E_{fs} = \frac{(Z\alpha)^4}{2} \frac{mc^2}{n^3} \left(\frac{3}{4n} - \frac{2}{2j+1} \right) \tag{18}$$

where $j = |l \pm s| = |l \pm 1/2|$ since we have $s = 1/2$ for hydrogenic atoms [18]. Notice that these perturbations are scaled by a factor of $(Z\alpha)^2$ in comparison to Eq. (17) such that for atoms with a sufficiently large atomic number, these approximate energy shifts arrived at by perturbation theory break down.

There exist yet smaller energy shifts due to the interaction between the internal magnetic fields of the nucleus and the orbital motion of the electron along with its spin [25]. The hyperfine energy splitting term is given as

$$\Delta E_{hfs} = \frac{A}{2} \left[F(F+1) - I(I+1) - j(j+1) \right] \tag{19}$$

where A is known as the hyperfine constant, F is the total angular momentum quantum number with operator $\hat{F} = \hat{I} + \hat{S}$, I is the nuclear spin quantum number, and j is defined as before [18][25].

Spectroscopic notation is used to describe atomic levels. It is expressed

$$n^{2s+1}L_j \tag{20}$$

where n, s, and j are atomic numbers as we have defined, $2S+1$ is the multiplicity of states, L is the atomic term symbol $(S, P, D, F, \ldots)$ [19]. Notice that for a hydrogenic atom we will always have multiplicity of $2s+1=2$ which correspond to the 2 hyperfine states $F=1$ and $F=2$

2. *Atomic transitions*

In a hydrogenic atom, the transitions between states are mediated by excitations in the electromagnetic field (i.e. photons). It is crucial in atomic physics to understand the probabilities of these atomic state transitions.

The interaction between an atom and the electromagnetic field is not a linear interaction, creating difficulty in finding an exact solution. To obtain approximate behaviour, first order perturbation theory can be applied to dominating effects of this interaction [19]. From this, one obtains the transition dipole matrix with elements

$$\vec{d}_{ji} = \langle j|\hat{\vec{d}}|i\rangle = e\langle j|\hat{\vec{r}}|i\rangle \tag{21}$$

where $\hat{\vec{d}} = e\hat{\vec{r}}$ is the dipole transition operator and $\hat{\vec{r}}$ is the position operator [19].

If $\vec{d}_{ji} = 0$, then the transition between states $|i\rangle$ and $|j\rangle$ is a 'forbidden' dipole transition, otherwise the transition is an allowed dipole transition. This criterion enforces the dipole selection rules that [19]

$$\Delta l = \pm 1 \tag{22}$$

$$\Delta m_l = 0, \pm 1 \tag{23}$$

$$\Delta F = 0, \pm 1 \tag{24}$$

$$F = 0 \leftrightarrow F = 0 \text{ is forbidden.} \tag{25}$$

It is important to note that these transition rules are not always perfectly enforced. Since perturbation theory was applied to obtain them, there exist transitions which disobey these rules, however, such transitions occur with significantly decreased probability.

3. *Three-level atoms*

In order to meaningfully utilize the power of quantum information via quantum states, it is beneficial to work with simplified, yet somewhat abstract, systems. It would be ideal if we could realize any imaginary quantum systems that exactly match our mathematical descriptions. Reality is not so convenient, however. We must work within the constraints enforced by the available quantum systems in terms of allowed states and transitions between those states.

With enough ingenuity, we can use existing systems to approximate useful mathematical abstractions. Enter, the three-level (hydrogenic) atom. All hydrogenic atoms, in theory, have infinitely many states up to ionization ($n \in \mathbb{N}$), and transition probabilities between many of these states can be non-zero. Though by finding a subset of states with desired characteristics and leveraging transition rules where any undesired state transitions are suppressed, a physical atom can be used to approximate a model system. This becomes especially useful in the limit of a large number of atoms where, statistically, the desired transitions are the most probable, and any undesired transitions are vastly outnumbered, such that the system tends towards the desired behaviour in the limit of many particles.

A three-level atom has three distinct states with some set of allowed transitions between those states. There exist Λ-systems, V-systems , and Ξ-systems, each depicted in Fig. 1. An common example of a Λ-system exists in ^{87}Rb with the collection of hyperfine split states $5^2S_{1/2}$ for $F=1$ and $F=2$ with energy difference given according to (19) so that they take on the roles of states $|1\rangle$ and $|2\rangle$, respectively, and state $5^2P_{1/2}$ for $F=1$ acts as the excited state $|3\rangle$ [17].

Some points of note in this system are as follows. State $5^2S_{1/2}$ $F=1$ is the absolute ground state of the system. The nuclear spin of ^{87}Rb is $3/2$ [26], so states $|1\rangle$ and $|2\rangle$ differ by the direction of electron spin $s = \pm 1/2$, so the energy difference between these states is the result of hyperfine splitting. State $|3\rangle$ has the same principle quantum number of $n=5$ as states $|1\rangle$ and $|2\rangle$ but a different angular momentum quantum number, indicating that this state is due to fine structure (with a hyperfine energy correction). Since states $|1\rangle$ and $|2\rangle$ have the same l value, the transition between these levels is forbidden as per Eqs. (22), (23), (24), and (25).

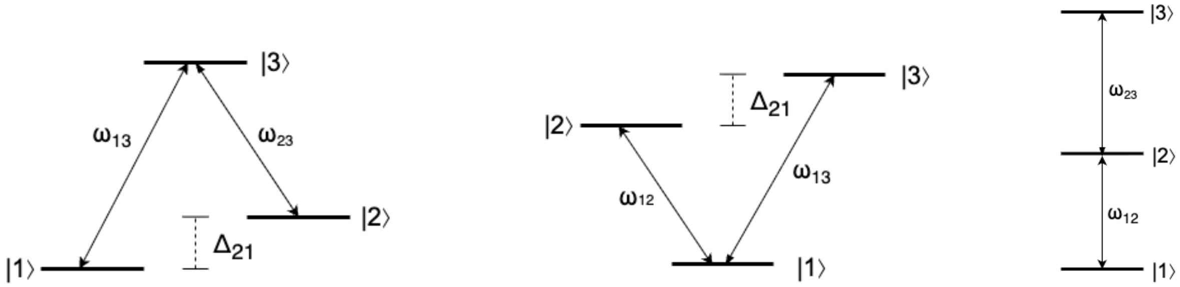

FIG. 1: Three-level atomic configurations: (left) Λ-system, (middle) V-system, and (right) Ξ-system.

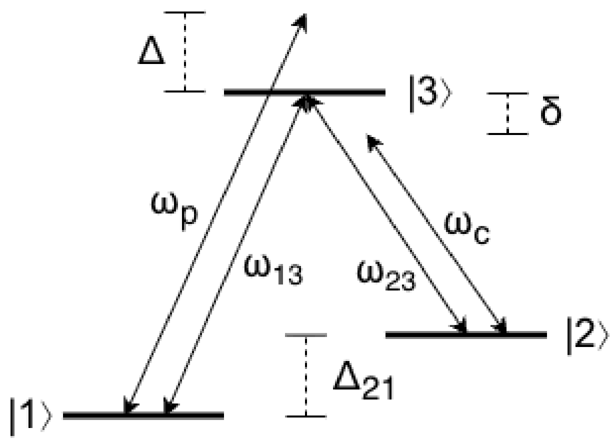

FIG. 2: A thee-level Λ-system with natural transition frequencies ω_{13} and ω_{23} which is driven near-resonance by detuned beams operating at ω_p and ω_c

Of course, a transition could place the electron into some state outside this state subset (for example, the $5^2P_{1/2}$ $F = 2$ state), so to promote population only between these states driving lasers tuned near the transition frequencies are shone on the system to induce only the desired transitions. Then in the limit of Avogadro's number of atoms the desired system behaviour becomes statistically enforced.

In this Λ-system example, the transition amplitude between states $|1\rangle$ and $|2\rangle$ is suppressed, so if the system is in state $|2\rangle$, then in order to transition back to state $|1\rangle$ the system will likely need to transition through the excited state $|3\rangle$. This makes $|2\rangle$ a quasi-stable state. If there were a way to suppress all transitions to state $|3\rangle$, then we would obtain a system in which the atom is bound to states $|1\rangle$ and $|2\rangle$ such that any photons which would stimulate a transition to the excited state cannot be absorbed.

There is, in fact, a way to do just this. By driving the system with external electromagnetic fields (lasers) at the transition frequencies, destructive quantum interference between the two fields along with specific initial conditions results in zero probability for the system to be in the excited state [27]. This state in which there is no probability of occupying $|3\rangle$ is then called the 'dark state'. In practice, since perfect precision in real-world systems is difficult, both lasers are detuned from the exact transition frequency. The probe laser stimulates transition 13 $\omega_p = \omega_{13} + \Delta$, $\Delta \ll \omega_{13}$, and the control laser stimulates transition 23 $\omega_c = \omega_{23} + \delta$, $\delta \ll \omega_{23}$ (see Fig. 2).

The main idea is that the system then becomes effectively 'transparent' to the probe laser frequency. This is how EIT can be produced in practice.

EIT can also be achieved in a Ξ-system of rubidium between the $|1\rangle = 5^2S_{1/2}$, $|2\rangle = 5^2P_{3/2}$, and $|3\rangle = 5^2D_{5/2}$ states by resonantly driving the $|1\rangle \leftrightarrow |2\rangle$ and $|2\rangle \leftrightarrow |3\rangle$ transitions such that transition pathways interfere to induce transparency [28]. This is sometimes called 'cascade' EIT.

EIT in V-systems is less widely applied and studied in comparison to Λ and cascade EIT since there is no stable dark state in V-systems [29]. Since two excited states may decay to ground, decoherence effects decrease the efficiency of producing EIT.

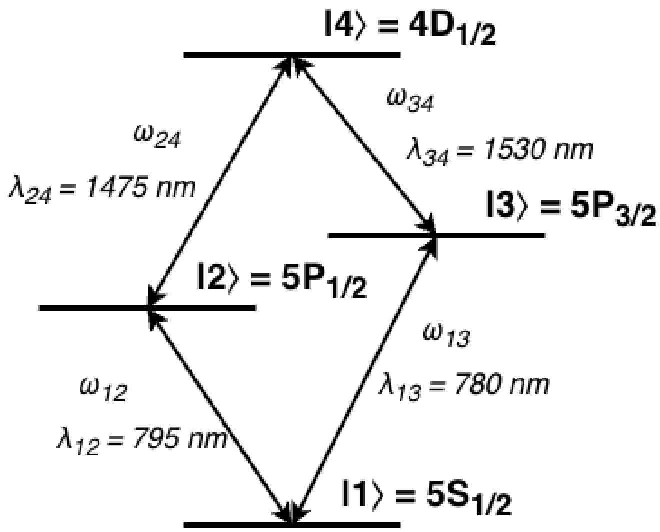

FIG. 3: A diamond configuration atomic system with non-degenerate energy levels $|1\rangle, |2\rangle, |3\rangle, |4\rangle$. The system has allowed transitions $|1\rangle \leftrightarrow |2\rangle$, $|1\rangle \leftrightarrow |3\rangle$, $|2\rangle \leftrightarrow |4\rangle$, and $|3\rangle \leftrightarrow |4\rangle$ with respective energies $\hbar\omega_{12}$, $\hbar\omega_{13}$, $\hbar\omega_{24}$, and $\hbar\omega_{34}$ and corresponding wavelengths λ_{12}, λ_{13}, λ_{24}, and λ_{34}.

4. Diamond configuration four-level atoms

In order to convert electromagnetically encoded quantum information to telecom wavelengths we require a structure with a transition wavelength in the telecom band. Once candidate of note is the diamond transition structure of ^{87}Rb depicted in Fig. 3. It is composed of the $5S_{1/2} - 5P_{1/2} - 5P_{3/2} - 4D_{1/2}$ energy substructure, representing states $|1\rangle$, $|2\rangle$, $|3\rangle$, and $|4\rangle$, respectively. This transition structure can be through of like 2 parallel Ξ-systems which meet at the $|1\rangle$ and $|4\rangle$ states, or as a V-system connected with a Λ-system sharing states $|2\rangle$ and $|3\rangle$.

It has been shown that high efficiency frequency conversion to the original band (O-band) using EIT is possible with an alternative diamond configuration transition structure [15]. Chen et al. report that cascade-type and V-type EIT is present in their diamond configuration system, resulting in higher efficiency conversion.

We are instead concerned with conversion to the C-band of range 1530 nm to 1565 nm [30]. Hence, we consider the alternative diamond configuration transition structure depicted in Fig. 3, of which, the $|2\rangle \leftrightarrow |4\rangle$ transition has a wavelength within the C-band. Applying slight detunings to the driving lasers can allow the conversion wavelength to be slightly altered such that minimal losses may be achieved within the C-band window.

5. Electromagnetically induced transparency in hot vapours

Consider a system composed of Avogadro's number of three-level or four-level atoms. In the ideal scenario, all atoms would have the exact same transition frequencies such that when the probe and control beams are fired through the cloud. For hot atomic vapours, this is not the case. When the system is in a gaseous excited state: the electromagnetic fields of nearby atoms may interact to induce Stark and Zeeman shifts in their spectra, atoms can collide resulting in state decoherence, and atoms can experience Doppler shifts in the observed frequency of light resulting in a distribution of Doppler shifts throughout the medium in a process known as Doppler broadening. In addition, the laser used will have some bandwidth of emission frequencies. These effects combine to cause spectral broadening such that the transparency window allowed by EIT is suppressed [27].

However, it is known that in Λ-systems composed of 'hyperfine-split metastable states' (as described in Section II A 3) are unaffected by Doppler broadening if the two applied fields are copropagating [27]. This is because for a copropagating pump and control beam (i.e. beams traveling in opposite directions) the Doppler shift observed by any one atom with respect to the pump is equal and opposite to that of the control beam, so they cancel out. This type of system is called Doppler insensitive.

Improved behaviour is achievable for an ultracold atomic system where temperatures are cooled to near-zero-kelvin in order to minimize [27]. This is possible with advanced hardware such as a magneto optical trap [31], however deploying such technology is currently costly and labor-intensive, so application to large scale networks is not

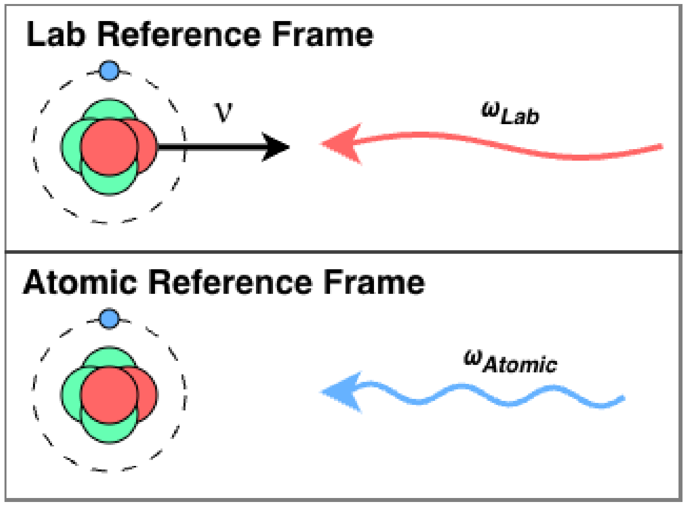

FIG. 4: (Top) the lab frame in which an atom moves towards an electromagnetic wave with velocity v and (bottom) the atomic reference frame in which the atom observes a shift in the field frequency via a Doppler shift.

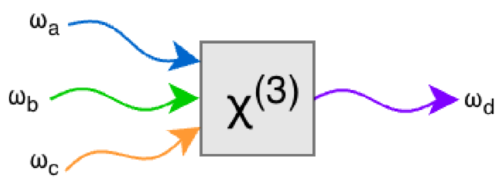

FIG. 5: A cartoon depiction of FWM in a $\chi^{(3)}$ medium involving three beams of frequencies ω_a, ω_b, and ω_c which mix to produce a field of frequency ω_d.

economically viable in the near-term. Therefore, for telecom transmission purposes, it would be favorable to deploy hot atomic frequency converters as opposed to ultracold ones.

If it can be shown that a hot atomic vapour composed of four-level diamond-configuration atoms can be made to behave as Doppler insensitive, then this would be a step towards actualizing a hot C-band frequency converter for electromagnetically encoded quantum information. This is because EIT can be applied to improve FWM efficiency [17], and FWM can be used to generate squeezed light [16].

B. Four Wave Mixing

FWM is a nonlinear optical process in which three electromagnetic fields of distinct frequencies interact with some medium to produce quantum coherence and interference effects such that an additional distinct field is produced [32]. In an atomic medium, FWM is a $\chi^{(3)}$ process [33] in which energy conservation and momentum conservation are upheld [9].

$$\hbar\omega_a + \hbar\omega_b = \hbar\omega_c + \hbar\omega_d \tag{26}$$

$$\vec{k}_a + \vec{k}_b + \vec{k}_c + \vec{k}_d = 0 \tag{27}$$

This allows FWM to be used for optical field frequency conversion. We will be considering FWM in a diamond-configuration atom (see Fig. 3) where transitions $|1\rangle \leftrightarrow |2\rangle$, $|1\rangle \leftrightarrow |3\rangle$, and $|2\rangle \leftrightarrow |4\rangle$ are driven by near-resonant beams to produce the emission of photons via the $|3\rangle \leftrightarrow |4\rangle$ transition, as previously discussed.

For an atomic vapour, the $\chi^{(2)}$ term is zero valued such that the polarization becomes

$$\vec{P} \approx \epsilon_0\chi^{(1)}\vec{E} + \epsilon_0\chi^{(3)}\vec{E}\vec{E}\vec{E} \tag{28}$$

and with three driving fields we have

$$\vec{E} = \vec{E}_a e^{-i\omega_a t} + \vec{E}_b e^{-i\omega_b t} + \vec{E}_c e^{-i\omega_c t} + c.c. \tag{29}$$

such that the nonlinear third order polarization response of the medium becomes

$$\vec{P}^{(3)} = \epsilon_0 \chi^{(3)} \left(\vec{E}_a e^{-i\omega_a t} + \vec{E}_b e^{-i\omega_b t} + \vec{E}_c e^{-i\omega_c t} + c.c. \right)^3 \tag{30}$$

and so it involves the term

$$\epsilon_0 \chi^{(3)} \left(\vec{E}_a \vec{E}_b \vec{E}_c^* e^{-i(\omega_a + \omega_b - \omega_c)t} + c.c. \right) \tag{31}$$

so by the conservation equations Eqs. (26), (27), the polarization response of the medium produces a wave of frequency ω_d and wavevector $\vec{k}_d$ [22]. Of course, there are other terms involved in the expansion as well. For example, consider term $\chi^{(3)} \vec{E}_a \vec{E}_b \vec{E}_c e^{-i(\omega_a + \omega_b + \omega_c)t}$, which generates a polarization response at frequency $\omega_a + \omega_b + \omega_c$ in a process known as sum-frequency generation, or term $\chi^{(3)} \vec{E}_a \vec{E}_a \vec{E}_a e^{-i3\omega_a t}$ which produces a field at frequency $3\omega_a$ in a process called third-harmonic generation [22].

So why is the ω_d field produced so important? What about all the other fields? Well, a photon is emitted when a charge accelerates, so for an electromagnetic field to be produced the valance electron must transition between states. By cleverly selecting the level structure and driving beam frequencies, many undesired transitions pathways can be suppressed and a transition near frequency ω_d can be driven [22]. Rogue transitions may produce photons outside the set of frequencies $\{\omega_a, \omega_b, \omega_c, \omega_d\}$, but their number will be negligible in comparison to that of the ω_d field.

However, this is more of a theoretical hope than a scientific fact. The probability of any given competing nonlinear processes is likely to be diminished, but there are many of them. All of these additional nonlinear effects compete with FWM [34], turning FWM generation into a parameter optimization problem in which the system must be set up to support the FWM process while other maximally suppressing any other nonlinear mixing processes such as third-harmonic generation [35].

For the four-level diamond configuration atom of consideration (see Fig. 3), driving fields of transition frequency ω_{12}, ω_{13}, and ω_{24} will be applied to induce a near-ω_{34} transition within the telecom C-band frequency range.

1. *Squeezed states*

A significant property of FWM is that the additional field generated can be 'squeezed'. Optical squeezing gives a way to decrease the noise (i.e. loss in quantum information) of the resulting beam, making it a crucial factor to consider. A frequency converter which produces a signal with excessive noise is not useful in quantum information applications. According to Heisenberg's uncertainty principle, the product of the standard deviations of two canonically conjugate observables, $\hat{x}$ and $\hat{p}$ where $[\hat{x}, \hat{p}] = i\hbar$, must satisfy

$$\Delta\hat{x}\Delta\hat{p} \geq \frac{\hbar}{2}. \tag{32}$$

For light of sufficient intensity (i.e. for sufficiently many photons), one can consider its amplitude and phase as canonically conjugate variables [16]. For a generically produced beam, one would expect the standard deviations of amplitude and phase to be roughly equal. If measurement were to be performed in a single quadrature (amplitude for example), then reducing the uncertainty in that quadrature would result in a more precise overall reading.

When quantum information is stored in the amplitude and phase quadratures of light, it is called continuous variable quantum information [36]. Continuous variable measurements are typically performed with homodyne detectors, compare some input beam against a controlled probe beam and takes the difference between their readings [37].

This leads to the idea of a squeezed state in which one quadrature has its uncertainty decreased at the expense of the other, such that the uncertainty principle is still observed. It has been shown that optical squeezed states can be produced through a FWM process in optical cavities [16][32].

The operators for amplitude $\hat{X}$ and phase $\hat{P}$ are given as

$$\hat{X} = \frac{1}{2}\left(\hat{a} + \hat{a}^\dagger\right) \tag{33}$$

$$\hat{P} = \frac{1}{i2}\left(\hat{a} - \hat{a}^\dagger\right) \tag{34}$$

with associated uncertainties

$$\Delta\hat{X} = \frac{1}{2}\sqrt{1 + 2\delta\left(\hat{a}^\dagger\hat{a}\right) + 2Re\left(\delta\left(\hat{a}\hat{a}\right)\right)} \tag{35}$$

$$\Delta\hat{P} = \frac{1}{2}\sqrt{1 + 2\delta\left(\hat{a}^\dagger\hat{a}\right) - 2Re\left(\delta\left(\hat{a}\hat{a}\right)\right)} \tag{36}$$

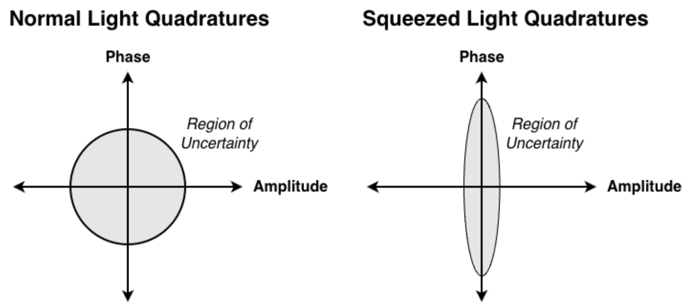

FIG. 6: Cartoon depiction of optical squeezing in the conjugate phase and amplitude quadratures with (left) unmodified light and (right) squeezed light.

where $\delta(\hat{x}\hat{y}) = \langle\hat{x}\hat{y}\rangle - \langle\hat{x}\rangle\langle\hat{y}\rangle$ [33]. This tells us that the amount of squeezing produced between quadratures is a function of $Re(\delta(\hat{a}\hat{a})) = Re(\langle\hat{a}\hat{a}\rangle - \langle\hat{a}\rangle\langle\hat{a}\rangle)$.

There are actually two forms of squeezing: single-mode and two-mode squeezing. Single-mode squeezing is typically the result of a phase relationship between emitted photon pairs of the same frequency, while in two-mode squeezing the two photons produces are of different frequencies such that there are correlations between the phase and amplitude of the two beams [32]. We are specifically concerned with single-mode squeezed states, and so any mention of squeezing will be in reference to single-mode squeezing.

It is crucial to note that any practical implementation of a system in which the production of squeezed light is attempted, that 'amplitude squeezing is maintained only if classical pump relative-intensity noise (RIN) is suppressed below shot noise' [33].

2. The effects of optical depth on four wave mixing

Recall the Beer-Lambert law of Eq. (10), which says that the field within a medium decays exponentially with $Im(\chi)$. There is another convention for expressing field decay within a medium, however. It is conventional to take the exponent as the optical depth OD such that

$$e^{-OD} = e^{-Im\left(\chi^{(1)}\right)kz/2} \tag{37}$$

such that the optical depth determines the amount of absorption by the medium [23]. However, there is an alternative definition of optical depth which is far more coarse grained. Some define optical depth as

$$OD = \nu s l \tag{38}$$

where ν is the number density of atoms, s is the resonant absorption cross-section of the medium, and l is the length of the medium [38]. By this definition, Zhang et al. state that 'an increase in [optical depth] indicates an increase in the number density of atoms, which implies an enhanced probability of FWM process occurring between optical fields and atoms' [38]. The relationship presented is that an increase in the optical depth (by increasing atomic density) is strongly correlated with an increase in the conversion efficiency of FWM.

Atomic density alkali metal vapour in a cell has the well-defined relationship with temperature T

$$\nu = \frac{1}{T}10^{A-B/T} \tag{39}$$

where A and B are material dependent constants (for rubidium vapour these are $A = 31.178$ and $B = 4040$) [39]. Therefore, there is a way to significantly increase this measure of optical depth which may be correlated with increased FWM conversion efficiency for our system of interest.

3. Four wave mixing and electromagnetically induced absorption

A less talked about, but relevant phenomenon is that of electromagnetically induced absorption (EIA): the process by which absorption of a medium is significantly increased via coherent quantum interference between driving fields and an atomic medium [40]. It has been observed that there is a correlation between EIA and increased FWM efficiency [41][42].

The theory behind EIA is out of the scope of this work, but we may apply the physics intuition built up this far to approximate it. By the Beer-Lambert law (see Eq. (10)), absorption exponentially increases with the imaginary component of electric susceptibility $Im(\chi)$. So by maximizing $Im(\chi)$, it may be possible to find approximate system behaviour corresponding to EIA. However, the spike in absorption is typically quite narrow with respect to a parameter such as field detuning [41][42], so searching for such conditions may be difficult.

Interestingly, the statement that EIA results in greater FWM conversion efficiency is in line with the statement that optical depth has a strong positive correlation with conversion efficiency [38], taking the definition of optical depth to be that which is given be Eq. (37) rather than Eq. (38) here. This supports the ladder statement across both conventions of defining optical depth, as consistent results have been found for each.

4. Four wave mixing and electromagnetically induced transparency

FWM and EIT tend to be regarded as complementary phenomena since EIT can enhance FWM [17]. If transparency is induced for the generated transmission field, then it can coherently propagate through the medium with minimal losses. In the presence of both phenomena, the polarization response of the medium would oscillate at the desired fourth frequency to coherently produce a field. In theory, quantum information imposed onto one of the input beams could be coherently absorbed by the medium and re-applied to its response field (this idea is also used in rubidium quantum memories) [32].

One dimension of consideration is as follows. Light is quantized, and any beam containing quantum information is typically weak. Individual photons arrive at the detector in a Poisson process, resulting in shot noise for the detector [1]. The random rate at which photons arrive at the detector causes signal noise. It would be highly desirable for the input beam to follow sub-Poissonian statistics (with the same mean as a Poisson distribution but a lower variance) [1] and for the beam to display photon anti-bunching (arrival events are distributed more evenly as opposed to having intensity spikes) [43].

It has been shown that applying cavity EIT to three-level atoms can produce sub-Poissonian light [1], and that applying FWM to three-level systems can produce photon anti-bunching effects [43]. Combining these techniques theoretically allows for the information of continuous variable beams to be more reliably transmitted to some measurement device, such that readings fall below the shot-noise level.

This makes the combination of FWM and EIT processes in hot atomic vapour composed of four-level diamond configuration atoms an attractive possibility. If achievable, then its implementation could lead to an efficient frequency conversion and transmission method for quantum information.

Overall, if EIA can be induced with respect to the input signal field to strengthen the output transmission field produced via FWM, and EIT can be induced to decrease the self-absorption of the transmission field, then conversion efficiency could theoretically be maximized. That is, assuming all these phenomena can occur simultaneously in a Doppler broadened medium.

C. Quantum Methods

In order to make any quantitative statements about the systems and phenomena we have been considering thus far, we must have a mathematical framework for describing them. To this end, we may apply the Lindblad master equation to find steady-state solutions for our system, and the Maxwell-Bloch equations to find the electric field strength within the medium. We may algorithmically combine these approaches to find the steady state of each region within a theoretical sample of ^{87}Rb by propagating the electromagnetic fields throughout.

1. Lindblad Master equation

For any quantum system, the total amount of information must be conserved, yet when dealing with atomic systems states can decay and the corresponding photon can be lost to the environment. This type of loss must be accounted for and is not included in typical quantum mechanical descriptions of systems unless the entire environment (i.e. the rest of the universe) is also described. This information would all be contained in the von Neumann density matrix time evolution equation

$$\frac{d\hat{\rho}_T}{dt} = -i\left[\hat{H}_T, \hat{\rho}_T\right] \tag{40}$$

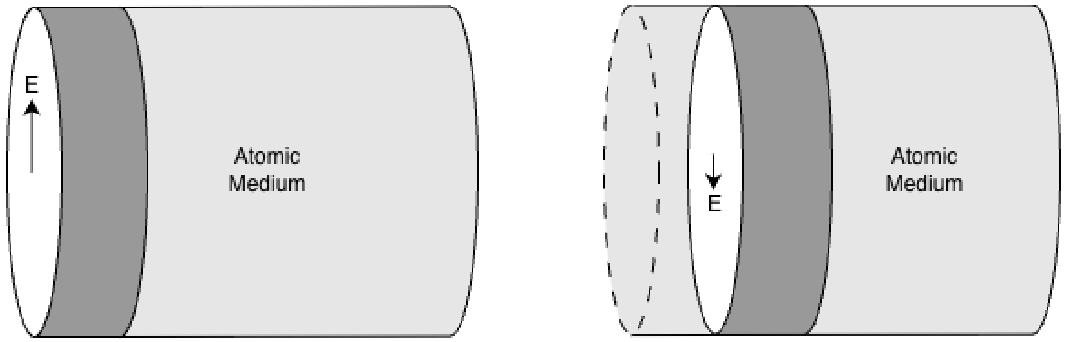

FIG. 7: (Left) the electric field experienced by a point within a region of an atomic medium and (right) a different electric field experienced by a neighboring region.

where $\hat{\rho}_T$ is the total system density matrix and $\hat{H}_T$ is the total Hamiltonian [44]. Needless to say, solving this is not an option. By tracing out the environment to isolate the subsystem $\hat{\rho}_S = Tr_E\left[\hat{\rho}_T\right]$, applying the Born approximation where all interactions between the system and environment are assumed to be weak such that they may be traced out for simplification, applying the Markov approximation assuming that the system is memoryless (past states do not affect future states), then finally applying the rotating wave approximation where rapidly evolving components are averaged out and neglected (since time evolution can be factored in terms of complex exponentials $e^{-i\omega t}$) such that the resulting density matrix is strictly non-negative [44][45]. Additional simplification yields the Lindblad master equation as

$$\frac{d\hat{\rho}_S}{dt} = -i\left[\hat{H}_S, \hat{\rho}_S\right] + \sum_\alpha \gamma_\alpha \left(\hat{L}_\alpha \hat{\rho}_S \hat{L}_\alpha^\dagger - \frac{1}{2}\left\{\hat{L}_\alpha^\dagger \hat{L}_\alpha, \hat{\rho}_S\right\}\right) \tag{41}$$

where $\hat{L}_\alpha$ is the α^{th} dissipative operator (or jump operator), γ_α is the corresponding decay rate, and $\left\{\hat{a}, \hat{b}\right\} = \hat{a}\hat{b} + \hat{b}\hat{a}$ is the anti-commutator [46].

By solving for $d\hat{\rho}_S/dt = 0$ one may find the steady state density matrix of the system. For large systems this is typically done via computational methods. We are interested in the steady state regime as we will assume that the timescale of laser pulses is much greater than the transient response time of the medium. For short laser pulses, considering transient dynamics is important [23], but as a first step towards frequency conversion of continuous variable quantum information longer pulses should be used. With refinement, shorter pulse durations may be achievable, and then transient dynamics would become relevant, but that is a future engineering challenge.

2. *Maxwell-Bloch equations*

There is one problem with solving for the steady state density matrix of an atomic sample alone: it neglects propagation effects [23][47]. Even though the entire system may be in a steady state that does not mean that the field strengths are uniform throughout. Any neighboring regions within the sample may experience different fields. Assume that the system can be broken up into cross-sectional regions which all experience approximately the same field strength, which would be the case if the beams were co-propagating or counter-propagating plane waves under the paraxial approximation [48]. Then one could consider the field strength of neighboring slices, see Fig. 7.

The Maxwell-Bloch equations can be derived and applied to account for propagation through the medium. Under the assumptions that the medium is non-magnetic, has no free charges, and no currents due to free charges, we may use Maxwell's equations to obtain the expression

$$\nabla \times \nabla \times \vec{E} + \frac{n^2}{c^2}\frac{\partial^2}{\partial t^2}\vec{E} = -\mu_0 \frac{\partial^2}{\partial t^2}\vec{P} \tag{42}$$

where n is the index of refraction [22][48]. We may express $\nabla \times \nabla \times \vec{E} = \nabla\left(\nabla \cdot \vec{E}\right) - \nabla^2\vec{E}$, and apply the slowly varying amplitude approximation such that

$$\vec{E} = \vec{\varepsilon}(z,t)e^{-i(\omega t - kz)} + c.c. \tag{43}$$

where the electromagnetic amplitude envelope $\vec{\varepsilon}(z,t)$ varies in time and space much slower than the carrier wave such that [22][48]

$$\left|\frac{\partial^2}{\partial t^2}\vec{\varepsilon}(z,t)\right| \ll \left|\omega\frac{\partial}{\partial t}\vec{\varepsilon}(z,t)\right| \tag{44}$$

$$\left|\frac{\partial^2}{\partial z^2}\vec{\varepsilon}(z,t)\right| \ll \left|k\frac{\partial}{\partial z}\vec{\varepsilon}(z,t)\right| \tag{45}$$

$$\tag{46}$$

In this regime, the $\nabla\left(\nabla\cdot\vec{E}\right)$ term then becomes negligibly small such that we may write [22]

$$\left(\nabla^2 - \frac{n^2}{c^2}\frac{\partial^2}{\partial t^2}\right)\vec{E} = \mu_0\frac{\partial^2}{\partial t^2}\vec{P}. \tag{47}$$

Then since we are assuming plane wave propagation in the paraxial approximation we have

$$\left(\frac{\partial^2}{\partial z^2} - \frac{n^2}{c^2}\frac{\partial^2}{\partial t^2}\right)\vec{E} = \mu_0\frac{\partial^2}{\partial t^2}\vec{P}. \tag{48}$$

By evaluating our derivatives we find that

$$\frac{\partial^2}{\partial t^2}\vec{E} = \left(-2i\omega\frac{\partial}{\partial t}\vec{\varepsilon}(z,t) + \frac{\partial^2}{\partial t^2}\vec{\varepsilon}(z,t) - \omega^2\vec{\varepsilon}(z,t)\right)e^{-i(\omega t-kz)} + c.c. \tag{49}$$

$$\frac{\partial^2}{\partial z^2}\vec{E} = \left(2ik\frac{\partial}{\partial z}\vec{\varepsilon}(z,t) + \frac{\partial^2}{\partial z^2}\vec{\varepsilon}(z,t) - k^2\vec{\varepsilon}(z,t)\right)e^{-i(\omega t-kz)} + c.c. \tag{50}$$

and by the slowly varying amplitude approximation we have that the $\partial_t^2\vec{\varepsilon}(z,t)$ $\partial_z^2\vec{\varepsilon}(z,t)$ terms are negligible. So we have

$$\left(2ik\frac{\partial}{\partial z}\vec{\varepsilon}(z,t) - k^2\vec{\varepsilon}(z,t)\right)e^{-i(\omega t-kz)} - \frac{n^2}{c^2}\left(-2i\omega\frac{\partial}{\partial t}\vec{\varepsilon}(z,t) - \omega^2\vec{\varepsilon}(z,t)\right)e^{-i(\omega t-kz)} + c.c. = \mu_0\frac{\partial^2}{\partial t^2}\vec{P}$$

$$\left(\frac{2in^2\omega}{c^2}\frac{\partial}{\partial t}\vec{\varepsilon}(z,t) + 2ik\frac{\partial}{\partial z}\vec{\varepsilon}(z,t)\right)e^{-i(\omega t-kz)} + \left(\frac{n^2\omega^2}{c^2} - k^2\right)\vec{\varepsilon}(z,t)e^{-i(\omega t-kz)} + c.c. = \mu_0\frac{\partial^2}{\partial t^2}\vec{P}.$$

The phase velocity of light may be expressed as

$$v_p = \frac{\omega}{k} = \frac{c}{n} \tag{51}$$

so we have the relation

$$k = \frac{n\omega}{c} \tag{52}$$

$$\frac{n^2\omega^2}{c^2} - k^2 = 0 \tag{53}$$

thus our expression simplifies to

$$\left(\frac{2ink}{c}\frac{\partial}{\partial t}\vec{\varepsilon}(z,t) + 2ik\frac{\partial}{\partial z}\vec{\varepsilon}(z,t)\right)e^{-i(\omega t-kz)} + c.c. = \frac{1}{\epsilon_0 c^2}\frac{\partial^2}{\partial t^2}\vec{P}$$

$$\left(\frac{\partial}{\partial t}\vec{\varepsilon}(z,t) + \frac{c}{n}\frac{\partial}{\partial z}\vec{\varepsilon}(z,t)\right)e^{-i(\omega t-kz)} + c.c. = -i\frac{1}{2nk\epsilon_0 c}\frac{\partial^2}{\partial t^2}\vec{P}$$

We can then also apply the slowly varying envelope approximation to $\vec{P}$ such that

$$\vec{P} = \vec{\pi}(z,t)e^{-i(\omega t-kz)} + c.c. \tag{54}$$

$$\frac{\partial^2}{\partial t^2}\vec{P} = \left(-2i\omega\frac{\partial}{\partial t}\vec{\pi}(z,t) + \frac{\partial^2}{\partial t^2}\vec{\pi}(z,t) - \omega^2\vec{\pi}(z,t)\right)e^{-i(\omega t-kz)} + c.c. \tag{55}$$

then we have that

$$\left|\frac{\partial^2}{\partial t^2}\vec{\pi}(z,t)\right| \ll \left|\omega\frac{\partial}{\partial t}\vec{\pi}(z,t)\right| \ll \left|\omega^2\vec{\pi}(z,t)\right| \tag{56}$$

so we may neglect the other terms to obtain

$$\frac{\partial^2}{\partial t^2}\vec{P} \approx -\omega^2\vec{\pi}(z,t)e^{-i(\omega t-kz)} + c.c. \tag{57}$$

Thus, we may write

$$\frac{\partial}{\partial t}\vec{\varepsilon}(z,t) + v_p\frac{\partial}{\partial z}\vec{\varepsilon}(z,t) = i\frac{\omega^2}{2nk\epsilon_0 c}\vec{\pi}(z,t).$$

All vectors will be aligned in an isotropic gaseous media and $k = n\omega/c$, so we can write

$$\begin{aligned}
\frac{\partial}{\partial t}\varepsilon(z,t) + v_p\frac{\partial}{\partial z}\varepsilon(z,t) &= i\frac{\omega}{2n^2\epsilon_0}\pi(z,t) \\
\frac{1}{v_p}\frac{\partial}{\partial t}\varepsilon(z,t) + \frac{\partial}{\partial z}\varepsilon(z,t) &= i\frac{1}{v_p}\frac{\omega}{2n^2\epsilon_0}\pi(z,t) \\
\frac{1}{v_p}\frac{\partial}{\partial t}\varepsilon(z,t) + \frac{\partial}{\partial z}\varepsilon(z,t) &= i\frac{\omega}{2n\epsilon_0 c}\pi(z,t).
\end{aligned} \tag{58}$$

This can be simplified even further by moving to a co-moving frame via the transformation $\zeta = z$, $\tau = t - z/v_p$ to obtain [23]

$$\frac{\partial}{\partial \zeta}\varepsilon(\zeta,\tau) = i\frac{\omega}{2n\epsilon_0 c}\pi(\zeta,\tau) \tag{59}$$

Finding an expression for $\partial_\zeta\varepsilon(\zeta,\tau)$ for each relevant field gives a set of equations dubbed the Maxwell-Bloch equations. By finding a way to express the polarization strength in terms of the electric field strength for the relevant system, then we may use it to find the electric field strength in the neighboring region, which can then be used to find the polarization in said region, and so on. This would allow us to simulate field propagation through the medium and find the output strength.

3. *The iterative Lindblad-Maxwell-Bloch solution method*

The Lindblad master equation (Eq. (41)) can be used in tandem with the Maxwell-Bloch equations (Eq. (59)) to solve for the steady state density matrix, which may be applied to find the expected polarization, then find the electric field strength at the next cross-sectional region. This process can be performed iteratively throughout the medium via computational methods in order to find the desired field strength at the end of the medium. For an idler field, this could then tell us how much field intensity is produced as output via FWM. Due to the use of a co-moving frame, no time evolution is required. This all comes together in Alg. 1

Algorithm 1 The Lindblad-Maxwell-Bloch Algorithm

1: Initialize the electric field strength, frequency, and propagation direction for each field at location $\zeta = 0$ along the atomic medium. Also initialize step size $\Delta\zeta$ and sample length Z (stopping condition).
2: Solve the Lindblad master equation Eq. (41) for the steady state density matrix satisfying $d\hat{\rho}/dt = 0$.
3: Use the steady state density matrix $\hat{\rho}$ to find the expected polarization amplitude required in Eq. (59) and use it to update the relevant electric field at $\zeta + \Delta\zeta$ as $\varepsilon \to \varepsilon + \partial_\zeta\varepsilon$ via approximation.
4: Update $\zeta \to \zeta + \Delta\zeta$.
5: Repeat steps 2 through 4 until $\zeta \geq Z$.
6: Return ε and $\hat{\rho}$.

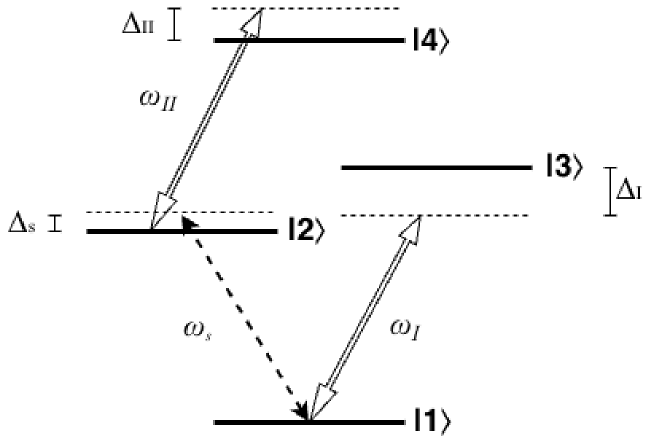

FIG. 8: A four-level diamond configuration atom driven by three external fields where fields ω_I and ω_{II} are strong pump fields and field ω_s is the weak input signal field. These fields have respective detunings Δ_I, Δ_{II}, and Δ_s.

III Problem Definition

The overarching goal of this research is to find out whether a hot gas composed of four-level diamond configuration atoms (see Fig. 3) can be resonantly driven by three external electromagnetic fields (one of which is a weak signal beam encoded with continuous variable quantum information) such that a FWM process occurs to produce a fourth field with a frequency in the telecom C-band range of 1530 nm to 1565 nm. Specifically, the field produced must be optically squeezed via the FWM process such that quantum information may be converted then transmitted with minimal noise. It would be ideal if the medium experiences EIT with respect to the frequency of the weak signal beam and/or the output telecom beam.

By selecting some system parameterization (such as initial beam strengths, detunings, gas temperature, cell length, etc.) with the output fields strength initially set to zero, then applying the Maxwell-Bloch equations (Eq. (59)) to propagate the fields through the medium in the steady-state limit such that the Lindbladian may be used to find the overall system state (Eq. (41)), the output field strength and conversion efficiency of the input beam may be found. Via the same process, the density matrix describing the output state may be found then applied to obtain a quantifiable measure of squeezing from Eqs. (35), (36). The density matrix may also be used to solve for the dipole moment of the medium, from which the $\chi^{(1)}$ susceptibility can be obtained with respect to a select frequency (Eq. (9)), which can then be applied to the Beer-Lambert law (Eq. (10)) to find the field decay within the cavity. A relatively small value of $Im\left(\chi^{(1)}\right)$ indicates that there is little decay in the relevant field such that EIT may be present, while a large $Im\left(\chi^{(1)}\right)$ value gives evidence to the contrary.

A. Doppler Broadening and Doppler Insensitivity

It would be highly advantageous if we could show that the diamond-configuration system can be Doppler insensitive. When driving the three transitions such that a telecom wavelength field is promoted we can depict our system as in Fig. 8. This is essentially the combination of a V-system and a ladder system, or at least that is how some interpret it [42]. Lee et al. [42] state that the two-photon coherence of the V-type and Ξ-type subsystems can be made Doppler insensitive when the (ω_s) and strong pump I (ω_I) co-propagate (for V-type) and when the weak signal (ω_s) and strong pump II (ω_{II}) beams counter-propagate (for Ξ-type). They extend this to line of reasoning to state that both conditions should hold in order to obtain Doppler insensitive three-photon coherence in a Doppler broadened medium. They found that Doppler-free three-photon electromagnetically induced absorption (EIA) of the weak signal beam was present under these conditions. Additionally, they found that increased EIA intensity was strongly correlated with increased FWM field production. It should be noted that Lee et al. used a slightly different atomic configuration in the ^{87}Rb spectrum.

Thus, there is evidence to suggest that the system may be Doppler insensitive by aligning $\vec{k}_s$ and $\vec{k}_I$ to be parallel while aligning $\vec{k}_{II}$ to be antiparallel to both $\vec{k}_s$ and $\vec{k}_I$. The momentum conservation equation then tells us that we must have

$$\left|\vec{k}_i\right| = \left|\vec{k}_s\right| + \left|\vec{k}_I\right| - \left|\vec{k}_{II}\right| \tag{60}$$

where $\vec{k}_i$ is the idler beam field which we wish to induce via FWM. This also tells us that $\vec{k}_i$ would need to co-propagate with $\vec{k}_{II}$.

This setup is mutually beneficial for Maxwell-Bloch solution steps. Consider solving for the change in electric field strength through a slice of an atomic cell. By taking $\vec{k}_s$ (or any k-vector for that matter) as the normal vector of the slice plane we can justify the approximation that the signal, pump I, and pump II field strengths are all uniform about the plane. Additionally, for strong pump I and pump II fields, we can assume that their intensities do not change significantly through propagation (approximately fixed beam strength). Therefore, we can justifiably state that we know the field strength of all three input beams at the location of the first slice. This allows us to then find the field strength for all subsequent slices of the medium in an iterative process.

However, if it is not possible to enforce Eq. (60), then Doppler broadening must be modelled. To do this, consider the velocity vector of the m^{th} atom, $\vec{v}^{(m)}$. This velocity will be pulled from a multi-variate Gaussian distribution as

$$\vec{v}^{(m)} = v_x^{(m)}\hat{x} + v_y^{(m)}\hat{y} + v_z^{(m)}\hat{z} \tag{61}$$

$$v_x^{(m)} \sim \mathcal{N}(0, \sigma_x) \tag{62}$$

$$v_y^{(m)} \sim \mathcal{N}(0, \sigma_y) \tag{63}$$

$$v_z^{(m)} \sim \mathcal{N}(0, \sigma_z) \tag{64}$$

$$\tag{65}$$

where $\mathcal{N}(\mu, \sigma)$ represents a Gaussian distribution of mean μ and standard deviation σ, where we will take

$$\sigma_x = \sigma_y = \sigma_z = \sqrt{\frac{m}{k_b T}} \tag{66}$$

for our isotropic medium with k_b being the Boltzmann constant. We then have the PDF as

$$f(v)\Big|_{\mu=0,\sigma=\sqrt{\frac{m}{k_bT}}} = \sqrt{\frac{m}{2\pi k_b T}} \exp\left(-\frac{mv^2}{2k_v T}\right). \tag{67}$$

Consider also any electromagnetic field of lab frame wavevector $\vec{k}$ from the perspective of the m^{th} atom. If the field is of frequency ω, then the Doppler shift experienced by the atom is given by

$$\omega' = \omega - \vec{k} \cdot \vec{v} \tag{68}$$

Consider that the index of refraction is given by $n(\omega) = c/v_p(\omega)$ and we have the phase velocity as $v_p(\omega) = \omega/k$. Combining these equations gives us the dispersion relation

$$\vec{k} = \frac{n(\omega)\omega}{c} \frac{\vec{k}}{|\vec{k}|} \tag{69}$$

such that we then have the Doppler shift as

$$\omega' = \omega - \frac{n(\omega)\omega}{c} \frac{\vec{k}}{|\vec{k}|} \cdot \vec{v} \tag{70}$$

So if a frequency dependent dispersion relation for our system can be derived, then an accurate Doppler shift may be applied. However, we instead take the index of refraction to be approximately constant in this context as to obtain

$$\omega' \approx \omega - \frac{n\omega}{c} \frac{\vec{k}}{|\vec{k}|} \cdot \vec{v} \tag{71}$$

but note that a derivation of $n(\omega)$ for inclusion in this equation would improve the accuracy of any Doppler broadening results. This derivation is out of the scope of this thesis.

This single atom will have a different Doppler shift with respect to each beam. The shift is velocity direction dependent so all three velocity coordinates must be sampled with the dot product being taken with respect to each $\vec{k}$.

Suppose that we apply the Doppler shift to an atom and then obtain its steady-state density matrix. Each atom experiences a different Doppler shift, so each will need to be calculated based on its sampled velocity vector. All of these density matrices can be added together and averaged as

$$\overline{\rho} = \frac{1}{N} \sum_{m=1}^{N} \hat{\rho}^{(m)}(v_x, v_y, v_z) \tag{72}$$

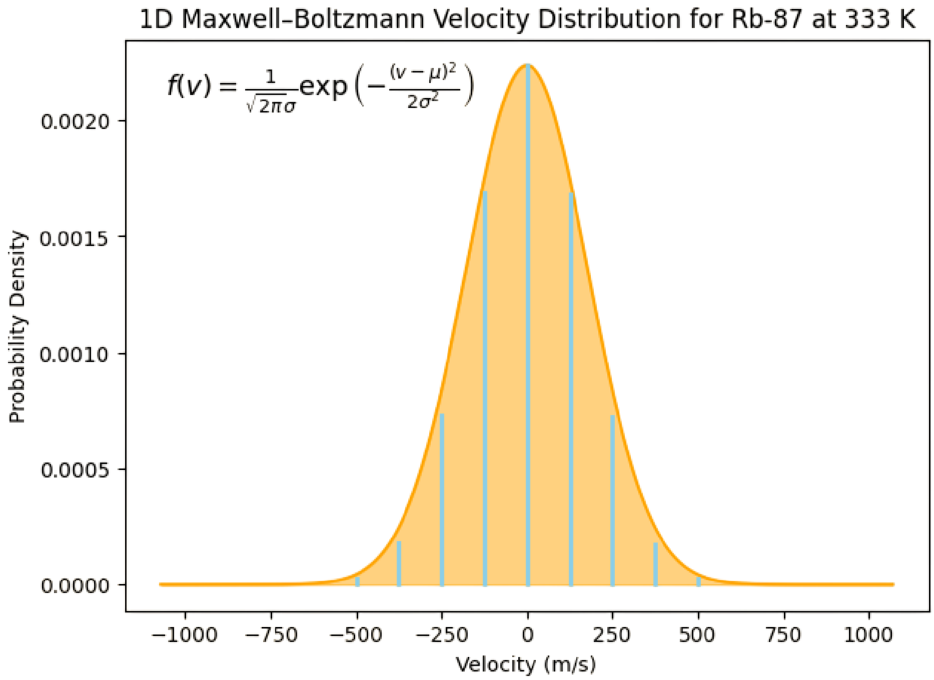

FIG. 9: Gaussian velocity distribution separated into 9 velocity subclasses marked by blue lines. Calculated for ^{87}Rb at 333 K.

However, sampling Avagadro's number of atoms is not feasible. Instead, consider the average density matrix in the continuous limit and apply the Gaussian density function

$$\overline{\rho} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(v_x) f(v_y) f(v_z) \hat{\rho}^{(m)}(v_x, v_y, v_z) dv_x dv_y dv_z \tag{73}$$

Each density function can be broken up into a weighted slice such that we can approximately compute the mean density matrix as

$$\overline{\rho} \approx \sum_{v_x, v_y, v_z} w(v_x) w(v_y) w(v_z) \hat{\rho}^{(m)}(v_x, v_y, v_z) \tag{74}$$

where the range of velocities is broken down into a sequence of M evenly spaced velocity subclasses $v_x^{(1)} \leq v_x^{(2)} \leq \cdots \leq v_x^{(M)}$ with $v_x^{(i+1)} - v_x^{(i)} = \eta$, $\forall i$, such that $w\left(v_x^{(i)}\right) = f\left(v_x^{(i)}\right)\eta$. Similarly, for v_y and v_z. Note that unevenly spaced velocity subclasses can also be useful, but is problem specific.

This gives a computational speed up, yet M^3 steady state density matrices must be computed, which is still quite costly. If we, however, assume that all beams are propagating along the same axis as is the case in Eq. (60), then a single velocity value can be sampled (as opposed to three) as $v \sim \mathcal{N}(0, \sigma)$ could be used for all fields. We then need to calculate just M steady state density matrices.

$$\overline{\rho} \approx \sum_{v} w(v) \hat{\rho}^{(m)}(v) \tag{75}$$

Similarly, if beams are separated into two sets, such that all wavevectors of one set are parallel with one another, but any wavevectors from different sets are not parallel, then M^2 density matrices would need to be computed.

Note that any density matrix obtained by this method should be renormalized.

B. System Hamiltonian

In Appendix A we derive the Hamiltonian for our four-level atom by following a similar procedure to that of Du et al. [9]. The Hamiltonian is obtained as

$$\tilde{H} = \hbar \begin{pmatrix} 0 & \Omega_s(t)\hat{a}^\dagger & \Omega_I(t) & 0 \\ \Omega_s^*(t)\hat{a} & \Delta_s & 0 & \Omega_{II}(t) \\ \Omega_I^*(t) & 0 & \Delta_I & \Omega_i(t)\hat{b}^\dagger \\ 0 & \Omega_{II}^*(t) & \Omega_i^*(t)\hat{b} & \Delta_{II}+\Delta_s \end{pmatrix} + \hbar\Delta_a \hat{a}^\dagger\hat{a} + \hbar\Delta_b \hat{b}^\dagger\hat{b} \tag{76}$$

where $\Omega_x(t),\ x \in \{s, i, I, II\}$ are the Rabi frequencies for each, and we have the detunings defined as

$$\Delta_s = \omega_{12} - \omega_s \tag{77}$$
$$\Delta_I = \omega_{13} - \omega_I \tag{78}$$
$$\Delta_{II} = \omega_{24} - \omega_{II} \tag{79}$$
$$\Delta_i = \omega_{34} - \omega_i \tag{80}$$
$$\Delta_a = \omega_a - \omega_s \tag{81}$$
$$\Delta_b = \omega_b - \omega_i \tag{82}$$

where ω_s and ω_i are the weak signal and idler beams frequencies, respectively, ω_I and ω_{II} are the strong pump beam frequencies, and ω_a and ω_b are the resonant cavity frequencies. This Hamiltonian couples the 4-state atomic basis with the distinct electromagnetic field modes of $\hat{a}$ and $\hat{b}$ at respective frequencies ω_a (tuned near ω_{12}) and ω_b (tuned near ω_{34}), respectively. Note that we choose these frequencies for the $\hat{a}$ and $\hat{b}$ field modes because we cannot simply assume that the cavity is perfectly on resonance with the driving field or the relevant atomic transition frequency.

We will be working with this Hamiltonian in the steady-state limit such that we can assume that each $\Omega_x,\ x \in \{s, i, I, II\}$ is constant for all times t. These values will, however, vary in space. Our Hamiltonian may then be expressed

$$\tilde{H} = \hbar \begin{pmatrix} 0 & \Omega_s\hat{a}^\dagger & \Omega_I & 0 \\ \Omega_s^*\hat{a} & \Delta_s & 0 & \Omega_{II} \\ \Omega_I^* & 0 & \Delta_I & \Omega_i\hat{b}^\dagger \\ 0 & \Omega_{II}^* & \Omega_i^*\hat{b} & \Delta_{II}+\Delta_s \end{pmatrix} + \hbar\Delta_a \hat{a}^\dagger\hat{a} + \hbar\Delta_b \hat{b}^\dagger\hat{b} \tag{83}$$

It is important to note that some of the definitions involved in this Hamiltonian deviate from the conventions. To see this, first consider a real-valued electromagnetic field

$$\vec{E}(z,t) = \vec{\varepsilon}_0 \cos(\omega t - kz)$$
$$\vec{E}(z,t) = \frac{\vec{\varepsilon}_0}{2}\left(e^{-i(\omega t - kz)} + e^{i(\omega t - kz)}\right).$$

where $\vec{\varepsilon}_0$ acts as a slowly-varying envelope which changes slowly on the timescale of oscillation to allow for steady-state dynamics to occur. In Appendix A, each $\vec{\varepsilon}$ was taken to include the factor of 2 to simplify calculations. Also by convention, weak quantized fields are expressed with the factor of 2 absorbed where the symbol g takes the place of Ω to represent the atom-field coupling strength. For consistency, we opt to not apply these conventions to avoid confusion, but make a distinct note of it.

There are still alternative Hamiltonians which may be applicable to this problem. Notice that the signal and idler beams do not populate the cavity mode in the current form. To handle this, Souza et al. include a field population term ($\varepsilon\hat{a} + h.c.$) to their Hamiltonian [1], which introduces a degree of realistic noise. We can do the same for our two cavity modes, yielding

$$\tilde{H}_{ab} = \hbar \begin{pmatrix} 0 & \Omega_s\hat{a}^\dagger & \Omega_I & 0 \\ \Omega_s^*\hat{a} & \Delta_s & 0 & \Omega_{II} \\ \Omega_I^* & 0 & \Delta_I & \Omega_i\hat{b}^\dagger \\ 0 & \Omega_{II}^* & \Omega_i^*\hat{b} & \Delta_{II}+\Delta_s \end{pmatrix} + \hbar\Delta_a \hat{a}^\dagger\hat{a} + \hbar\Delta_b \hat{b}^\dagger\hat{b} + \hbar\left(\varepsilon_s\hat{a} + \varepsilon_s^*\hat{a}^\dagger\right) + \hbar\left(\varepsilon_i\hat{b} + \varepsilon_i^*\hat{b}^\dagger\right) \tag{84}$$

On top of this, we must also take the use of cavities into consideration and ask: is it practical to use resonant cavities for this system? Recall that the condition for Doppler insensitivity is that $\vec{k}_s$ and $\vec{k}_I$ must co-propagate, $\vec{k}_i$ and $\vec{k}_{II}$ must co-propagate, and $\vec{k}_s$ and $\vec{k}_i$ must counter-propagate such that all four beams travel along the same line. But how does one place two resonant cavities of distinct frequencies along the same line? It is possible that there is a way to cleverly engineer a system to alleviate this issue, but considering such a problem is out of the scope of this report. Instead, we present the other possible Hamiltonians obtained by de-quantizing one of the two fields.

$$\tilde{H}_a = \hbar \begin{pmatrix} 0 & \Omega_s \hat{a}^\dagger & \Omega_I & 0 \\ \Omega_s^* \hat{a} & \Delta_s & 0 & \Omega_{II} \\ \Omega_I^* & 0 & \Delta_I & \Omega_i \\ 0 & \Omega_{II}^* & \Omega_i^* & \Delta_{II} + \Delta_s \end{pmatrix} + \hbar \Delta_a \hat{a}^\dagger \hat{a} + \hbar \left(\varepsilon_s \hat{a} + \varepsilon_s^* \hat{a}^\dagger \right) \tag{85}$$

$$\tilde{H}_b = \hbar \begin{pmatrix} 0 & \Omega_s & \Omega_I & 0 \\ \Omega_s^* & \Delta_s & 0 & \Omega_{II} \\ \Omega_I^* & 0 & \Delta_I & \Omega_i \hat{b}^\dagger \\ 0 & \Omega_{II}^* & \Omega_i^* \hat{b} & \Delta_{II} + \Delta_s \end{pmatrix} + \hbar \Delta_b \hat{b}^\dagger \hat{b} + \hbar \left(\varepsilon_i \hat{b} + \varepsilon_i^* \hat{b}^\dagger \right) \tag{86}$$

$\tilde{H}_{ab}$, $\tilde{H}_a$, and $\tilde{H}_b$, are the main systems of concern. Technically, we could de-quantize both weak fields, but then we would have no meaningful way to quantify FWM or squeezing. The system described by $\tilde{H}_b$ may be the most relevant since it allows us to quantify the amount of squeezing via expectation values, and it realistically involves only a single cavity. System $\tilde{H}_{ab}$ could be applicable, but (likely) only if the possible benefit of Doppler insensitivity is abandoned.

C. Lindbladian and Maxwell-Bloch Equations

The only step left in expressing the Lindbladian for our system is specifying the decay operators. We can obtain them by simply considering each decay cannel with an associated decay rate. Consider $\tilde{H}_{ab}$. Atomic state $|4\rangle$ can decay to states $|2\rangle$ or $|3\rangle$, $|3\rangle$ can decay to $|1\rangle$, and $|2\rangle$ can decay to $|1\rangle$. Let these decay operators be denoted as $\hat{L}_{34} = \hat{\sigma}_{34}$, $\hat{L}_{24} = \hat{\sigma}_{24}$, $\hat{L}_{13} = \hat{\sigma}_{13}$, and $\hat{L}_{12} = \hat{\sigma}_{12}$, respectively, each with decay rates Γ_{34}, Γ_{24}, Γ_{13}, and Γ_{12}, respectively. We must also consider cavity decay for both quantized fields. The decay operator for the cavity field of frequency ω_a is $\hat{L}_a = \hat{a}$ with decay rate Γ_a and that for the cavity field of frequency ω_b is $\hat{L}_b = \hat{b}$ with decay rate Γ_b.

With these and $\tilde{H}_{ab}$ we have all the components required to fully express the Lindblad master equation Eq. (41). We may also specify that of $\tilde{H}_a$ by simply dropping $\hat{L}_b = \hat{b}$ and Γ_b; or $\tilde{H}_b$ by dropping $\hat{L}_a = \hat{a}$ and Γ_a. However, we will not do so here as our goal is not to produce an analytic solution. Instead, we will opt to use highly efficient computational methods to solve for the steady state density matrix.

The mean Doppler broadened density matrix can be obtained via Eq. (74) in general or by Eq. (75) when the Doppler insensitive phase matching condition of Eq. (60) is enforced.

As discussed in Sec. II C 3, this density matrix will be obtained in an iterative process then used in the Maxwell-Bloch equations to solve for the change in Rabi frequency at the next point within the medium. The intuition behind the Maxwell-Bloch equations was given discussed in Sec. II C 2, but the specific equations for the system we are considering were derived in Appendix B to be

$$\partial_\zeta \Omega_s(\zeta, \tau) \approx i \frac{\omega_s \nu d_{12}^2}{2 v_g \epsilon_0 \hbar} \overline{\rho}_{12} \tag{87}$$

$$\partial_\zeta \Omega_s(\zeta, \tau) \approx i \frac{\omega_i \nu d_{34}^2}{2 v_g \epsilon_0 \hbar} \overline{\rho}_{34} \tag{88}$$

where v_g is group velocity, ν is atomic density, and we are in the co-moving frame with $\zeta = z$ and $\tau = t - z/v_g$. It was noted in the appendix that these equations hold whether or not the noise term is included in the Hamiltonian.

By propagating the field throughout the entire cell, the field strength of the expectation value of the number operator $\hat{a}^\dagger \hat{a}$ for each field could be obtained using the final mean density matrix, the quadrature noise and the amount of squeezing can be obtained using Eqs. (35),(36).

IV Computational Methods

In order to handle the massive amount of information involved with such a quantized system, computational methods will be applied. Introductory computations with the QuTip python package will be performed to reproduce known results and investigate how various phenomena, then a set of computational tools based on the contents of Sec. III will be introduced.

A. Learning to use QuTiP

QuTiP, standing for quantum toolbox in python, is an open source library for simulating quantum systems [49]. Our main uses of this library will be for creating operators of the `Qobj` class in the form of finite-dimensional tensors, applying the Lindblad master equation to find the steady state density matrix satisfying $\partial_t \hat{\rho} = 0$ via the `steadystate` function, and taking the expectation value of various operators $\langle \hat{A} \rangle = Tr(\rho \hat{A})$ via the `expect` function.

In order to work up to simulating a four-level atom with multiple Fock bases for photons of distinct frequency modes with dissipative Lindbladian operators, we will begin with a simpler model and reproduce known results, then gradually increase complexity towards the target system.

1. *The Jaynes-Cummings model*

In the Jaynes-Cummings model we consider a two-level atom within a resonant cavity (see Fig. 10) [18]. The field of a single photon within the cavity is quite significant as it will bounce back and forth between cavity walls, building up a non-negligible field. So we must quantize photons in the Fock basis. The cavity is finely tuned to a resonant frequency ω_c such that any other frequency of light destructively interferes with itself. This leads to a simplification for our Fock basis which may be restricted to photons of a single frequency. There can be a detuning between the frequencies, but we will consider the case of perfect resonance here $\omega = \omega_a = \omega_c$.

FIG. 10: Depiction of the system for a Jaynes-Cummings model with (left) a resonant cavity containing a single atom in the presence of a driving electromagnetic field resonant with the cavity frequency of ω, with (right) the atomic system depicted as a two-level atom of transition frequency ω_a.

The states of our system live within the combined Hilbert space $\mathcal{H}_{atom} \otimes \mathcal{H}_{photon}$, and so states are described by $|atom\rangle \otimes |photons\rangle$ where $|atom\rangle$ has two states and $|photon\rangle$ describes all n-photon states. Consider a state containing N photons worth of energy. Then the possible states of the system are $|{\downarrow}\rangle \otimes |N\rangle$ and $|{\uparrow}\rangle \otimes |N-1\rangle$.

The Hamiltonian describing this system is given by [18][50]

$$\hat{H}_{JC} = \hat{H}_{Atom} + \hat{H}_{Photon} + \hat{H}_{Interaction}. \tag{89}$$

The atomic term is given by

$$\begin{aligned} \hat{H}_{Atom} &= \frac{\hbar\omega}{2}\hat{\sigma}_z \\ &= \frac{\hbar\omega}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \end{aligned} \tag{90}$$

Code Block 1: Python code for the Jaynes-Cummings Model

```
# Initial Arguments
w_a = 2.0
w_c = 2.0
W = 0.02
gamma = 0.005
xi = 0.0025
eta = 0.05
h = 1.0
N = 10 # Truncate the Fock basis to N photons
times = np.linspace(0, 2500, 10000)

# Set up operators in the relevant basis
a = tensor(qeye(2), destroy(N)) # Acts on photon Fock basis with N states as
    lowering operator
s_minus = tensor(destroy(2), qeye(N)) # Acts on atomic basis with 2 states as
    lowering operator
s_atom = tensor(Qobj([[0, 0], [0, 1]]), qeye(N)) # Acts on atomic basis with a
    shifted ground states

# Set up the Hamiltonian
H = (h * w_a) * s_atom + (h * w_c) * a.dag() * a + (h * W)/2 * (s_minus.dag() *
    a + s_minus * a.dag())

# Set up dissipative operator for a lossy cavity
cavity_decay = np.sqrt(gamma) * a
c_ops = [cavity_decay]

# Set up the expectation value operators
number_operator = a.dag() * a
atomic_state_operator = s_minus.dag() * s_minus
e_ops = [number_operator, atomic_state_operator]

# Start the system in an initially excited state with 0 photons
psi_0 = tensor(basis(2,1), basis(N, 0))

# Time evolve the system using qutip mesolve
result = mesolve(H, psi_0, times, c_ops=c_ops, e_ops=e_ops)
photons = result.expect[0]
```

The photonic term is given by

$$\hat{H}_{Photon} = \hbar\omega\left(\hat{a}^{\dagger}\hat{a} + \frac{1}{2}\right).$$

where we may shift by the zero-point energy since it is a constant term to obtain

$$\hat{H}_{Photon} = \hbar\omega\hat{a}^{\dagger}\hat{a}$$

Then finally, the interaction term is given by

$$\begin{aligned}\hat{H}_{Interaction} &= -\frac{\hbar\Omega}{2}\left(\hat{\sigma}_{+}\otimes\hat{a} + \hat{\sigma}_{-}\otimes\hat{a}^{\dagger}\right) \\ &= \frac{\hbar\Omega}{2}\begin{pmatrix} 0 & \hat{a} \\ \hat{a}^{\dagger} & 0 \end{pmatrix}\end{aligned} \tag{91}$$

where Ω is the Rabi frequency, $\hat{\sigma}_{\pm}$ are the atomic raising and lowering operators, and we have changed the state convention such that we have $-\sigma_z$ as opposed to σ_z in the original work [50].

In order to combine the Hilbert spaces for these two separate quantizes systems, block-operator notation is applied. Notice that $\hat{H}_{Atom}$ acts on the two-state atomic basis, while $\hat{H}_{Photon}$ acts on the infinite Fock basis of photons. We then must combine the two bases in $\hat{H}_{Interaction}$ which describes the creation of a photon in the Fock basis upon atomic emission, and the annihilation of a photon upon absorption. So this Hamiltonian should not be thought of as a two-by-two matrix acting on the atomic basis, but rather an infinite matrix enumerating the interactions between all possible states. All together, the final Hamiltonian is then [50]

$$\begin{aligned}\hat{H}_{JC} &= -\frac{\hbar\omega}{2}\hat{\sigma}_z + \hbar\omega\hat{a}^\dagger\hat{a} + \frac{\hbar\Omega}{2}\left(\hat{\sigma}_+ \otimes \hat{a} + \hat{\sigma}_- \otimes \hat{a}^\dagger\right) \\ &= \frac{\hbar}{2}\begin{pmatrix} -\omega & \Omega\hat{a} \\ \Omega\hat{a}^\dagger & \omega \end{pmatrix} + \hbar\omega\hat{a}^\dagger\hat{a}\end{aligned} \tag{92}$$

We may then choose to arbitrarily shift the ground state as to obtain

$$\hat{H}_{JC} = \frac{\hbar}{2}\begin{pmatrix} 0 & \Omega\hat{a} \\ \Omega\hat{a}^\dagger & 2\omega \end{pmatrix} + \hbar\omega\hat{a}^\dagger\hat{a}. \tag{93}$$

This is not the whole story, however. Any real cavity has some photonic decay rate, meaning we must include dissipative operators and apply Lindbladian dynamics to solve for a steady state via the master equation. Our dissipative operator then becomes $L = \sqrt{\gamma}\hat{a}$ where γ gives the decay rate such that our master equation is [50]

$$\dot{\rho} = -i\left[\hat{H}_{JC}, \rho\right] + \gamma\left(\hat{a}\rho\hat{a}^\dagger - \frac{1}{2}\hat{a}^\dagger\hat{a}\rho - \frac{1}{2}\rho\hat{a}^\dagger\hat{a}\right). \tag{94}$$

To learn the QuTiP package, the Jaynes-Cummings model was coded (see Code Block. IV A 1) based on an implementation by Robert Johansson [51]. This code differs slightly from the above description in that the atomic Hamiltonian basis is shifted such that it is in the rotating frame $\hat{H}_{Atom} = \hbar\omega|e\rangle\langle e|$.

This code uses the `mesolve` function for time evolving a quantum system such that we may inspect the evolution of the state. For an initial state where the atom is excited and there are no photons present in the cavity, we obtain decaying Rabi oscillations as expected (see Fig. 11).

This code will be a base for all quantum code going forward, but before we can dive any deeper into quantized systems, we must take a step back to look at a semi-classical system.

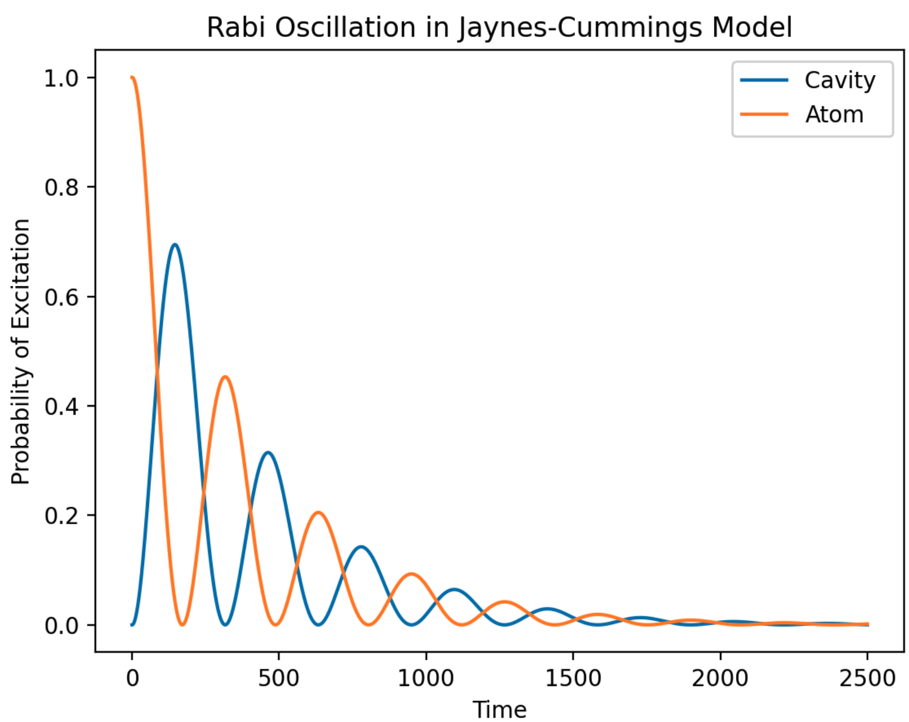

FIG. 11: Decaying Rabi oscillation produced by the Jaynes-Cummings model with initial state $|\psi\rangle_0 = |e\rangle \otimes |0\rangle$.

2. Electromagnetically induced transparency in a Λ-system with classical driving fields

Consider a quantized 3-level Λ-system of states driven by strong external electromagnetic fields. We consider the fields to be so strong that the quantum mechanical nature of the beams is reduced to classical behaviour in accordance with Ehrenfest's theorem (i.e. the correspondence principle). This makes the system semi-classical since the atomic system is quantized while the driving fields are treated classically.

The Hamiltonian for this system in the atomic basis can be obtained by writing down the diagonal atomic Hamiltonian and shifting it such that $\langle 1|\hat{H}|1\rangle = 0$, then adding on the effects of the electromagnetic fields which drive the allowed transitions $|1\rangle \leftrightarrow |3\rangle$ and $|2\rangle \leftrightarrow |3\rangle$ at frequencies $\omega_s \approx \omega_{13}$ and $\omega_c \approx \omega_{23}$. The rotating wave approximation may then be applied to simplify the off-diagonal transitional entries as Rabi frequencies Ω_s, Ω_c, and a unitary transformation may be applied to write the diagonal entries in terms of the detunings due to the driving beams $\Delta_s = \omega_{13} - \omega_s$ and $\Delta_c = \omega_{23} - \omega_c$. A similar procedure is applied to obtain the Hamiltonian for the 4-level system of focus in Appendix A. The Λ-system Hamiltonian in this form is well known to be [27]

$$\hat{H} = -\frac{\hbar}{2}\begin{pmatrix} 0 & 0 & \Omega_s \\ 0 & -2\delta & \Omega_c \\ \Omega_s^* & \Omega_c^* & 2\Delta \end{pmatrix} \tag{95}$$

where $\delta = \Delta_s - \Delta_c$ and for the sake of simplicity we redefine $\Delta = \Delta_s$.

In order to consider the effect of the environment for Lindbladian time evolution, we must mathematically express the decay operators of the system. We have decay operator Γ_{13} which acts on transition σ_{13} and Γ_{23} which acts on σ_{23}. We can then follow the template given by the Jaynes-Cummings model in Code Block IV A 1, changing out the `mesolve` time evolution function for the `steadystate` function which numerically solves the Lindblad master equation for $\hat{\rho}$ when $d\hat{\rho}/dt = 0$.

Now consider the linear electric susceptibility in Eq. (9). It tells us that $\chi^{(1)} \propto \langle\hat{d}\rangle$. We may expand $\langle\hat{d}\rangle$ as

$$\langle\hat{d}\rangle = \sum_{i,j} d_{ij}\langle\hat{\sigma}_{ji}\rangle \tag{96}$$

where removing the zero-valued d_{ij} terms and noticing that $\rho_{ji} = \langle\hat{\sigma}_{ji}\rangle$ leaves us with

$$\langle\hat{d}\rangle = d_{13}\rho_{31} + d_{31}\rho_{13} + d_{23}\rho_{32} + d_{32}\rho_{23}. \tag{97}$$

This expression can then be re-expressed fully in terms of d_{13} as [27]

$$\chi^{(1)} = \frac{N|d_{13}|^2}{V\hbar\epsilon_0}\left[\frac{4\delta(|\Omega_c| - 4\delta\Delta) - 4\Delta\Gamma_{21}^2}{||\Omega_c|^2 + (\Gamma_{31} + i2\Delta)(\Gamma_{21} + i2\delta)|^2} + i\frac{8\delta^2\Gamma_{31} + 2\Gamma_{21}(|\Omega_c|^2 + \Gamma_{21}\Gamma_{31})}{||\Omega_c|^2 + (\Gamma_{31} + i2\Delta)(\Gamma_{21} + i2\delta)|^2}\right] \tag{98}$$

where V is volume. However, we can get an intuitive sense for how the system behaves making some approximations. Fleischhauer et al. [27] give an expression where $\rho_{32} \propto \rho_{12}$ (with a prefactor that is not particularly large under typical circumstances), but the $|1\rangle \leftrightarrow |2\rangle$ transition is forbidden such that in the steady-state regime where the system is only driven by beams tuned near the transition resonances ω_{13} and ω_{23} we can expect ρ_{12} to be near-zero valued. This then leads us to suspect that we may have $\rho_{23} \approx 0$. Assume this is the case. Then we may write the much simpler expression

$$\langle\hat{d}\rangle \approx d_{13}\rho_{31} + d_{31}\rho_{13} \tag{99}$$

such that $\chi^{(1)} \underset{\sim}{\propto} \rho_{31}$.

By setting Δ_c to be fixed, we can scan over Δ values to find the optimal conditions for EIT at the given Δ_c (and any other necessary parameters). Solving for the steady state and taking the corresponding ρ_{31} value at each detuning produces fig. 12. This plot depicts the real and imaginary components of $\chi^{(1)}$ over a range of detunings where we see $Im(\chi^{(1)})$ decreases on resonance indicating EIT, and in the same region $Re(\chi^{(1)})$ has a large slope indicating slowed light propagation.

We may also check the validity of our approximation $\rho_{12} \approx \rho_{23} \approx 0$ by inspecting the density matrix elements at each detuning. For all detunings applied, both values are near-zero valued, thereby giving further justification to our approximation.

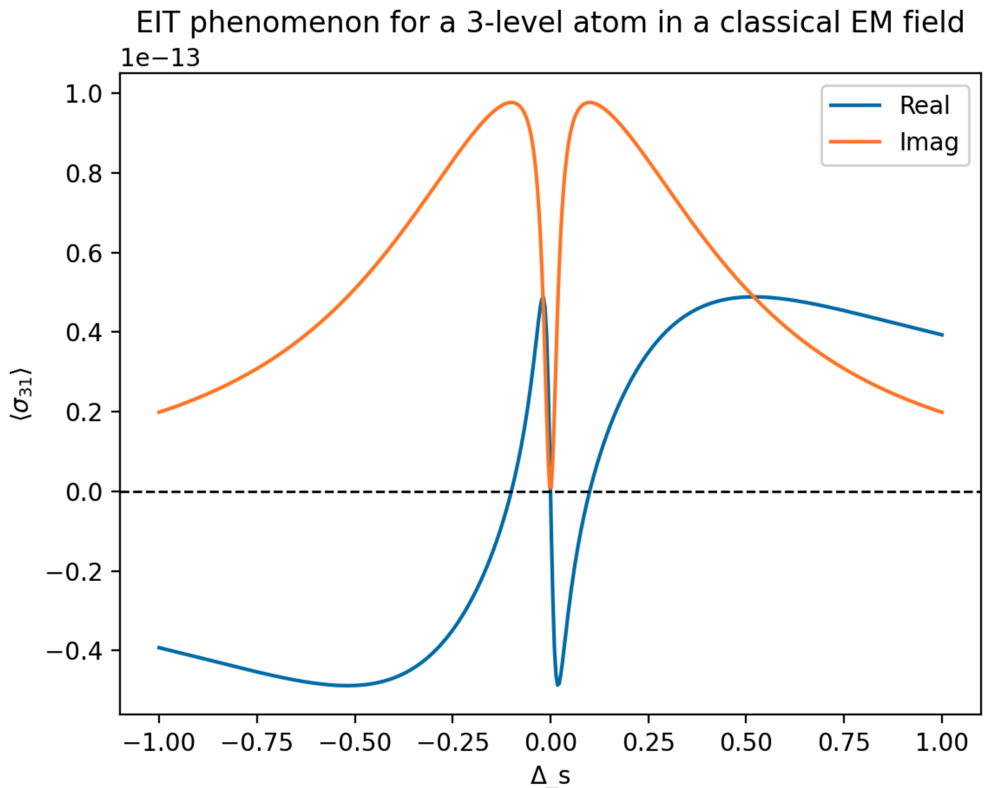

FIG. 12: The real and imaginary components of the density matrix element $\rho_{31} = \langle \hat{\sigma} \rangle$ over a range of detunings Δ_s. The response indicates the presence of EIT and slowed light propagation at resonance.

3. *Cavity electromagnetically induced transparency in a* Λ*-system with quantized driving fields*

The final stepping stone towards developing system-specific code comes with reproducing a known result of a more advanced and modern system. In the paper 'Coherent Control of Quantum Fluctuations Using Cavity Electromagnetically Induced Transparency' by Souza et al. they describe a three-level Λ-system within a resonant cavity of frequency ω_{cav} in the presence of a strong classical pump field of frequencyω_c and a weak quantized probe beam of frequency ω_p and solve for steady state solutions via the Lindblad master equation [1]. This combines all the techniques seen thus far. They give their Hamiltonian as

$$\begin{aligned}\hat{H} &= \Delta\hat{\sigma}_{33} + (\Delta_1 - \Delta_2)\hat{\sigma}_{22} - \Delta\hat{\sigma}_{11} + \Delta\hat{a}^\dagger\hat{a} + (\varepsilon\hat{a} + g\hat{a}\hat{\sigma}_{31} + \Omega_{32} + h.c.) \\ &= \begin{pmatrix} -\Delta & 0 & g^*\hat{a}^\dagger \\ 0 & (\Delta_1 - \Delta_2) & \Omega_c^* \\ g\hat{a} & \Omega_c & \Delta_1 \end{pmatrix} + \varepsilon\hat{a} + \varepsilon^*\hat{a}^\dagger + \Delta\hat{a}^\dagger\hat{a}\end{aligned} \tag{100}$$

where g is the atom-field coupling strength for the weak pump field, Ω_c is the strong pump field dependent Rabi frequency, ε is the weak probe field strength, $\Delta = \omega_{cav} - \omega_p$, $\Delta_1 = \omega_{31} - \omega_{cav}$, and $\Delta_2 = \omega_{32} - \omega_c$.

There is one particular term of note here which has not yet shown up: $\varepsilon\hat{a} + \varepsilon^*\hat{a}^\dagger$. To understand this term, we may take a step back to the Jaynes-Cummings model. By including the $\varepsilon\hat{a} + \varepsilon^*\hat{a}^\dagger$ term to our Hamiltonian with $\varepsilon = 0.5$ we find the introduction of noise to our model as the probe field populates the cavity in Fig. 13. Notice also, that the total probability of having an excitation in the field and the cavity surpasses 1 since we have introduced an external source.

They then define decay operators for their cavity field mode $\sqrt{\kappa}\hat{a}$, and their atomic states $\sqrt{\Gamma_{13}}\hat{\sigma}_{13}$, $\sqrt{\Gamma_{23}}\hat{\sigma}_{23}$. Implementing this Hamiltonian with QuTiP and solving for the steady-state density matrix via the Lindblad mater equation with the specified decay operators allowed us to reproduce some of the results presented by Souza et al. [1]. Fig. 14 reproduces the portion of figure 1(c) depicting the transmission profile over a range of Δ detunings with cavity EIT (CEIT). Fig. 15 reproduces figure 2(b) depicting the state expectation values over the range of detunings, where we have opted to split the original figure into two to avoid overcrowding. Additionally, the expectation value of atomic

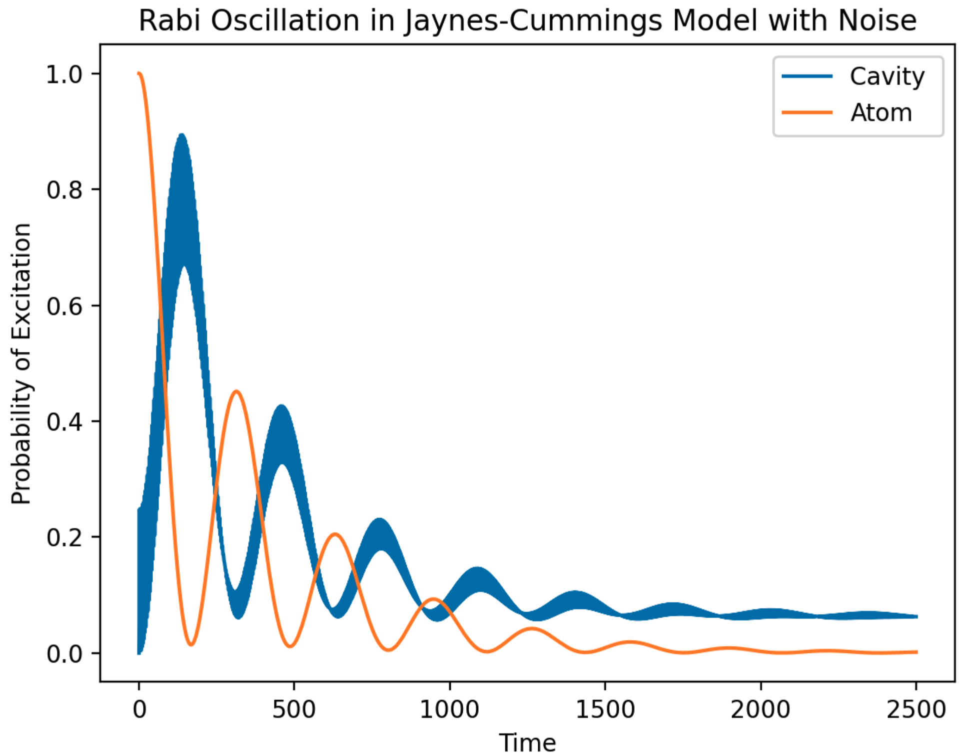

FIG. 13: Decaying Rabi oscillation produced by the Jaynes-Cummings model with initial state $|\psi\rangle_0 = |e\rangle \otimes |0\rangle$ with the introduction of noise by including $\varepsilon\hat{a} + \varepsilon^*\hat{a}^\dagger$ to the Hamiltonian in Eq. (93).

state $|3\rangle$ was added, where we observe a sharp dip in state population on resonance. This indicates the presence of a dark state, giving evidence to the presence of EIT. Fig. 14 also suggests that the cavity mode becomes more populated on resonance, supporting evidence of EIT; with increases also observed at other detunings corresponding to alternative transition frequencies introduced by 'Autler-Townes-like' energy level splitting [1].

B. Computational Tools

A set of computational tools were developed specifically to model the four-level diamond configuration atomic system. The code for these tools is presented in Appendix C. The main function is defined as `iterative_solve` taking inputs as the initial Rabi frequencies Ω_s, Ω_i, Ω_I, Ω_{II}, the detunings Δ_s, Δ_i, Δ_I, Δ_{II}, Δ_a, Δ_b, temperature T, cell length l, step size δ, the number of velocity subclasses M, and the desired Hamiltonian function.

It is closely aligned with the algorithm outlined in Alg. 1, adding the additional step of including Doppler broadening. For the purposes of this code, the phase matching condition given by Eq. (60) is applied. Otherwise, the algorithmic time complexity would increase drastically.

To perform the Doppler broadening procedure, M evenly spaced velocities are generated from Eq. (67) about the mean extending 2 standard deviations on either side. For each velocity, the Doppler shift with respect to each field is calculated and added to its respective detuning. This is used to form the Hamiltonian. Each of the 3 Hamiltonians $\tilde{H}_{ab}$, $\tilde{H}_a$, and $\tilde{H}_b$ from Eqs. (84),(85),(86) have been implemented as functions called `form_Hamiltonian_ab`, `form_Hamiltonian_a`, and `form_Hamiltonian_b`, each with input arguments Ω_s, Ω_i, Ω_I, Ω_{II}, Δ_s, Δ_i, Δ_I, Δ_{II}, Δ_a, and Δ_b.

This Hamiltonian is then passed into the QuTiP `steadystate` function along with the decay operators listed in Sec. III C to find the steady state density matrix for a single velocity. This is repeated for all velocity subclasses, then the mean density matrix is calculated according to Eq. (75). The result is renormalized to unity.

With this, the Maxwell-BLoch equations of Eqs. (87),(88) are calculated and the Rabi frequencies are updated as $\Omega_s \leftarrow \Omega_s + \partial_z\Omega_s \cdot \delta$ and $\Omega_i \leftarrow \Omega_i + \partial_z\Omega_i \cdot \delta$. This entire process is repeated for every step until the end of the cell is reached and the final density matrix $\hat{\rho}$, signal field Rabi frequency Ω_s, and idler field Rabi frequency Ω_i are returned.

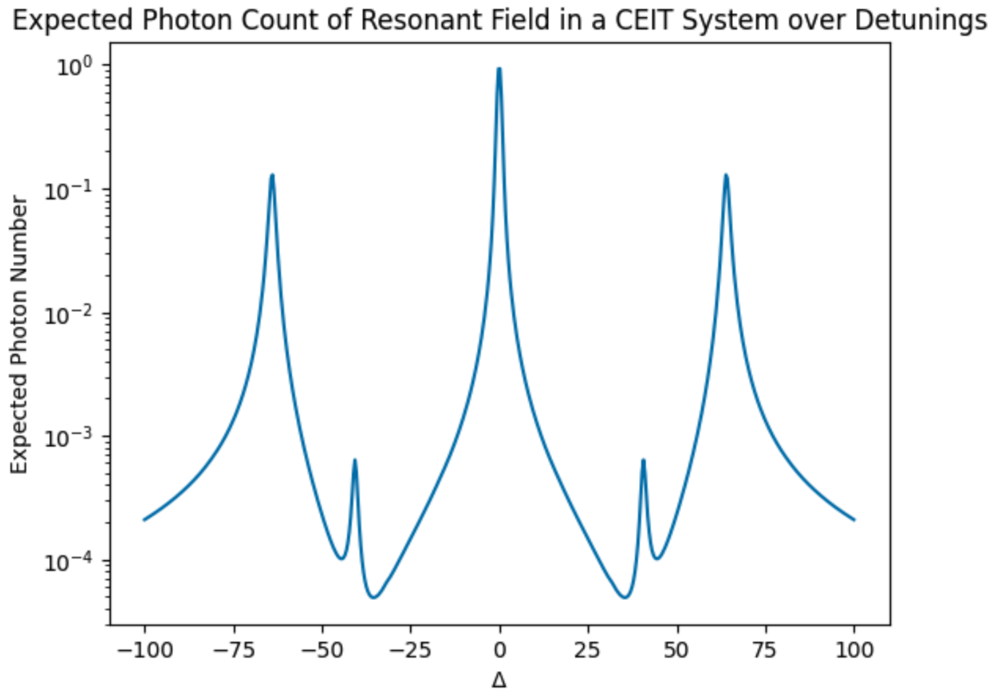

FIG. 14: Reproduction of a portion of figure 1(c) from 'Coherent Control of Quantum Fluctuations Using Cavity Electromagnetically Induced Transparency' by Souza et al. [1] depicting the expected photon count within the cavity over a range of frequency detunings Δ.

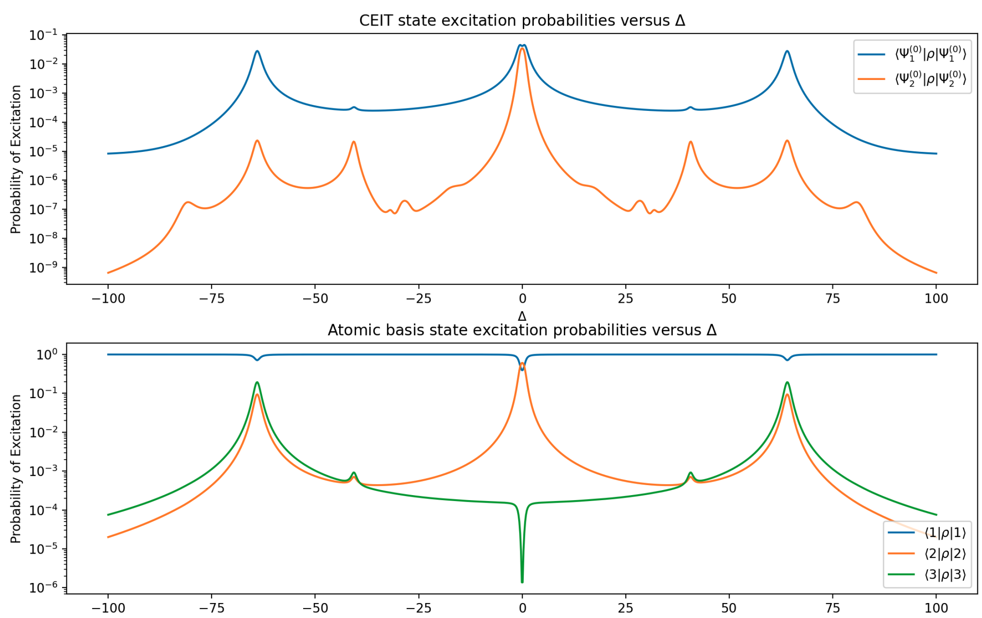

FIG. 15: Reproduction of a portion of figure 1(c) from 'Coherent Control of Quantum Fluctuations Using Cavity Electromagnetically Induced Transparency' by Souza et al. [1] — which has been slightly modified and added upon for clarity — depicting (top) the state expectation values of states in an energy split and rotated basis over a range of detunings and (bottom) the state expectation values of states in the atomic basis over a range of detunings.

This code makes the following approximations and assumptions: it assumes the mono-axial phase matching condition of Eq. (60) such that Eq. (75) may be applied to reduce computational complexity; the frequency-dependent index of refraction involved in calculating the Doppler shift from Eq. (70) was taken to be constant as in Eq. (71) since we have not investigated a way to accurately calculate this; and the group velocity v_g in the maxwell bloch equations (87),(88) is taken to be the speed of light c since we have not yet investigated a well-defined way of computationally calculating v_g. Each of these points can be improved upon in future iteration.

Additionally, within this code constants such as c, ϵ_0, $\hbar$, etc. are all set to 1 as placeholders. In fact, all constants within the `CONSTANTS, BASIS STATES, AND OPERATORS` section of the code in Sec. C are placeholders. Some of their values are set to give intuition as to how they should be used, but are meant to be modified. This is because many of these constant values are often expressed using unit conventions that result in them differing by orders of magnitude. Multiplication of such floating point numbers results in floating point errors. With the amount of computations performed in this algorithm and the prevalence of these constants, the choice of an ill-suited unit convention would be detrimental to any computational results as floating point errors would propagate throughout every step of calculation and compound.

Therefore, the code presented in Sec. C is meant to act as a template for further development and research. It is recommended that any choice of unit convention scales such constants to unity if possible, and that any other physically relevant parameters are defined in terms of some other parameter. The latter has been done in the presented code such that atomic decay rates are expressed in terms of the cavity decay rate, giving relative meaning to parameter choices.

V Discussion and Outlook

Progress has been made towards answering the question: can the four-level diamond configuration formed by the $5S_{1/2} - 5P_{1/2} - 5P_{3/2} - 4D_{1/2}$ energy level substructure in hot ^{87}Rb vapour be used to convert a weak laser field near 795 nm encoded with continuous variable quantum information to a field in the telecom C-band of 1530 nm to 1565 nm via an EIT (and perhaps EIA) enhanced FWM process while generating a single-mode squeezed state to reduce noise below the shot noise limit? If so, then it may be possible to rapidly scale quantum networks in an economically viable way by using existing telecom infrastructure to prepare secure sensitive information networks for a post-quantum world. In this pursuit, the theoretical connections between various nonlinear optical phenomena were formed and related to the problem at hand, the problem itself was given a well-defined definition with quantifiable performance metrics, various derivations were performed to obtain the Lindblad master equation and Maxwell-Bloch equations for the system, and a set of computational tools were developed for the specific problem and presented alongside reproductions of known computational results of similar physical systems to build intuition of how results may be obtained.

Application of the Lindblad master equation in conjunction with the Maxwell-Bloch equations as outlined in Alg. 1 gives a method to theoretically improve simulation accuracy over those which only apply the master equation by assuming uniform steady state propagation dynamics in a Doppler broadened atomic medium undergoing nonlinear optical processes. However, computational accuracy can still be improved over that of the methods presented in this body of work. The tools presented make the approximations that the Doppler free phase matching condition of Eq. (60) is enforced and simplifying assumptions were applied to some equations since exact equations which were compatible with the algorithmic solution method could not be derived. Whether this phase matching condition is practical to a real-world system must be investigated further. The code can be improved in these respects in future work to improve accuracy and robustness. It is important to note that the computational time complexity of any additional steps should be considered as the computation time of an algorithm can hinder its ability to practically produce results.

If it is found that efficient frequency conversion is viable with this method, then there would be evidence to support experimentation towards the development of a deployable quantum device. If the contrary is found, then the possibility of viability cannot be fully ruled out, but this would suggest evidence in favor of applying ultracold atomic frequency converters across quantum telecom networks. This ladder case would likely drastically increase the cost, implementation time, and maintenance requirement for such networks.

While the mathematical framework presented theoretically describes the system, its predictive utility is yet to be supported via simulation results. It is intended that the foundations built by this body of work will advance this pursuit. If future results support this approach, then the methods and algorithms described may be applied to other systems. The shortcomings and limitations of these methods may also be highlighted in this process, possibly leading to subsequent iteration and improvement. The equations presented may also be applied to theoretical works for analytical calculations. However, any application of them should recognize the various approximations applied (such as the rotating wave approximation or slowly varying envelope approximation) and confirm that they hold in the regime of consideration.

It is intended that this thesis may be used as a self-contained starting point and building block for subsequent collaborators. All the theory and examples included exist to build one's intuition with respect to the problem, and ideas between topics in nonlinear optics have been connected to help the reader understand the broad scale network of ideas.

VI Conclusion

Quantum technologies are advancing rapidly introducing new possibilities, however, they are dual use in that they can be used for both civilian and military purposes. Quantum methods can potentially crack conventional encryption algorithms, but quantum communication techniques like quantum key distribution are theoretically impervious to such attack. So it is considered a national interest to secure sensitive communication channels. It would be advantageous to use existing telecommunications infrastructure, but many techniques for producing optically encoded quantum information do not operate at comparable frequencies. For this purpose, a hot atomic vapour system composed of four-level diamond configuration ^{87}Rb is presented as a candidate for frequency conversion and noise reduction of optically encoded continuous variable quantum signals via EIT and FWM. A key hurdle to achieving these phenomena in a hot medium is the effect of Doppler broadening. If the system can be made Doppler insensitive, then FWM and EIT may be promoted.

By iteratively applying the Lindblad master equation and Maxwell-Bloch equations to a Doppler-broadened medium, it is theorized that signal propagation can be computationally modelled in order to obtain a measure of signal conversion efficiency and optical squeezing. Derivations and algorithms have been presented towards this end, although work remains. The code presented must be iterated upon, applied to obtain a computational result, then possibly verified against an experimental system. This body of work is intended to be a self-contained guide to aid future collaborators execute these goals.

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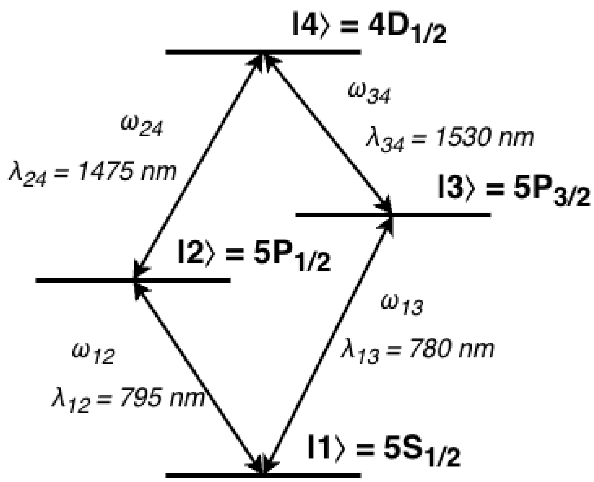
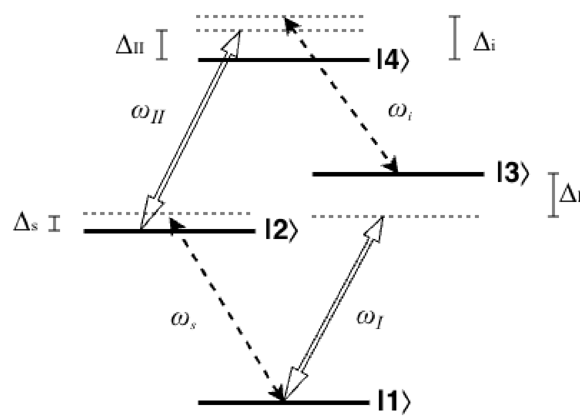
A Hamiltonian Formulation

The first step in Hamiltonian formulation is system definition. We are modelling a 4-level diamond system as depicted in Fig. 16a. For this system, the choice of the ground state energy is arbitrary, and all other energies can be defined in terms of their differences as $\omega_2 = \omega_1 + \omega_{12}$, $\omega_3 = \omega_1 + \omega_{13}$, and $\omega_4 = \omega_2 + \omega_{24} = \omega_3 + \omega_{34}$. These states can



(a)



(b)

FIG. 16: Diamond configuration atom diagrams. (Left) Duplicate of Fig. 3 placed for convenience. A diamond configuration atomic system with non-degenerate energy levels $|1\rangle, |2\rangle, |3\rangle, |4\rangle$. The system has allowed transitions $|1\rangle \leftrightarrow |2\rangle$, $|1\rangle \leftrightarrow |3\rangle$, $|2\rangle \leftrightarrow |4\rangle$, and $|3\rangle \leftrightarrow |4\rangle$ with respective energies $\hbar\omega_{12}$, $\hbar\omega_{13}$, $\hbar\omega_{24}$, and $\hbar\omega_{34}$ and corresponding wavelengths λ_{12}, λ_{13}, λ_{24}, and λ_{34}. (Right) A strong pump drives transition $|1\rangle \leftrightarrow |3\rangle$ at frequency ω_I with detuning $\Delta_I = \omega_I - \omega_{13}$, a separate strong pump drives transition $|2\rangle \leftrightarrow |4\rangle$ with detuning $\Delta_{II} = \omega_{II} - \omega_{24}$, a weak signal beam drives transition $|1\rangle \leftrightarrow |2\rangle$ with detuning $\Delta_s = \omega_s - \omega_{12}$, and a weak idler field (undriven) is induced by the presence of the other fields through a FWM process, resulting in transition $|3\rangle \leftrightarrow |4\rangle$ with emission detuning $\Delta_i = \omega_i - \omega_{34}$.

be described without any interactions by the simple Hamiltonian

$$\begin{aligned}\hat{H}_0 &= \hbar\omega_1\hat{\sigma}_{11} + \hbar\omega_2\hat{\sigma}_{22} + \hbar\omega_3\hat{\sigma}_{33} + \hbar\omega_4\hat{\sigma}_{44} \\ &= \hbar \begin{pmatrix} \omega_1 & 0 & 0 & 0 \\ 0 & \omega_2 & 0 & 0 \\ 0 & 0 & \omega_3 & 0 \\ 0 & 0 & 0 & \omega_4 \end{pmatrix}\end{aligned} \tag{A1}$$

The reason we are attempting to obtain optical squeezing in the up-converted idler field is to transmit quantum information at telecom wavelengths. Therefore, we care about the quantized fields of the signal (input carrier of quantum information) and idler (output of frequency-converted quantum information) beams. Apply the theoretical resonant cavity approximation such that we consider photons in the Fock bases of a monochromatic cavity tuned near the signal field frequency ω_a where $\omega_a - \omega_s \ll \omega_s$ and another cavity tuned near the idler frequency ω_b where $\omega_b - \omega_i \ll \omega_i$ to obtain the Hamiltonian

$$\hat{H}_{photon} = \hbar\omega_a\hat{a}^\dagger\hat{a} + \hbar\omega_b\hat{b}^\dagger\hat{b}. \tag{A2}$$

We then have the Hamiltonian describing these separate systems as

$$\begin{aligned}\hat{H}_S &= \hat{H}_0 + \hat{H}_{photon} \\ &= \hbar\omega_1\hat{\sigma}_{11} + \hbar\omega_2\hat{\sigma}_{22} + \hbar\omega_3\hat{\sigma}_{33} + \hbar\omega_4\hat{\sigma}_{44} \\ &\quad + \hbar\omega_a\hat{a}^\dagger\hat{a} + \hbar\omega_b\hat{b}^\dagger\hat{b}.\end{aligned} \tag{A3}$$

Each transition is driven by a laser as depicted in Fig. 16b. These driving fields cause significant coupling between states, requiring the definition of an interaction Hamiltonian. Under the dipole approximation following the procedure used in [9], this interaction Hamiltonian takes the form

$$\hat{H}_{Int} = -\hat{\vec{d}} \cdot \vec{E} \tag{A4}$$

$$\hat{\vec{d}} = \vec{d}_{12}\hat{\sigma}_{12} + \vec{d}_{13}\hat{\sigma}_{13} + \vec{d}_{24}\hat{\sigma}_{24} + \vec{d}_{34}\hat{\sigma}_{34} + h.c. \tag{A5}$$

$$\vec{E} = \vec{E}_s + \vec{E}_I + \vec{E}_{II} + \vec{E}_i \tag{A6}$$

where $\vec{d}_{ij}$ is the transition dipole moment between states $|i\rangle$ and $|j\rangle$, the strong pump fields $\vec{E}_I$ and $\vec{E}_{II}$ are treated as classical fields, and the weak fields $\vec{E}_s$ and $\vec{E}_i$ are treated as quantized fields. Each field can be separated into a rapidly oscillating component and a slowly-varying envelope $\vec{\varepsilon}(t)$ as

$$\vec{E}_s = \vec{\varepsilon}_s(t)\hat{a}e^{i\vec{k}_s\cdot\vec{r}} + h.c. \tag{A7}$$

$$\vec{E}_I = \vec{\varepsilon}_I(t)e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}} + c.c. \tag{A8}$$

$$\vec{E}_{II} = \vec{\varepsilon}_{II}(t)e^{-i\omega_{II} t + i\vec{k}_{II}\cdot\vec{r}} + c.c. \tag{A9}$$

$$\vec{E}_i = \vec{\varepsilon}_i(t)\hat{b}e^{i\vec{k}_i\cdot\vec{r}} + h.c. \tag{A10}$$

where $\hat{a}$ is the annihilation operator for the signal field, $\hat{b}$ is that of the idler field, $\vec{k}_j$ is the wavevector for each plane wave, and $\vec{r}$ is the position at which the field is experienced by the atom.

1. Classical Field Simplification via Rotating Wave Approximation

Consider some coupled classical electric field $\vec{E}_c$ and dipole operator $\hat{\vec{d}}$

$$\vec{E}_c = \vec{\varepsilon}(e^{i\phi} + e^{-i\phi}), \qquad \phi = kz - \omega t \tag{A11}$$

$$\hat{\vec{d}} = \vec{d}_{ij}\hat{\sigma}_{ij} + \vec{d}_{ji}\hat{\sigma}_{ji} \tag{A12}$$

where we may arbitrarily take $\vec{\varepsilon} = \vec{\varepsilon}^{\,*}$ and $\vec{d}_{ij} = \vec{d}_{ji}$ (real-valued vectors). We can then expand

$$\vec{E}_c \cdot \hat{\vec{d}} = \vec{\varepsilon} \cdot \vec{d}_{ij}\left(\hat{\sigma}_{ij}e^{i\phi} + \hat{\sigma}_{ij}e^{-i\phi} + \hat{\sigma}_{ji}e^{i\phi} + \hat{\sigma}_{ji}e^{-i\phi}\right). \tag{A13}$$

Go to the interaction picture where we have transition operators evolve as

$$\hat{\sigma}_{ij}(t) = \hat{\sigma}_{ij}e^{-i\omega_0 t} \tag{A14}$$

$$\hat{\sigma}_{ji}(t) = \hat{\sigma}_{ji}e^{i\omega_0 t} \tag{A15}$$

where ω_0 is the atomic transition frequency. We then have

$$\vec{E}_c \cdot \hat{\vec{d}} = \vec{\varepsilon} \cdot \vec{d}_{ij}\left(\hat{\sigma}_{ij}e^{-i\omega_0 t}e^{i\phi} + \hat{\sigma}_{ij}e^{-i\omega_0}e^{-i\phi} + \hat{\sigma}_{ji}e^{i\omega_0 t}e^{i\phi} + \hat{\sigma}_{ji}e^{i\omega_0 t}e^{-i\phi}\right)$$

$$\vec{E}_c \cdot \hat{\vec{d}} = \vec{\varepsilon} \cdot \vec{d}_{ij}\left(\hat{\sigma}_{ij}e^{i(kz-(\omega+\omega_0)t)} + \hat{\sigma}_{ij}e^{i(-kz+(\omega-\omega_0)t)} + \hat{\sigma}_{ji}e^{i(kz-(\omega-\omega_0)t)} + \hat{\sigma}_{ji}e^{i(-kz+(\omega+\omega_0)t)}\right). \tag{A16}$$

Assuming that the driving field of frequency ω is near-resonant with the transition, we may apply the rotating wave approximation. Oscillations at frequency $\omega+\omega_0$ are so rapid in comparison to those at frequency $\omega-\omega_0$ that we may neglect them as they average to zero on optical timescales. Thus, we have

$$\vec{E}_c \cdot \hat{\vec{d}} \approx \vec{\varepsilon} \cdot \vec{d}_{ij}\left(\hat{\sigma}_{ij}e^{i(-kz+(\omega-\omega_0)t)} + \hat{\sigma}_{ji}e^{i(kz-(\omega-\omega_0)t)}\right). \tag{A17}$$

We may then go back to the Schrödinger picture of state evolution and write

$$\vec{E}_c \cdot \hat{\vec{d}} \approx \vec{\varepsilon} \cdot \vec{d}_{ij}\left(\hat{\sigma}_{ji}e^{i(kz-\omega t)} + \hat{\sigma}_{ij}e^{i(-kz+\omega t)}\right). \tag{A18}$$

2. Quantum Field Simplification via Rotating Wave Approximation

Similar to the classical field approach, consider some coupled quantized electric field $\vec{E}_q$ and dipole operator $\hat{\vec{d}}$

$$\begin{aligned}\vec{E}_q \cdot \hat{\vec{d}} &= \vec{\varepsilon}\left(\hat{a}e^{i\vec{k}\cdot\vec{r}} + \hat{a}^\dagger e^{-i\vec{k}\cdot\vec{r}}\right) \cdot \left(\vec{d}_{ij}\hat{\sigma}_{ij} + \vec{d}_{ji}\hat{\sigma}_{ji}\right) \\ &= \vec{\varepsilon} \cdot \vec{d}_{ij}\left(\hat{a}\hat{\sigma}_{ij}e^{i\vec{k}\cdot\vec{r}} + \hat{a}^\dagger\hat{\sigma}_{ij}e^{-i\vec{k}\cdot\vec{r}} + \hat{a}\vec{\sigma}_{ji}e^{i\vec{k}\cdot\vec{r}} + \hat{a}^\dagger\vec{\sigma}_{ji}e^{-i\vec{k}\cdot\vec{r}}\right)\end{aligned} \tag{A19}$$

We again go to the interaction picture in which we have operators evolve as

$$\hat{a}(t) = \hat{a}e^{-i\omega t} \tag{A20}$$

$$\hat{a}^\dagger(t) = \hat{a}^\dagger e^{i\omega t} \tag{A21}$$

$$\hat{\sigma}_{ij}(t) = \hat{\sigma}_{ij}e^{-i\omega_0 t} \tag{A22}$$

$$\hat{\sigma}_{ji}(t) = \hat{\sigma}_{ji}e^{i\omega_0 t} \tag{A23}$$

so we have

$$\begin{aligned}\vec{E}_q \cdot \hat{\vec{d}} &= \vec{\varepsilon} \cdot \vec{d}_{ij}\left(\hat{a}e^{-i\omega t}e^{-i\omega_0 t}\hat{\sigma}_{ij}e^{i\vec{k}\cdot\vec{r}} + \hat{a}^\dagger e^{i\omega t}e^{-i\omega_0 t}\hat{\sigma}_{ij}e^{-i\vec{k}\cdot\vec{r}} + \hat{a}e^{-i\omega t}e^{i\omega_0 t}\vec{\sigma}_{ji}e^{i\vec{k}\cdot\vec{r}} + \hat{a}^\dagger e^{i\omega t}e^{i\omega_0 t}\vec{\sigma}_{ji}e^{-i\vec{k}\cdot\vec{r}}\right) \\ &= \vec{\varepsilon} \cdot \vec{d}_{ij}\left(\hat{a}e^{-i(\omega+\omega_0)t}\hat{\sigma}_{ij}e^{i\vec{k}\cdot\vec{r}} + \hat{a}^\dagger e^{-i(\omega-\omega_0)t}\hat{\sigma}_{ij}e^{-i\vec{k}\cdot\vec{r}} + \hat{a}e^{-i(\omega-\omega_0)t}\vec{\sigma}_{ji}e^{i\vec{k}\cdot\vec{r}} + \hat{a}^\dagger e^{i(\omega+\omega_0)t}\vec{\sigma}_{ji}e^{-i\vec{k}\cdot\vec{r}}\right).\end{aligned} \tag{A24}$$

Then, again, by the rotating wave approximation we have

$$\vec{E}_q \cdot \hat{\vec{d}} = \vec{\varepsilon} \cdot \vec{d}_{ij}\left(\hat{a}^\dagger e^{-i(\omega-\omega_0)t}\hat{\sigma}_{ij}e^{-i\vec{k}\cdot\vec{r}} + \hat{a}e^{-i(\omega-\omega_0)t}\vec{\sigma}_{ji}e^{i\vec{k}\cdot\vec{r}}\right). \tag{A25}$$

Going back to Schrödinger picture then yields

$$\vec{E}_q \cdot \hat{\vec{d}} = \vec{\varepsilon} \cdot \vec{d}_{ij}\left(\hat{a}^\dagger\hat{\sigma}_{ij}e^{-i\vec{k}\cdot\vec{r}} + \hat{a}\vec{\sigma}_{ji}e^{i\vec{k}\cdot\vec{r}}\right). \tag{A26}$$

3. Hamiltonian Simplification

Putting this all together, we obtain our interaction Hamiltonian as

$$\begin{aligned}\hat{H}_{Int} = &\left(\vec{d}_{12} \cdot \vec{\varepsilon}_s(t)\hat{\sigma}_{12}\hat{a}^\dagger e^{i\vec{k}_s\cdot\vec{r}} + h.c.\right) \\ &+ \left(\vec{d}_{13} \cdot \vec{\varepsilon}_I(t)\hat{\sigma}_{13}e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}} + c.c.\right) \\ &+ \left(\vec{d}_{24} \cdot \vec{\varepsilon}_{II}(t)\hat{\sigma}_{24}e^{-i\omega_{II} t + i\vec{k}_{II}\cdot\vec{r}} + c.c.\right) \\ &+ \left(\vec{d}_{34} \cdot \vec{\varepsilon}_i(t)\hat{\sigma}_{34}\hat{b}^\dagger e^{i\vec{k}_i\cdot\vec{r}} + h.c.\right)\end{aligned} \tag{A27}$$

The Rabi frequency (scaled by a factor of 2 here for simplicity) is the rate at which probability amplitudes between states oscillates in the presence of an electromagnetic field [18] and is defined as

$$\Omega = \frac{\vec{d} \cdot \vec{\varepsilon}}{\hbar} \tag{A28}$$

so we have

$$\begin{aligned}\hat{H}_{Int} = &\left(\hbar\Omega_s(t)\hat{\sigma}_{12}\hat{a}^\dagger e^{-i\vec{k}_s\cdot\vec{r}} + h.c.\right) \\ &+ \left(\hbar\Omega_I(t)\hat{\sigma}_{13}e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} + c.c.\right) \\ &+ \left(\hbar\Omega_{II}(t)\hat{\sigma}_{24}e^{i\omega_{II} t - i\vec{k}_{II}\cdot\vec{r}} + c.c.\right) \\ &+ \left(\hbar\Omega_i(t)\hat{\sigma}_{34}\hat{b}^\dagger e^{-i\vec{k}_i\cdot\vec{r}} + h.c.\right)\end{aligned} \tag{A29}$$

where the Rabi frequencies changes on a much slower timescale than the rate of its corresponding carrier frequency, ω, or the Rabi frequency, Ω, itself. To clarify, there is the rate of the carrier frequency (fast), the rate of Rabi oscillations (moderate), and the rate of change in the Rabi frequency (slow). This is because $\Omega \propto |\varepsilon|$ and, recall that, the field amplitude is the carrier of quantum information. This line of reasoning allows us to leverage the slowly-varying envelope approximation.

Combining this with $\hat{H}_S$ we obtain a Hamiltonian coupling atomic states with the weak signal and idler fields as

$$\begin{aligned}\hat{H} &= \hat{H}_S + \hat{H}_{Int} \\ &= \hbar\omega_1\hat{\sigma}_{11} + \hbar\omega_2\hat{\sigma}_{22} + \hbar\omega_3\hat{\sigma}_{33} + \hbar\omega_4\hat{\sigma}_{44} \\ &\quad + \hbar\omega_a\hat{a}^\dagger\hat{a} + \hbar\omega_b\hat{b}^\dagger\hat{b} \\ &\quad + \left(\hbar\Omega_s(t)\hat{\sigma}_{12}\hat{a}^\dagger e^{-i\vec{k}_s\cdot\vec{r}} + h.c.\right) \\ &\quad + \left(\hbar\Omega_I(t)\hat{\sigma}_{13}e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} + c.c.\right) \\ &\quad + \left(\hbar\Omega_{II}(t)\hat{\sigma}_{24}e^{i\omega_{II} t - i\vec{k}_{II}\cdot\vec{r}} + c.c.\right) \\ &\quad + \left(\hbar\Omega_i(t)\hat{\sigma}_{34}\hat{b}^\dagger e^{-i\vec{k}_i\cdot\vec{r}} + h.c.\right). \end{aligned} \tag{A30}$$

We may then define the unitary transformation which rotates with state $|1\rangle$ as reference

$$\hat{U}_{atom} = \hat{\sigma}_{11} + e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{22} + e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{33} + +e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44} \tag{A31}$$

and the unitary transform rotating at the driving field frequencies which acts on the Fock photon bases

$$\hat{U}_{photon} = e^{-i(\omega_s\hat{a}^\dagger\hat{a} + \omega_i\hat{b}^\dagger\hat{b})t}. \tag{A32}$$

The total unitary transformation of the coupled system is then

$$\begin{aligned}\hat{U} &= \hat{U}_{atom} \otimes \hat{U}_{photon} \\ &= \left(\hat{\sigma}_{11} + e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{22} + e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{33} + e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\right) \otimes e^{-i(\omega_s\hat{a}^\dagger\hat{a} + \omega_i\hat{b}^\dagger\hat{b})t}. \end{aligned} \tag{A33}$$

The total Hamiltonian can then be unitarily transformed as

$$\tilde{H} = \hat{U}^\dagger\hat{H}\hat{U} + i\hbar(\partial_t\hat{U}^\dagger)\hat{U}. \tag{A34}$$

Since the bases are independent in this transformation, we may compute the corresponding components independently

$$\hat{U}_{atom} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}} & 0 & 0 \\ 0 & 0 & e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}} & 0 \\ 0 & 0 & 0 & e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}} \end{pmatrix} \tag{A35}$$

$$\begin{aligned}\partial_t\hat{U}^\dagger_{atom} &= \partial_t \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & e^{i\omega_s t - i\vec{k}_s\cdot\vec{r}} & 0 & 0 \\ 0 & 0 & e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} & 0 \\ 0 & 0 & 0 & e^{i(\omega_{II}+\omega_s)t - i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}} \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & i\omega_s U^*_{22} & 0 & 0 \\ 0 & 0 & i\omega_I U^*_{33} & 0 \\ 0 & 0 & 0 & i(\omega_{II}+\omega_s)U^*_{44} \end{pmatrix} \end{aligned} \tag{A36}$$

$$i\hbar(\partial_t \hat{U}^\dagger_{atom})\hat{U}_{atom} = i\hbar \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & i\omega_s U^*_{22}U_{22} & 0 & 0 \\ 0 & 0 & i\omega_I U^*_{33}U_{33} & 0 \\ 0 & 0 & 0 & i(\omega_{II}+\omega_s)U^*_{44}U_{44} \end{pmatrix}$$
$$= \hbar \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -\omega_s & 0 & 0 \\ 0 & 0 & -\omega_I & 0 \\ 0 & 0 & 0 & -(\omega_{II}+\omega_s) \end{pmatrix}$$
$$= -\hbar\omega_s\hat{\sigma}_{22} - \hbar\omega_I\hat{\sigma}_{33} - \hbar(\omega_{II}+\omega_s)\hat{\sigma}_{44} \tag{A37}$$

$$\partial_t \hat{U}^\dagger_{photon} = i(\omega_s \hat{a}^\dagger\hat{a} + \omega_i \hat{b}^\dagger\hat{b})e^{i(\omega_s\hat{a}^\dagger\hat{a}+\omega_i\hat{b}^\dagger\hat{b})t} \tag{A38}$$

$$i\hbar(\partial_t \hat{U}^\dagger_{photon})\hat{U}_{photon} = i^2\hbar(\omega_s \hat{a}^\dagger\hat{a} + \omega_i \hat{b}^\dagger\hat{b})e^{i(\omega_s\hat{a}^\dagger\hat{a}+\omega_i\hat{b}^\dagger\hat{b})t}e^{-i(\omega_s\hat{a}^\dagger\hat{a}+\omega_i\hat{b}^\dagger\hat{b})t}$$
$$= -\hbar(\omega_s \hat{a}^\dagger\hat{a} + \omega_i \hat{b}^\dagger\hat{b}) \tag{A39}$$

$$i\hbar(\partial_t \hat{U}^\dagger)\hat{U} = -\hbar\omega_s\hat{\sigma}_{22} - \hbar\omega_I\hat{\sigma}_{33} - \hbar(\omega_{II}+\omega_s)\hat{\sigma}_{44} - \hbar\omega_s\hat{a}^\dagger\hat{a} - \hbar\omega_i\hat{b}^\dagger\hat{b} \tag{A40}$$

Then expanding the other term we have

$$\begin{aligned}
\hat{U}^\dagger\hat{H}\hat{U} = &\left(\hat{\sigma}_{11} + e^{i\omega_s t - i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{22} + e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{33} + e^{i(\omega_{II}+\omega_s)t - i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\right) \otimes e^{i(\omega_s\hat{a}^\dagger\hat{a}+\omega_i\hat{b}^\dagger\hat{b})t} \\
&\cdot \Big[\hbar\omega_1\hat{\sigma}_{11} + \hbar\omega_2\hat{\sigma}_{22} + \hbar\omega_3\hat{\sigma}_{33} + \hbar\omega_4\hat{\sigma}_{44} \\
&\quad + \hbar\omega_a\hat{a}^\dagger\hat{a} + \hbar\omega_b\hat{b}^\dagger\hat{b} \\
&\quad + \hbar\left(\Omega_s(t)\hat{\sigma}_{12}\hat{a}^\dagger e^{-i\vec{k}_s\cdot\vec{r}} + \Omega^*_s(t)\hat{\sigma}_{21}\hat{a}e^{i\vec{k}_s\cdot\vec{r}}\right) \\
&\quad + \hbar\left(\Omega_I(t)\hat{\sigma}_{13}e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} + \Omega^*_I(t)\hat{\sigma}_{31}e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\right) \\
&\quad + \hbar\left(\Omega_{II}(t)\hat{\sigma}_{24}e^{i\omega_{II} t - i\vec{k}_{II}\cdot\vec{r}} + \Omega^*_{II}(t)\hat{\sigma}_{42}e^{-i\omega_{II} t + i\vec{k}_{II}\cdot\vec{r}}\right)\Big] \\
&\quad + \hbar\left(\Omega_i(t)\hat{\sigma}_{34}\hat{b}^\dagger e^{-i\vec{k}_i\cdot\vec{r}} + \Omega^*_i(t)\hat{\sigma}_{43}\hat{b}e^{i\vec{k}_i\cdot\vec{r}}\right) \\
&\cdot \left(\hat{\sigma}_{11} + e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{22} + e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{33} + e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\right) \otimes e^{-i(\omega_s\hat{a}^\dagger\hat{a}+\omega_i\hat{b}^\dagger\hat{b})t}
\end{aligned} \tag{A41}$$

Since $\hat{U}_{atom}$ commutes with $\hat{U}_{photon}$, and $\hat{U}_{atom}$ commutes with $\hat{H}_{photon}$ such that

$$\hat{U}^\dagger_{atom}\hat{H}_{photon}\hat{U}_{atom} = \hat{U}^\dagger_{atom}\hat{U}_{atom}\hat{H}_{photon}$$
$$= \hat{H}_{photon} \tag{A42}$$

we have

$$
\begin{aligned}
\frac{\hat{U}^\dagger \hat{H} \hat{U}}{\hbar} &= e^{i(\omega_s \hat{a}^\dagger \hat{a} + \omega_i \hat{b}^\dagger \hat{b})t} \\
&\otimes \left(\hat{\sigma}_{11} + e^{i\omega_s t - i\vec{k}_s\cdot\vec{r}} \hat{\sigma}_{22} + e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} \hat{\sigma}_{33} + e^{i(\omega_{II}+\omega_s)t - i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}} \hat{\sigma}_{44} \right) \\
&\cdot \Big[\omega_1 \hat{\sigma}_{11} + \omega_2 \hat{\sigma}_{22} + \omega_3 \hat{\sigma}_{33} + \omega_4 \hat{\sigma}_{44} \\
&\quad + \omega_a \hat{a}^\dagger \hat{a} + \omega_b \hat{b}^\dagger \hat{b} \\
&\quad + \left(\Omega_s(t) \hat{\sigma}_{12} \hat{a}^\dagger e^{-i\vec{k}_s\cdot\vec{r}} + \Omega_s^*(t) \hat{\sigma}_{21} \hat{a} e^{i\vec{k}_s\cdot\vec{r}} \right) \\
&\quad + \left(\Omega_I(t) \hat{\sigma}_{13} e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} + \Omega_I^*(t) \hat{\sigma}_{31} e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}} \right) \\
&\quad + \left(\Omega_{II}(t) \hat{\sigma}_{24} e^{i\omega_{II} t - i\vec{k}_{II}\cdot\vec{r}} + \Omega_{II}^*(t) \hat{\sigma}_{42} e^{-i\omega_{II} t + i\vec{k}_{II}\cdot\vec{r}} \right) \\
&\quad + \left(\Omega_i(t) \hat{\sigma}_{34} \hat{b}^\dagger e^{-i\vec{k}_i\cdot\vec{r}} + \Omega_i^*(t) \hat{\sigma}_{43} \hat{b} e^{i\vec{k}_i\cdot\vec{r}} \right) \Big] \\
&\cdot \left(\hat{\sigma}_{11} + e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}} \hat{\sigma}_{22} + e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}} \hat{\sigma}_{33} + e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}} \hat{\sigma}_{44} \right) \\
&\otimes e^{-i(\omega_s \hat{a}^\dagger \hat{a} + \omega_i \hat{b}^\dagger \hat{b})t} \\
&+ e^{i(\omega_s \hat{a}^\dagger \hat{a} + \omega_i \hat{b}^\dagger \hat{b})t} \left(\omega_a \hat{a}^\dagger \hat{a} + \omega_b \hat{b}^\dagger \hat{b} \right) e^{-i(\omega_s \hat{a}^\dagger \hat{a} + \omega_i \hat{b}^\dagger \hat{b})t}. \qquad \text{(A43)}
\end{aligned}
$$

Separate the terms as

$$
\begin{aligned}
\hat{A} &= \left(\hat{\sigma}_{11} + e^{i\omega_s t - i\vec{k}_s\cdot\vec{r}} \hat{\sigma}_{22} + e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} \hat{\sigma}_{33} + e^{i(\omega_{II}+\omega_s)t - i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}} \hat{\sigma}_{44} \right) \\
&\cdot \Big[\omega_1 \hat{\sigma}_{11} + \omega_2 \hat{\sigma}_{22} + \omega_3 \hat{\sigma}_{33} + \omega_4 \hat{\sigma}_{44} \\
&\quad + \omega_a \hat{a}^\dagger \hat{a} + \omega_b \hat{b}^\dagger \hat{b} \\
&\quad + \left(\Omega_s(t) \hat{\sigma}_{12} \hat{a}^\dagger e^{-i\vec{k}_s\cdot\vec{r}} + \Omega_s^*(t) \hat{\sigma}_{21} \hat{a} e^{i\vec{k}_s\cdot\vec{r}} \right) \\
&\quad + \left(\Omega_I(t) \hat{\sigma}_{13} e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} + \Omega_I^*(t) \hat{\sigma}_{31} e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}} \right) \\
&\quad + \left(\Omega_{II}(t) \hat{\sigma}_{24} e^{i\omega_{II} t - i\vec{k}_{II}\cdot\vec{r}} + \Omega_{II}^*(t) \hat{\sigma}_{42} e^{-i\omega_{II} t + i\vec{k}_{II}\cdot\vec{r}} \right) \\
&\quad + \left(\Omega_i(t) \hat{\sigma}_{34} \hat{b}^\dagger e^{-i\vec{k}_i\cdot\vec{r}} + \Omega_i^*(t) \hat{\sigma}_{43} \hat{b} e^{i\vec{k}_i\cdot\vec{r}} \right) \Big] \\
&\cdot \left(\hat{\sigma}_{11} + e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}} \hat{\sigma}_{22} + e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}} \hat{\sigma}_{33} + e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}} \hat{\sigma}_{44} \right) \qquad \text{(A44)}
\end{aligned}
$$

$$
\hat{B} = e^{i(\omega_s \hat{a}^\dagger \hat{a} + \omega_i \hat{b}^\dagger \hat{b})t} \hat{A} e^{-i(\omega_s \hat{a}^\dagger \hat{a} + \omega_i \hat{b}^\dagger \hat{b})t} \qquad \text{(A45)}
$$

$$
\hat{C} = \left(e^{i(\omega_s \hat{a}^\dagger \hat{a} + \omega_i \hat{b}^\dagger \hat{b})t} \right) \left(\omega_a \hat{a}^\dagger \hat{a} + \omega_b \hat{b}^\dagger \hat{b} \right) \left(e^{-i(\omega_s \hat{a}^\dagger \hat{a} + \omega_i \hat{b}^\dagger \hat{b})t} \right). \qquad \text{(A46)}
$$

Let us compute $\hat{A}$ using the facts that $\hat{\sigma}_{ij}\hat{\sigma}_{kl} = \hat{\sigma}_{il}\delta_{jk}$, $\hat{\sigma}_{ij}\hat{\sigma}_{jk}\hat{\sigma}_{kl} = \hat{\sigma}_{il}$, and $[\hat{\sigma}_{ij}, \hat{a}] = [\hat{\sigma}_{ij}, \hat{a}^\dagger] = [\hat{\sigma}_{ij}, \hat{b}] = [\hat{\sigma}_{ij}, \hat{b}^\dagger] = 0$. First notice that

$$
\begin{aligned}
&\left(\hat{\sigma}_{11} + e^{i\omega_s t - i\vec{k}_s\cdot\vec{r}} \hat{\sigma}_{22} + e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} \hat{\sigma}_{33} + e^{i(\omega_{II}+\omega_s)t - i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}} \hat{\sigma}_{44} \right) \\
&\cdot \left[\omega_1 \hat{\sigma}_{11} + \omega_2 \hat{\sigma}_{22} + \omega_3 \hat{\sigma}_{33} + \omega_4 \hat{\sigma}_{44} \right] \\
&\cdot \left(\hat{\sigma}_{11} + e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}} \hat{\sigma}_{22} + e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}} \hat{\sigma}_{33} + e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}} \hat{\sigma}_{44} \right) \\
&\qquad = \left[\omega_1 \hat{\sigma}_{11} + \omega_2 \hat{\sigma}_{22} + \omega_3 \hat{\sigma}_{33} + \omega_4 \hat{\sigma}_{44} \right]. \qquad \text{(A47)}
\end{aligned}
$$

So we have

$$
\begin{aligned}
\hat{A} &= \omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44} \\
&\quad + \left(\hat{\sigma}_{11} + e^{i\omega_s t - i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{22} + e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{33} + e^{i(\omega_{II}+\omega_s)t - i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\right) \\
&\quad \cdot \Bigg[\left(\Omega_s(t)\hat{\sigma}_{12}\hat{a}^\dagger e^{-i\vec{k}_s\cdot\vec{r}} + \Omega_s^*(t)\hat{\sigma}_{21}\hat{a}e^{i\vec{k}_s\cdot\vec{r}}\right) \\
&\qquad + \left(\Omega_I(t)\hat{\sigma}_{13}e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} + \Omega_I^*(t)\hat{\sigma}_{31}e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\right) \\
&\qquad + \left(\Omega_{II}(t)\hat{\sigma}_{24}e^{i\omega_{II} t - i\vec{k}_{II}\cdot\vec{r}} + \Omega_{II}^*(t)\hat{\sigma}_{42}e^{-i\omega_{II} t + i\vec{k}_{II}\cdot\vec{r}}\right) \\
&\qquad + \left(\Omega_i(t)\hat{\sigma}_{34}\hat{b}^\dagger e^{-i\vec{k}_i\cdot\vec{r}} + \Omega_i^*(t)\hat{\sigma}_{43}\hat{b}e^{i\vec{k}_i\cdot\vec{r}}\right)\Bigg] \\
&\quad \cdot \left(\hat{\sigma}_{11} + e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{22} + e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{33} + e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\right) \\
&= \omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44} \\
&\quad + \left(\Omega_s(t)\hat{\sigma}_{11}\hat{\sigma}_{12}\hat{\sigma}_{22}e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}}\hat{a}^\dagger e^{-i\vec{k}_s\cdot\vec{r}} + \Omega_s^*(t)e^{i\omega_s t - i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{22}\hat{\sigma}_{21}\hat{\sigma}_{11}\hat{a}e^{i\vec{k}_s\cdot\vec{r}}\right) \\
&\quad + \left(\Omega_I(t)\hat{\sigma}_{11}\hat{\sigma}_{13}\hat{\sigma}_{33}e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}} + \Omega_I^*(t)e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{33}\hat{\sigma}_{31}\hat{\sigma}_{11}e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\right) \\
&\quad + \Big(\Omega_{II}(t)e^{i\omega_s t - i\vec{k}_s\cdot\vec{r}}\hat{\sigma}_{22}\hat{\sigma}_{24}\hat{\sigma}_{44}e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}e^{i\omega_{II} t - i\vec{k}_{II}\cdot\vec{r}} \\
&\qquad + \Omega_{II}^*(t)e^{i(\omega_{II}+\omega_s)t - i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\hat{\sigma}_{42}\hat{\sigma}_{22}e^{-i\omega_s t + i\vec{k}_s\cdot\vec{r}}e^{-i\omega_{II} t + i\vec{k}_{II}\cdot\vec{r}}\Big) \\
&\quad + \Big(\Omega_i(t)e^{i\omega_I t - i\vec{k}_I\cdot\vec{r}}\hat{\sigma}_{33}\hat{\sigma}_{34}\hat{\sigma}_{44}e^{-i(\omega_{II}+\omega_s)t + i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{b}^\dagger e^{-i\vec{k}_i\cdot\vec{r}} \\
&\qquad + \Omega_i^*(t)e^{i(\omega_{II}+\omega_s)t - i(\vec{k}_{II}+\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{44}\hat{\sigma}_{43}\hat{\sigma}_{33}e^{-i\omega_I t + i\vec{k}_I\cdot\vec{r}}\hat{b}e^{i\vec{k}_i\cdot\vec{r}}\Big) \\
&= \omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44} \\
&\quad + \left(\Omega_s(t)e^{-i\omega_s t}\hat{\sigma}_{12}\hat{a}^\dagger + \Omega_s^*(t)e^{i\omega_s t}\hat{\sigma}_{21}\hat{a}\right) \\
&\quad + \left(\Omega_I(t)\hat{\sigma}_{13} + \Omega_I^*(t)\hat{\sigma}_{31}\right) \\
&\quad + \left(\Omega_{II}(t)\hat{\sigma}_{24} + \Omega_{II}^*(t)\hat{\sigma}_{42}\right) \\
&\quad + \left(\Omega_i(t)e^{i(\omega_I-\omega_{II}-\omega_s)t - i(\vec{k}_I+\vec{k}_i-\vec{k}_{II}-\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{34}\hat{b}^\dagger + \Omega_i^*(t)e^{-i(\omega_{II}+\omega_s-\omega_I)t + i(\vec{k}_I+\vec{k}_i-\vec{k}_{II}-\vec{k}_s)\cdot\vec{r}}\hat{\sigma}_{43}\hat{b}\right).
\end{aligned} \tag{A48}
$$

We can then impose the energy conservation condition and phase match condition (in that order as follows)

$$\omega_I + \omega_i = \omega_s + \omega_{II} \tag{A49}$$

$$k_I + k_i = k_s + k_{II} \tag{A50}$$

$$
\begin{aligned}
\hat{A} &= \omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44} \\
&\quad + \left(\Omega_s(t)e^{-i\omega_s t}\hat{\sigma}_{12}\hat{a}^\dagger + \Omega_s^*(t)e^{i\omega_s t}\hat{\sigma}_{21}\hat{a}\right) \\
&\quad + \left(\Omega_I(t)\hat{\sigma}_{13} + \Omega_I^*(t)\hat{\sigma}_{31}\right) \\
&\quad + \left(\Omega_{II}(t)\hat{\sigma}_{24} + \Omega_{II}^*(t)\hat{\sigma}_{42}\right) \\
&\quad + \left(\Omega_i(t)e^{-i\omega_i t}\hat{\sigma}_{34}\hat{b}^\dagger + \Omega_i^*(t)e^{i\omega_i t}\hat{\sigma}_{43}\hat{b}.\right)
\end{aligned} \tag{A51}
$$

Now we may compute $\hat{B}$ from Eq. (A45) as

$$
\begin{aligned}
\hat{B} &= e^{i(\omega_s \hat{a}^\dagger \hat{a} + \omega_i \hat{b}^\dagger \hat{b})t} \hat{A} e^{-i(\omega_s \hat{a}^\dagger \hat{a} + \omega_i \hat{b}^\dagger \hat{b})t} \\
&= e^{i(\omega_s \hat{a}^\dagger \hat{a} + \omega_i \hat{b}^\dagger \hat{b})t} \\
&\quad \cdot \Big[\omega_1 \hat{\sigma}_{11} + \omega_2 \hat{\sigma}_{22} + \omega_3 \hat{\sigma}_{33} + \omega_4 \hat{\sigma}_{44} \\
&\qquad + \left(\Omega_s(t) e^{-i\omega_s t} \hat{\sigma}_{12} \hat{a}^\dagger + \Omega_s^*(t) e^{i\omega_s t} \hat{\sigma}_{21} \hat{a} \right) \\
&\qquad + \left(\Omega_I(t) \hat{\sigma}_{13} + \Omega_I^*(t) \hat{\sigma}_{31} \right) \\
&\qquad + \left(\Omega_{II}(t) \hat{\sigma}_{24} + \Omega_{II}^*(t) \hat{\sigma}_{42} \right) \\
&\qquad + \left(\Omega_i(t) e^{-i\omega_i t} \hat{\sigma}_{34} \hat{b}^\dagger + \Omega_i^*(t) e^{i\omega_i t} \hat{\sigma}_{43} \hat{b} \right) \Big] \\
&\quad \cdot e^{-i(\omega_s \hat{a}^\dagger \hat{a} + \omega_i \hat{b}^\dagger \hat{b})t}
\end{aligned}
\tag{A52}
$$

which, via commutation relations, simplifies to

$$
\begin{aligned}
\hat{B} = \omega_1 \hat{\sigma}_{11} &+ \omega_2 \hat{\sigma}_{22} + \omega_3 \hat{\sigma}_{33} + \omega_4 \hat{\sigma}_{44} \\
&+ \left(\Omega_I(t) \hat{\sigma}_{13} + \Omega_I^*(t) \hat{\sigma}_{31} \right) \\
&+ \left(\Omega_{II}(t) \hat{\sigma}_{24} + \Omega_{II}^*(t) \hat{\sigma}_{42} \right) \\
&+ e^{i\omega_s \hat{a}^\dagger \hat{a} t} \left(\Omega_s(t) e^{-i\omega_s t} \hat{\sigma}_{12} \hat{a} + \Omega_s^*(t) e^{i\omega_s t} \hat{\sigma}_{21} \hat{a}^\dagger \right) e^{-i\omega_s \hat{a}^\dagger \hat{a} t} \\
&+ e^{i\omega_i \hat{b}^\dagger \hat{b} t} \left(\Omega_i(t) e^{-i\omega_i t} \hat{\sigma}_{34} \hat{b} + \Omega_i^*(t) e^{i\omega_i t} \hat{\sigma}_{43} \hat{b}^\dagger \right) e^{-i\omega_i \hat{b}^\dagger \hat{b} t}.
\end{aligned}
\tag{A53}
$$

We must then compute

$$
e^{i\omega_s \hat{a}^\dagger \hat{a} t} \hat{a} e^{-i\omega_s \hat{a}^\dagger \hat{a} t} \tag{A54}
$$

which we can do via the Baker-Campbell-Hausdorff formula

$$
e^{\hat{X}} \hat{Y} e^{-\hat{X}} = \sum_{n=0}^{\infty} \frac{[X, Y]_n}{n!} = Y + [X, Y] + \frac{1}{2!}[X, [X, Y]] + \cdots \tag{A55}
$$

where

$$
[X, Y]_n = \underbrace{[X, [X, \ldots, [X, [X}_{\text{n times}}, Y]], \ldots]] \tag{A56}
$$

$$
\begin{aligned}
e^{i\omega_s \hat{a}^\dagger \hat{a} t} \hat{a} e^{-i\omega_s \hat{a}^\dagger \hat{a} t} &= \sum_{n=0}^{\infty} \frac{[i\omega_s \hat{a}^\dagger \hat{a} t, \hat{a}]_n}{n!} \\
e^{i\omega_s \hat{a}^\dagger \hat{a} t} \hat{a} e^{-i\omega_s \hat{a}^\dagger \hat{a} t} &= \sum_{n=0}^{\infty} \frac{[i\omega_s \hat{a}^\dagger \hat{a} t, \hat{a}]_n}{n!} \\
&= \hat{a} + [i\omega_s \hat{a}^\dagger \hat{a} t, \hat{a}] + \frac{1}{2!}[i\omega_s \hat{a}^\dagger \hat{a} t, [i\omega_s \hat{a}^\dagger \hat{a} t, \hat{a}]] + \frac{1}{3!}[i\omega_s \hat{a}^\dagger \hat{a} t, [i\omega_s \hat{a}^\dagger \hat{a} t, [i\omega_s \hat{a}^\dagger \hat{a} t, \hat{a}]]] + \cdots \\
&= \hat{a} + i\omega_s t[\hat{a}^\dagger \hat{a}, \hat{a}] + \frac{(i\omega_s t)^2}{2!}[\hat{a}^\dagger \hat{a}, [\hat{a}^\dagger \hat{a}, \hat{a}]] + \frac{(i\omega_s t)^3}{3!}[\hat{a}^\dagger \hat{a}, [\hat{a}^\dagger \hat{a}, [\hat{a}^\dagger \hat{a}, \hat{a}]]] + \cdots
\end{aligned}
\tag{A57}
$$

To move forward we must then compute the commutators

$$
[\hat{a}^\dagger \hat{a}, \hat{a}] = \hat{a}^\dagger \hat{a} \hat{a} - \hat{a} \hat{a}^\dagger \hat{a}
$$

use the commutation relation

$$
\begin{aligned}
[\hat{a}, \hat{a}^\dagger] = \hat{a}\hat{a}^\dagger - \hat{a}^\dagger \hat{a} &= 1 \\
\hat{a}\hat{a}^\dagger &= 1 + \hat{a}^\dagger \hat{a}
\end{aligned}
\tag{A58}
$$

to obtain

$$\begin{aligned}[\hat{a}^\dagger\hat{a},\hat{a}] &= \hat{a}^\dagger\hat{a}\hat{a} - (1+\hat{a}^\dagger\hat{a})\hat{a} \\ &= \hat{a}^\dagger\hat{a}\hat{a} - \hat{a} - \hat{a}^\dagger\hat{a}\hat{a} \\ &= -\hat{a}. \end{aligned} \tag{A59}$$

We then have that

$$\begin{aligned}[\hat{a}^\dagger\hat{a},[\hat{a}^\dagger\hat{a},\hat{a}]] &= [\hat{a}^\dagger\hat{a},-\hat{a}] \\ &= -[\hat{a}^\dagger\hat{a},\hat{a}] \\ &= \hat{a}. \end{aligned} \tag{A60}$$

Assume this pattern holds until the k^{th} application

$$[\hat{a}^\dagger\hat{a},\hat{a}]_k = (-1)^k\hat{a} \qquad \text{(Induction Hypothesis)} \tag{A61}$$

then consider the $(k+1)^{th}$ step

$$\begin{aligned}[\hat{a}^\dagger\hat{a},\hat{a}]_{k+1} &= [\hat{a}^\dagger\hat{a},[\hat{a}^\dagger\hat{a},\hat{a}]_k] \\ &= [\hat{a}^\dagger\hat{a},(-1)^k\hat{a}] \\ &= (-1)^k[\hat{a}^\dagger\hat{a},\hat{a}] \\ &= (-1)^k(-\hat{a}) \\ &= (-1)^{k+1}\hat{a}. \end{aligned} \tag{A62}$$

Thus, by induction we have that

$$[\hat{a}^\dagger\hat{a},\hat{a}]_n = (-1)^n\hat{a}. \tag{A63}$$

So we may write

$$\begin{aligned} e^{i\omega_s\hat{a}^\dagger\hat{a}t}\hat{a}e^{-i\omega_s\hat{a}^\dagger\hat{a}t} &= \hat{a} + i\omega_s t[\hat{a}^\dagger\hat{a},\hat{a}] + \frac{(i\omega_s t)^2}{2!}[\hat{a}^\dagger\hat{a},[\hat{a}^\dagger\hat{a},\hat{a}]] + \frac{(i\omega_s t)^3}{3!}[\hat{a}^\dagger\hat{a},[\hat{a}^\dagger\hat{a},[\hat{a}^\dagger\hat{a},\hat{a}]]] + \cdots \\ &= \hat{a} + i\omega_s t(-\hat{a}) + \frac{(i\omega_s t)^2}{2!}(\hat{a}) + \frac{(i\omega_s t)^3}{3!}(-\hat{a}) + \cdots \\ &= \hat{a}\left(1 - i\omega_s t + \frac{(i\omega_s t)^2}{2!} - \frac{(i\omega_s t)^3}{3!} + \cdots\right) \\ &= \hat{a}\left(1 + (-i\omega_s t) + \frac{(-i\omega_s t)^2}{2!} + \frac{(-i\omega_s t)^3}{3!} + \cdots\right) \\ &= \hat{a}\sum_{n=0}^{\infty}\frac{(-i\omega_s t)^n}{n!} \\ &= \hat{a}e^{-i\omega_s t}. \end{aligned} \tag{A64}$$

Then we can find

$$\begin{aligned} \left(\hat{a}e^{-i\omega_s t}\right)^\dagger &= \left(e^{i\omega_s\hat{a}^\dagger\hat{a}t}\hat{a}e^{-i\omega_s\hat{a}^\dagger\hat{a}t}\right)^\dagger \\ \hat{a}^\dagger e^{i\omega_s t} &= \left(e^{-i\omega_s\hat{a}^\dagger\hat{a}t}\right)^\dagger\hat{a}^\dagger\left(e^{i\omega_s\hat{a}^\dagger\hat{a}t}\right)^\dagger \\ \hat{a}^\dagger e^{i\omega_s t} &= e^{i\omega_s\left(\hat{a}^\dagger\hat{a}\right)^\dagger t}\hat{a}^\dagger e^{i\omega_s\left(\hat{a}^\dagger\hat{a}\right)^\dagger t} \\ \hat{a}^\dagger e^{i\omega_s t} &= e^{i\omega_s\hat{a}^\dagger\left(\hat{a}^\dagger\right)^\dagger t}\hat{a}^\dagger e^{i\omega_s\hat{a}^\dagger\left(\hat{a}^\dagger\right)^\dagger\hat{a}t} \\ \hat{a}^\dagger e^{i\omega_s t} &= e^{i\omega_s\hat{a}^\dagger\hat{a}t}\hat{a}^\dagger e^{-i\omega_s\hat{a}^\dagger\hat{a}t}. \end{aligned} \tag{A65}$$

Similarly

$$e^{i\omega_i\hat{b}^\dagger\hat{b}t}\hat{b}e^{-i\omega_i\hat{b}^\dagger\hat{b}t} = \hat{b}e^{-i\omega_i t} \tag{A66}$$

$$e^{i\omega_i\hat{b}^\dagger\hat{b}t}\hat{b}^\dagger e^{-i\omega_i\hat{b}^\dagger\hat{b}t} = \hat{b}^\dagger e^{i\omega_i t}. \tag{A67}$$

So we can simplify Eq. (A53) to

$$\begin{aligned}
\hat{B} = {} & \omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44} \\
& + (\Omega_I(t)\hat{\sigma}_{13} + \Omega_I^*(t)\hat{\sigma}_{31}) \\
& + (\Omega_{II}(t)\hat{\sigma}_{24} + \Omega_{II}^*(t)\hat{\sigma}_{42}) \\
& + \left(\Omega_s(t)e^{-i\omega_s t}\hat{\sigma}_{12}\hat{a}^\dagger e^{i\omega_s t} + \Omega_s^*(t)e^{i\omega_s t}\hat{\sigma}_{21}\hat{a}e^{-i\omega_s t}\right) \\
& + \left(\Omega_i(t)e^{-i\omega_i t}\hat{\sigma}_{34}\hat{b}^\dagger e^{i\omega_i t} + \Omega_i^*(t)e^{i\omega_i t}\hat{\sigma}_{43}\hat{b}e^{-i\omega_i t}\right) \\
\\
= {} & \omega_1\hat{\sigma}_{11} + \omega_2\hat{\sigma}_{22} + \omega_3\hat{\sigma}_{33} + \omega_4\hat{\sigma}_{44} \\
& + \left(\Omega_s(t)\hat{a}^\dagger\hat{\sigma}_{12} + \Omega_s^*(t)\hat{a}\hat{\sigma}_{21}\right) \\
& + (\Omega_I(t)\hat{\sigma}_{13} + \Omega_I^*(t)\hat{\sigma}_{31}) \\
& + (\Omega_{II}(t)\hat{\sigma}_{24} + \Omega_{II}^*(t)\hat{\sigma}_{42}) \\
& + \left(\Omega_i(t)\hat{b}^\dagger\hat{\sigma}_{34} + \Omega_i^*(t)\hat{b}\hat{\sigma}_{43}\right) \\
\\
= {} & \begin{pmatrix} \omega_1 & \Omega_s(t)\hat{a}^\dagger & \Omega_I(t) & 0 \\ \Omega_s^*(t)\hat{a} & \omega_2 & 0 & \Omega_{II}(t) \\ \Omega_I^*(t) & 0 & \omega_3 & \Omega_i(t)\hat{b}^\dagger \\ 0 & \Omega_{II}^*(t) & \Omega_i^*(t)\hat{b} & \omega_4 \end{pmatrix}.
\end{aligned} \tag{A68}$$

Now consider the next component from Eq. (A46)

$$\begin{aligned}
\hat{C} &= \left(e^{i(\omega_s\hat{a}^\dagger\hat{a}+\omega_i\hat{b}^\dagger\hat{b})t}\right)\left(\omega_a\hat{a}^\dagger\hat{a} + \omega_b\hat{b}^\dagger\hat{b}\right)\left(e^{-i(\omega_s\hat{a}^\dagger\hat{a}+\omega_i\hat{b}^\dagger\hat{b})t}\right) \\
&= \omega_a e^{i\omega_s\hat{a}^\dagger\hat{a}t}\hat{a}^\dagger\hat{a}e^{-i\omega_s\hat{a}^\dagger\hat{a}} + \omega_b e^{\omega_i\hat{b}^\dagger\hat{b}t}\hat{b}^\dagger\hat{b}e^{-\omega_i\hat{b}^\dagger\hat{b}t}.
\end{aligned} \tag{A69}$$

Using the commutation relation for the distinct fields $\left[\hat{a}^\dagger\hat{a}, \hat{b}^\dagger\hat{b}\right] = 0$, we may expand the exponentials as

$$\begin{aligned}
e^{i\omega\hat{a}^\dagger\hat{a}t}\hat{a}^\dagger\hat{a}e^{-i\omega\hat{a}^\dagger\hat{a}t} &= \left[\sum_{n=0}^{\infty}\frac{\left(i\omega\hat{a}^\dagger\hat{a}t\right)^n}{n!}\right]\hat{a}^\dagger\hat{a}\left[\sum_{n=0}^{\infty}\frac{\left(-i\omega\hat{a}^\dagger\hat{a}t\right)^n}{n!}\right] \\
&= \left[\sum_{n=0}^{\infty}\frac{(i\omega t)^n\left(\hat{a}^\dagger\hat{a}\right)^{n+1}}{n!}\right]\left[\sum_{n=0}^{\infty}\frac{\left(-i\omega\hat{a}^\dagger\hat{a}t\right)^n}{n!}\right] \\
&= \hat{a}^\dagger\hat{a}\left[\sum_{n=0}^{\infty}\frac{\left(i\omega\hat{a}^\dagger\hat{a}t\right)^n}{n!}\right]\left[\sum_{n=0}^{\infty}\frac{\left(-i\omega\hat{a}^\dagger\hat{a}t\right)^n}{n!}\right] \\
&= \hat{a}^\dagger\hat{a}e^{i\omega\hat{a}^\dagger\hat{a}t}e^{-i\omega\hat{a}^\dagger\hat{a}t} \\
&= \hat{a}^\dagger\hat{a}
\end{aligned} \tag{A70}$$

which follows similarly for

$$e^{i\omega\hat{b}^\dagger\hat{b}t}\hat{b}^\dagger\hat{b}e^{-i\omega\hat{b}^\dagger\hat{b}t} = \hat{b}^\dagger\hat{b} \tag{A71}$$

yielding

$$\hat{C} = \omega_a\hat{a}^\dagger\hat{a} + \omega_b\hat{b}^\dagger\hat{b}. \tag{A72}$$

Thus, we have

$$
\begin{aligned}
\frac{\hat{U}^\dagger \hat{H} \hat{U}}{\hbar} &= \hat{B} + \hat{C} \\
&= \omega_1 \hat{\sigma}_{11} + \omega_2 \hat{\sigma}_{22} + \omega_3 \hat{\sigma}_{33} + \omega_4 \hat{\sigma}_{44} \\
&\quad + \left(\Omega_s(t) \hat{a}^\dagger \hat{\sigma}_{12} + \Omega_s^*(t) \hat{a} \hat{\sigma}_{21}\right) \\
&\quad + \left(\Omega_I(t) \hat{\sigma}_{13} + \Omega_I^*(t) \hat{\sigma}_{31}\right) \\
&\quad + \left(\Omega_{II}(t) \hat{\sigma}_{24} + \Omega_{II}^*(t) \hat{\sigma}_{42}\right) \\
&\quad + \left(\Omega_i(t) \hat{b}^\dagger \hat{\sigma}_{34} + \Omega_i^*(t) \hat{b} \hat{\sigma}_{43}\right) \\
&\quad + \omega_a \hat{a}^\dagger \hat{a} + \omega_b \hat{b}^\dagger \hat{b}.
\end{aligned}
\tag{A73}
$$

So we have the unitary transformation as

$$
\begin{aligned}
\tilde{H} &= \hat{U}^\dagger \hat{H} \hat{U} + i\hbar (\partial_t \hat{U}^\dagger) \hat{U} \\
&= \hbar \Big[\omega_1 \hat{\sigma}_{11} + \omega_2 \hat{\sigma}_{22} + \omega_3 \hat{\sigma}_{33} + \omega_4 \hat{\sigma}_{44} \\
&\qquad + \left(\Omega_s(t) \hat{a}^\dagger \hat{\sigma}_{12} + \Omega_s^*(t) \hat{a} \hat{\sigma}_{21}\right) \\
&\qquad + \left(\Omega_I(t) \hat{\sigma}_{13} + \Omega_I^*(t) \hat{\sigma}_{31}\right) \\
&\qquad + \left(\Omega_{II}(t) \hat{\sigma}_{24} + \Omega_{II}^*(t) \hat{\sigma}_{42}\right) \\
&\qquad + \left(\Omega_i(t) \hat{b}^\dagger \hat{\sigma}_{34} + \Omega_i^*(t) \hat{b} \hat{\sigma}_{43}\right) \\
&\qquad + \omega_a \hat{a}^\dagger \hat{a} + \omega_b \hat{b}^\dagger \hat{b} \Big] \\
&\quad + \hbar \Big[-\omega_s \hat{\sigma}_{22} - \omega_I \hat{\sigma}_{33} - (\omega_{II} + \omega_s) \hat{\sigma}_{44} - \omega_s \hat{a}^\dagger \hat{a} - \omega_i \hat{b}^\dagger \hat{b} \Big] \\
&= \hbar \Big[\omega_1 \hat{\sigma}_{11} + (\omega_2 - \omega_s) \hat{\sigma}_{22} + (\omega_3 - \omega_I) \hat{\sigma}_{33} + (\omega_4 - \omega_{II} + \omega_s) \hat{\sigma}_{44} \\
&\qquad + \left(\Omega_s(t) \hat{a}^\dagger \hat{\sigma}_{12} + \Omega_s^*(t) \hat{a} \hat{\sigma}_{21}\right) \\
&\qquad + \left(\Omega_I(t) \hat{\sigma}_{13} + \Omega_I^*(t) \hat{\sigma}_{31}\right) \\
&\qquad + \left(\Omega_{II}(t) \hat{\sigma}_{24} + \Omega_{II}^*(t) \hat{\sigma}_{42}\right) \\
&\qquad + \left(\Omega_i(t) \hat{b}^\dagger \hat{\sigma}_{34} + \Omega_i^*(t) \hat{b} \hat{\sigma}_{43}\right) \\
&\qquad + (\omega_a - \omega_s) \hat{a}^\dagger \hat{a} + (\omega_b - \omega_i) \hat{b}^\dagger \hat{b}. \Big]
\end{aligned}
\tag{A74}
$$

We can then choose to arbitrarily shift the energy levels of the Hamiltonian by the ground state energy $\hbar\omega_1$

$$
\begin{aligned}
\tilde{H} &\to \tilde{H} - \hbar\omega_1 \hat{\mathbb{1}} \\
&\to \hbar \Big[(\omega_2 - \omega_1 - \omega_s) \hat{\sigma}_{22} + (\omega_3 - \omega_1 - \omega_I) \hat{\sigma}_{33} + (\omega_4 - \omega_1 - \omega_{II} - \omega_s) \hat{\sigma}_{44} \\
&\qquad + \left(\Omega_s(t) \hat{a}^\dagger \hat{\sigma}_{12} + \Omega_s^*(t) \hat{a} \hat{\sigma}_{21}\right) \\
&\qquad + \left(\Omega_I(t) \hat{\sigma}_{13} + \Omega_I^*(t) \hat{\sigma}_{31}\right) \\
&\qquad + \left(\Omega_{II}(t) \hat{\sigma}_{24} + \Omega_{II}^*(t) \hat{\sigma}_{42}\right) \\
&\qquad + \left(\Omega_i(t) \hat{b}^\dagger \hat{\sigma}_{34} + \Omega_i^*(t) \hat{b} \hat{\sigma}_{43}\right) \\
&\qquad + (\omega_a - \omega_s) \hat{a}^\dagger \hat{a} + (\omega_b - \omega_i) \hat{b}^\dagger \hat{b}. \Big]
\end{aligned}
\tag{A75}
$$

Apply the relations

$$\omega_{12} = \omega_2 - \omega_1 \tag{A76}$$
$$\omega_{13} = \omega_3 - \omega_1 \tag{A77}$$
$$\omega_{24} = \omega_4 - \omega_2 \tag{A78}$$
$$\omega_{34} = \omega_4 - \omega_3 \tag{A79}$$
$$\Delta_s = \omega_{12} - \omega_s \tag{A80}$$
$$\Delta_I = \omega_{13} - \omega_I \tag{A81}$$
$$\Delta_{II} = \omega_{24} - \omega_{II} \tag{A82}$$
$$\Delta_i = \omega_{34} - \omega_i \tag{A83}$$
$$\Delta_a = \omega_a - \omega_s \tag{A84}$$
$$\Delta_b = \omega_b - \omega_i \tag{A85}$$

$$\begin{aligned} \omega_{34} + \omega_{13} &= (\omega_4 - \omega_3) + (\omega_3 - \omega_1) \\ &= \omega_4 - \omega_1 \end{aligned} \tag{A86}$$

to finally obtain our Hamiltonian as

$$\begin{aligned} \tilde{H} = \hbar \Big[& \Delta_s \hat{\sigma}_{22} + \Delta_I \hat{\sigma}_{33} + (\Delta_{II} + \Delta_s)\hat{\sigma}_{44} \\ & + \left(\Omega_s(t)\hat{a}^\dagger \hat{\sigma}_{12} + \Omega_s^*(t)\hat{a}\hat{\sigma}_{21}\right) \\ & + (\Omega_I(t)\hat{\sigma}_{13} + \Omega_I^*(t)\hat{\sigma}_{31}) \\ & + (\Omega_{II}(t)\hat{\sigma}_{24} + \Omega_{II}^*(t)\hat{\sigma}_{42}) \\ & + \left(\Omega_i(t)\hat{b}^\dagger \hat{\sigma}_{34} + \Omega_i^*(t)\hat{b}\hat{\sigma}_{43}\right) \\ & + \Delta_a \hat{a}^\dagger \hat{a} + \Delta_b \hat{b}^\dagger \hat{b} \Big] \end{aligned} \tag{A87}$$

$$= \hbar \begin{pmatrix} 0 & \Omega_s(t)\hat{a}^\dagger & \Omega_I(t) & 0 \\ \Omega_s^*(t)\hat{a} & \Delta_s & 0 & \Omega_{II}(t) \\ \Omega_I^*(t) & 0 & \Delta_I & \Omega_i(t)\hat{b}^\dagger \\ 0 & \Omega_{II}^*(t) & \Omega_i^*(t)\hat{b} & \Delta_{II} + \Delta_s \end{pmatrix} + \hbar\Delta_a \hat{a}^\dagger \hat{a} + \hbar\Delta_b \hat{b}^\dagger \hat{b} \tag{A88}$$

B Maxwell-Bloch Equations Derivation

Operator time evolution in the Heisenberg picture is given by

$$\frac{d}{dt}\hat{A}_H(t) = \frac{i}{\hbar}\left[\hat{H}_H(t), \hat{A}_H(t)\right] + \left(\frac{\partial \hat{A}_S}{\partial t}\right)_H \tag{B1}$$

To find the change in our electric field, we might consider applying this to a weak quantized electric field $\hat{\vec{E}}(\vec{r},t) = \vec{\varepsilon}(\vec{r},t)\hat{a}e^{i\vec{k}\cdot\vec{r}} + h.c.$, where $\vec{\varepsilon}(\vec{r},t)$ behaves as a slowly-varying envelope function, to obtain. However, we expect the carrier frequency and the vector direction $\vec{\varepsilon}/\varepsilon$ to remain constant, so it would be of more use to apply this equation directly to the envelope amplitude operator. Additionally, we may consider only positions along the $\vec{k}$ direction to have $\hat{\vec{E}}(z,t)$

$$\frac{d}{dt}\hat{\varepsilon}_H(z,t) = \frac{i}{\hbar}\left[\tilde{H}(t), \hat{\varepsilon}_H(z,t)\right] + \left(\frac{\partial \hat{\varepsilon}(z,t)}{\partial t} + h.c.\right)_H \tag{B2}$$

where $\hat{\varepsilon}$ is defined as the quantum electric field operator

$$\hat{\varepsilon} = \varepsilon\hat{a} + \varepsilon^*\hat{a}^\dagger. \tag{B3}$$

Note that a plane wave propagating through an atomic medium ε may be expressed

$$\varepsilon = \sqrt{\frac{\hbar\omega}{2\epsilon_0 V}} \tag{B4}$$

where V is the volume of the medium [52].

We may consider the system to always be in a pseudo-steady-state since $\hat{\varepsilon}(z,t)$ varies on such a large timescale in comparison to the driving frequencies that any transient responses are negligible [23]. Therefore, we may neglect $\partial_t\hat{\varepsilon}$ to obtain

$$\frac{d}{dt}\hat{\varepsilon}_H(z,t) = \frac{i}{\hbar}\left[\tilde{H}(t), \hat{\varepsilon}_H(z,t)\right] \tag{B5}$$

To work with the Hamiltonian obtained in Appendix A, take $\varepsilon = \varepsilon_s$ and since we expect dynamics to change with position, redefine Rabi frequencies as functions of both space and time $\Omega(z,t)$. Expanding we obtain

$$\begin{aligned}
\frac{d}{dt}\hat{\varepsilon}_H(z,t) = i &\begin{pmatrix} 0 & \Omega_s(z,t)\hat{a}^\dagger & \Omega_I(z,t) & 0 \\ \Omega_s^*(z,t)\hat{a} & \Delta_s & 0 & \Omega_{II}(z,t) \\ \Omega_I^*(z,t) & 0 & \Delta_I & \Omega_i(z,t)\hat{b}^\dagger \\ 0 & \Omega_{II}^*(z,t) & \Omega_i^*(z,t)\hat{b} & \Delta_{II}+\Delta_s \end{pmatrix}\hat{\varepsilon}_H(z,t) + i\Delta_a\hat{a}^\dagger\hat{a}\hat{\varepsilon}_H(z,t) + i\Delta_b\hat{b}^\dagger\hat{b}\hat{\varepsilon}_H(z,t) \\
&- i\hat{\varepsilon}_H(z,t)\begin{pmatrix} 0 & \Omega_s(z,t)\hat{a}^\dagger & \Omega_I(z,t) & 0 \\ \Omega_s^*(z,t)\hat{a} & \Delta_s & 0 & \Omega_{II}(z,t) \\ \Omega_I^*(z,t) & 0 & \Delta_I & \Omega_i(z,t)\hat{b}^\dagger \\ 0 & \Omega_{II}^*(z,t) & \Omega_i^*(z,t)\hat{b} & \Delta_{II}+\Delta_s \end{pmatrix} - i\hat{\varepsilon}_H(z,t)\Delta_a\hat{a}^\dagger\hat{a} - i\hat{\varepsilon}_H(z,t)\Delta_b\hat{b}^\dagger\hat{b} \\
= i &\begin{pmatrix} 0 & \Omega_s(z,t)\hat{a}^\dagger & \Omega_I(z,t) & 0 \\ \Omega_s^*(z,t)\hat{a} & \Delta_s & 0 & \Omega_{II}(z,t) \\ \Omega_I^*(z,t) & 0 & \Delta_I & \Omega_i(z,t)\hat{b}^\dagger \\ 0 & \Omega_{II}^*(z,t) & \Omega_i^*(z,t)\hat{b} & \Delta_{II}+\Delta_s \end{pmatrix}(\varepsilon_s(z,t)\hat{a} + \varepsilon_s^*(z,t)\hat{a}^\dagger) \\
&+ i\Delta_a\hat{a}^\dagger\hat{a}(\varepsilon_s(z,t)\hat{a} + \varepsilon_s^*(z,t)\hat{a}^\dagger) + i\Delta_b\hat{b}^\dagger\hat{b}(\varepsilon_s(z,t)\hat{a} + \varepsilon_s^*(z,t)\hat{a}^\dagger) \\
&- i(\varepsilon_s(z,t)\hat{a} + \varepsilon_s^*(z,t)\hat{a}^\dagger)\begin{pmatrix} 0 & \Omega_s(z,t)\hat{a}^\dagger & \Omega_I(z,t) & 0 \\ \Omega_s^*(z,t)\hat{a} & \Delta_s & 0 & \Omega_{II}(z,t) \\ \Omega_I^*(z,t) & 0 & \Delta_I & \Omega_i(z,t)\hat{b}^\dagger \\ 0 & \Omega_{II}^*(z,t) & \Omega_i^*(z,t)\hat{b} & \Delta_{II}+\Delta_s \end{pmatrix} \\
&- i(\varepsilon_s(z,t)\hat{a} + \varepsilon_s^*(z,t)\hat{a}^\dagger)\Delta_a\hat{a}^\dagger\hat{a} - i(\varepsilon_s(z,t)\hat{a} + \varepsilon_s^*(z,t)\hat{a}^\dagger)\Delta_b\hat{b}^\dagger\hat{b}.
\end{aligned} \tag{B6}$$

Removing all the zero-valued commutation relations, rearranging, and cleaning up notation gives

$$\begin{aligned}\frac{d}{dt}\hat{\varepsilon}_H &= i\left(\Omega_s\hat{a}^\dagger\hat{\sigma}_{12}+\Omega_s^*\hat{a}\hat{\sigma}_{21}\right)\left(\varepsilon_s\hat{a}+\varepsilon_s^*\hat{a}^\dagger\right) - i(\varepsilon_s\hat{a}+\varepsilon_s^*\hat{a}^\dagger)\left(\Omega_s\hat{a}^\dagger\hat{\sigma}_{12}+\Omega_s^*\hat{a}\hat{\sigma}_{21}\right)\\ &\quad + i\Delta_a\left(\hat{a}^\dagger\hat{a}(\varepsilon_s\hat{a}+\varepsilon_s^*\hat{a}^\dagger)-(\varepsilon_s\hat{a}+\varepsilon_s^*\hat{a}^\dagger)\hat{a}^\dagger\hat{a}\right)\\ \\ &= i\left(\Omega_s\hat{a}^\dagger\hat{\sigma}_{12}+\Omega_s^*\hat{a}\hat{\sigma}_{21}\right)\left(\varepsilon_s\hat{a}+\varepsilon_s^*\hat{a}^\dagger\right) - i(\varepsilon_s\hat{a}+\varepsilon_s^*\hat{a}^\dagger)\left(\Omega_s\hat{a}^\dagger\hat{\sigma}_{12}+\Omega_s^*\hat{a}\hat{\sigma}_{21}\right)\\ &\quad + i\Delta_a\left(\varepsilon_s\left(\hat{a}^\dagger\hat{a}\hat{a}-\hat{a}\hat{a}^\dagger\hat{a}\right)+\varepsilon_s^*\left(\hat{a}^\dagger\hat{a}\hat{a}^\dagger-\hat{a}^\dagger\hat{a}^\dagger\hat{a}\right)\right)\\ \\ &= i\left(\Omega_s\hat{a}^\dagger\hat{\sigma}_{12}+\Omega_s^*\hat{a}\hat{\sigma}_{21}\right)\left(\varepsilon_s\hat{a}+\varepsilon_s^*\hat{a}^\dagger\right) - i(\varepsilon_s\hat{a}+\varepsilon_s^*\hat{a}^\dagger)\left(\Omega_s\hat{a}^\dagger\hat{\sigma}_{12}+\Omega_s^*\hat{a}\hat{\sigma}_{21}\right)\\ &\quad + i\Delta_a\left(\varepsilon_s\left[\hat{a}^\dagger\hat{a},\hat{a}\right]+\varepsilon_s^*\left[\hat{a}^\dagger\hat{a},\hat{a}^\dagger\right]\right)\end{aligned} \tag{B7}$$

Similarly to Eq. A59, it can be shown that $\left[\hat{a}^\dagger\hat{a},\hat{a}^\dagger\right]=\hat{a}^\dagger$ so then

$$\begin{aligned}\frac{d}{dt}\hat{\varepsilon}_H &= i\left(\Omega_s\hat{a}^\dagger\hat{\sigma}_{12}+\Omega_s^*\hat{a}\hat{\sigma}_{21}\right)\left(\varepsilon_s\hat{a}+\varepsilon_s^*\hat{a}^\dagger\right) - i(\varepsilon_s\hat{a}+\varepsilon_s^*\hat{a}^\dagger)\left(\Omega_s\hat{a}^\dagger\hat{\sigma}_{12}+\Omega_s^*\hat{a}\hat{\sigma}_{21}\right)\\ &\quad + i\Delta_a\left(-\varepsilon_s\hat{a}+\varepsilon_s^*\hat{a}^\dagger\right)\\ \\ &= i\left(\Omega_s\varepsilon_s\hat{\sigma}_{12}\hat{a}^\dagger\hat{a}+\Omega_s^*\varepsilon_s\hat{\sigma}_{21}\hat{a}\hat{a}+\Omega_s\varepsilon_s^*\hat{\sigma}_{12}\hat{a}^\dagger\hat{a}^\dagger+\Omega_s^*\varepsilon_s^*\hat{\sigma}_{21}\hat{a}\hat{a}^\dagger\right)\\ &\quad - i\left(\Omega_s\varepsilon_s\hat{\sigma}_{12}\hat{a}\hat{a}^\dagger+\Omega_s^*\varepsilon_s^*\hat{\sigma}_{21}\hat{a}^\dagger\hat{a}+\Omega_s\varepsilon_s^*\hat{\sigma}_{12}\hat{a}^\dagger\hat{a}^\dagger+\Omega_s^*\varepsilon_s\hat{\sigma}_{21}\hat{a}\hat{a}\right)\\ &\quad + i\Delta_a\left(-\varepsilon_s\hat{a}+\varepsilon_s^*\hat{a}^\dagger\right)\\ \\ &= i\left(\Omega_s\varepsilon_s\hat{\sigma}_{12}\hat{a}^\dagger\hat{a}+\Omega_s^*\varepsilon_s^*\hat{\sigma}_{21}\hat{a}\hat{a}^\dagger\right) - i\left(\Omega_s\varepsilon_s\hat{\sigma}_{12}\hat{a}\hat{a}^\dagger+\Omega_s^*\varepsilon_s^*\hat{\sigma}_{21}\hat{a}^\dagger\hat{a}\right)\\ &\quad + i\Delta_a\left(-\varepsilon_s\hat{a}+\varepsilon_s^*\hat{a}^\dagger\right)\\ \\ &= i\Omega_s\varepsilon_s\hat{\sigma}_{12}[\hat{a}^\dagger,\hat{a}]+i\Omega_s^*\varepsilon_s^*\hat{\sigma}_{21}[\hat{a},\hat{a}^\dagger]+i\Delta_a\left(-\varepsilon_s\hat{a}+\varepsilon_s^*\hat{a}^\dagger\right)\end{aligned} \tag{B8}$$

Apply the commutation relations

$$[\hat{a},\hat{a}^\dagger]=1 \tag{B9}$$

$$[\hat{a}^\dagger,\hat{a}]=-1 \tag{B10}$$

$$\begin{aligned}\frac{d}{dt}\hat{\varepsilon}_H &= i\Omega_s\varepsilon_s\hat{\sigma}_{12}-i\Omega_s^*\varepsilon_s^*\hat{\sigma}_{21}+i\Delta_a\left(-\varepsilon_s\hat{a}+\varepsilon_s^*\hat{a}^\dagger\right)\\ &= i\varepsilon_s\left(\Omega_s\hat{\sigma}_{12}-\Delta_a\hat{a}\right)-i\varepsilon_s^*\left(\Omega_s^*\hat{\sigma}_{21}-\Delta_a\hat{a}^\dagger\right)\end{aligned} \tag{B11}$$

Let $\hat{\varepsilon}^{(+)}(z,t)=\varepsilon_s\hat{a}$ and $\hat{\varepsilon}^{(-)}(z,t)=\varepsilon_s^*\hat{a}^\dagger$ and express them as a Fourier transform over all optical modes

$$\hat{\varepsilon}^{(+)}(z,t)=\int f(\kappa)\hat{a}_\kappa(t)e^{i\kappa z}d\kappa \tag{B12}$$

where $f(\kappa)$ is some density function of optical modes. By $\omega\approx v_g k$ in a medium we then have that $\Delta_a\approx v_g\kappa_a$ where $\kappa_a=k_a-k_s$ is the phase mismatch between the cavity mode and signal mode. So then

$$\begin{aligned}i\Delta_a\hat{\varepsilon}^{(+)}(z,t) &\approx i\Delta_a\int f(\kappa)\hat{a}_\kappa(t)e^{i\kappa z}d\kappa\\ &\approx iv_g\kappa_a\int f(\kappa)\hat{a}_\kappa(t)e^{i\kappa z}d\kappa\end{aligned} \tag{B13}$$

Notice that

$$\partial_z\int f(\kappa)\hat{a}_\kappa(t)e^{i\kappa z}d\kappa=i\int\kappa f(\kappa)\hat{a}_\kappa(t)e^{i\kappa z}d\kappa \tag{B14}$$

then in the slowly varying envelope approximation take the frequency distribution to be of a single mode $f(\kappa) \approx \delta(\kappa - \kappa_a)$, as $f(\kappa)$ would be sharply peaked about the single mode κ_a, to obtain

$$\begin{aligned}
\partial_z \hat{\varepsilon}^{(+)}(z,t) &\approx \partial_z \int \delta(\kappa - \kappa_a)\hat{a}_\kappa(t)e^{i\kappa z}d\kappa \\
&\approx \partial_z \left(\hat{a}_{\kappa_a}(t)e^{i\kappa_a z}\right) \\
&\approx i\kappa_a \hat{a}_{\kappa_a}(t)e^{i\kappa_a z} \\
&\approx i\kappa_a \int \delta(\kappa - \kappa_a)\hat{a}_\kappa(t)e^{i\kappa z}d\kappa \\
&\approx i\kappa_a \int f(\kappa)\hat{a}_\kappa(t)e^{i\kappa z}d\kappa \\
&\approx i\kappa_a \hat{\varepsilon}^{(+)}(z,t)
\end{aligned} \tag{B15}$$

such that we may write

$$v_g \partial_z \hat{\varepsilon}^{(+)}(z,t) \approx i\Delta_a \hat{\varepsilon}^{(+)}(z,t) \tag{B16}$$

Then consider

$$\begin{aligned}
\hat{\varepsilon}^{(-)} &= \left(\hat{\varepsilon}^{(+)}\right)^\dagger \\
&= \int f^*(\kappa)\hat{a}_\kappa^\dagger(t)e^{-i\kappa z}d\kappa
\end{aligned} \tag{B17}$$

such that by a similar procedure we obtain

$$v_g \partial_z \hat{\varepsilon}^{(+)} \approx -i\Delta_a \hat{\varepsilon}^{(-)} \tag{B18}$$

and therefore

$$\begin{aligned}
&\frac{d}{dt}\hat{\varepsilon}_H \approx i\varepsilon_s\Omega_s\hat{\sigma}_{12} - i\varepsilon_s^*\Omega_s^*\hat{\sigma}_{21} - v_g\partial_z\hat{\varepsilon}^{(+)} - v_g\partial_z\hat{\varepsilon}^{(-)} \\
&\frac{d}{dt}\hat{\varepsilon}_H + v_g\partial_z\hat{\varepsilon}^{(+)} + v_g\partial_z\hat{\varepsilon}^{(-)} \approx i\varepsilon_s\Omega_s\hat{\sigma}_{12} - i\varepsilon_s^*\Omega_s^*\hat{\sigma}_{21}
\end{aligned} \tag{B19}$$

Then by approximating $\partial_t\hat{\varepsilon}_H \approx d\hat{\varepsilon}_H/dt$ we have

$$\begin{aligned}
&\partial_t\hat{\varepsilon}_H + v_g\partial_z\hat{\varepsilon}^{(+)} + v_g\partial_z\hat{\varepsilon}^{(-)} \approx i\varepsilon_s\Omega_s\hat{\sigma}_{12} - i\varepsilon_s^*\Omega_s^*\hat{\sigma}_{21} \\
&\partial_t\left(\hat{\varepsilon}^{(+)} + \hat{\varepsilon}^{(-)}\right) + v_g\partial_z\hat{\varepsilon}^{(+)} + v_g\partial_z\hat{\varepsilon}^{(-)} \approx i\varepsilon_s\Omega_s\hat{\sigma}_{12} - i\varepsilon_s^*\Omega_s^*\hat{\sigma}_{21} \\
&\left(\partial_t\hat{\varepsilon}^{(+)} + v_g\partial_z\hat{\varepsilon}^{(+)}\right) + \left(\partial_t\hat{\varepsilon}^{(-)} + v_g\partial_z\hat{\varepsilon}^{(-)}\right) \approx i\varepsilon_s\Omega_s\hat{\sigma}_{12} - i\varepsilon_s^*\Omega_s^*\hat{\sigma}_{21} \\
&\left(\partial_t\hat{\varepsilon}^{(+)} + v_g\partial_z\hat{\varepsilon}^{(+)}\right) + \left(\partial_t\hat{\varepsilon}^{(+)} + v_g\partial_z\hat{\varepsilon}^{(+)}\right)^\dagger \approx i\varepsilon_s\Omega_s\hat{\sigma}_{12} + \left(i\varepsilon_s\Omega_s\hat{\sigma}_{12}\right)^\dagger
\end{aligned} \tag{B20}$$

Change to the interaction picture where we have operator evolution as

$$\hat{\varepsilon}^{(+)} \to \hat{\varepsilon}^{(+)}e^{-i\omega t} \tag{B21}$$

$$\hat{\varepsilon}^{(-)} \to \hat{\varepsilon}^{(-)}e^{i\omega t} \tag{B22}$$

$$\hat{\sigma}_{12} \to \hat{\sigma}_{12}e^{-i\omega t} \tag{B23}$$

$$\hat{\sigma}_{21} \to \hat{\sigma}_{21}e^{i\omega t} \tag{B24}$$

so we have that

$$\begin{aligned}
&\left[\left(\partial_t\hat{\varepsilon}^{(+)} + v_g\partial_z\hat{\varepsilon}^{(+)}\right) - i\varepsilon_s\Omega_s\hat{\sigma}_{12}\right]e^{-i\omega t} \\
&\quad + \left[\left(\partial_t\hat{\varepsilon}^{(+)} + v_g\partial_z\hat{\varepsilon}^{(+)}\right)^\dagger - \left(i\varepsilon_s\Omega_s\hat{\sigma}_{12}\right)^\dagger\right]e^{i\omega t} \approx 0
\end{aligned} \tag{B25}$$

This is of the form

$$
\begin{aligned}
&(a-b)e^{-i\omega t} + (a^\dagger - b^\dagger)e^{i\omega t} = 0 \\
&(a-b)(x+iy) + (a^\dagger - b^\dagger)(x-iy) = 0 \\
&\text{where } x, y \in \mathbb{R}, \;\; x^2+y^2 = 1 \\
&(ax - bx + iay - iby) + (a^\dagger x - b^\dagger x - ia^\dagger y + ib^\dagger y) = 0 \\
&(ax - bx + a^\dagger x - b^\dagger x) + i(ay - by - a^\dagger y + b^\dagger y) = 0 \\
&((a+a^\dagger) - (b+b^\dagger))x + i((a-a^\dagger) - (b-b^\dagger))y = 0 \\
&\therefore a + a^\dagger = b + b^\dagger \;\text{ and }\; a - a^\dagger = b - b^\dagger \\
&\text{So it must be that } a = b
\end{aligned}
\tag{B26}
$$

By applying this line of reasoning, we have

$$
\left(\partial_t \hat{\varepsilon}^{(+)}(z,t) + v_g \partial_z \hat{\varepsilon}^{(+)}(z,t)\right) \approx i\varepsilon_s(z,t)\Omega_s(z,t)\hat{\sigma}_{12} \tag{B27}
$$

Then we may shift to the co-moving frame $\zeta = z$, $\tau = t - z/v_g$ to obtain

$$
v_g \partial_\zeta \hat{\varepsilon}^{(+)}(\zeta,\tau) \approx i\varepsilon_s(\zeta,\tau)\Omega_s(\zeta,\tau)\hat{\sigma}_{12} \tag{B28}
$$

Transitioning to the semi-classical picture take $\hat{\varepsilon}^{(+)}(z,t) \to \varepsilon(\zeta,\tau)$ such that we then have

$$
\Omega_s(\zeta,\tau) = \frac{d_{12}\varepsilon(\zeta,\tau)}{\hbar} \tag{B29}
$$

where it is assumed that $\vec{d} \parallel \vec{\varepsilon}$ since we are working with a gaseous isotropic medium. Applying this yields

$$
\begin{aligned}
&\frac{\hbar v_g}{d_{12}} \partial_\zeta \Omega_s(\zeta,\tau) \approx i\varepsilon_s(\zeta,\tau)\Omega_s(\zeta,\tau)\rho_{12} \\
&\partial_\zeta \Omega_s(\zeta,\tau) \approx i\frac{d_{12}}{\hbar v_g}\varepsilon_s(\zeta,\tau)\Omega_s(\zeta,\tau)\rho_{12}
\end{aligned}
\tag{B30}
$$

By Eq. (B4) this is

$$
\begin{aligned}
\partial_\zeta \Omega_s(\zeta,\tau) &\approx i\frac{d_{12}}{\hbar v_g}\sqrt{\frac{\hbar\omega_s}{2\epsilon_0 V}}\Omega_s(\zeta,\tau)\rho_{12} \\
&\approx i\frac{d_{12}}{\hbar v_g}\sqrt{\frac{\hbar\omega_s}{2\epsilon_0 V}}\left(\frac{d_{12}\varepsilon(\zeta,\tau)}{\hbar}\right)\rho_{12} \\
&\approx i\frac{d_{12}^2}{\hbar^2 v_g}\sqrt{\frac{\hbar\omega_s}{2\epsilon_0 V}}\sqrt{\frac{\hbar\omega_s}{2\epsilon_0 V}}\rho_{12} \\
&\approx i\frac{\omega_s d_{12}^2}{2v_g\epsilon_0\hbar V}\rho_{12}
\end{aligned}
\tag{B31}
$$

Each atom will have some velocity distributed as Maxwell-Boltzmann, so the state with respect to the m^{th} atom will be defined by some density matrix $\hat{\rho}^{(m)}$ which takes into account Doppler shifts any other interactions. We then have a notion for the mean density matrix and mean density matrix elements

$$
\overline{\rho}_{ij} = \frac{1}{N}\sum_{m=1}^{N}\rho_{ij}^{(m)} \tag{B32}
$$

Additionally, consider that in this derivation we considered only a single atom to obtain our expression for $\partial_\zeta \Omega_s(\zeta,\tau)$. To generalize this to the entire system we must consider all N atoms within the volume of considerations V. In this case, the total Hamiltonian would be $\tilde{H}_{total} = \tilde{H}_{photons} + \sum_{m=1}^{N}\left(\tilde{H}_{atom}^{(m)} + \tilde{H}_{interaction}^{(m)}\right)$. By extending our derivation to N atoms, we would obtain

$$\begin{aligned}
\partial_\zeta \Omega_s(\zeta,\tau) &\approx \sum_{m=1}^{N} i\frac{\omega_s d_{12}^2}{2v_g\epsilon_0\hbar V}\rho_{12}^{(m)} \\
&\approx i\frac{\omega_s d_{12}^2}{2v_g\epsilon_0\hbar V}\sum_{m=1}^{N}\rho_{12}^{(m)} \\
&\approx i\frac{\omega_s d_{12}^2}{2v_g\epsilon_0\hbar V}N\overline{\rho}_{12} \\
&\approx i\frac{\omega_s \nu d_{12}^2}{2v_g\epsilon_0\hbar}\overline{\rho}_{12}
\end{aligned} \tag{B33}$$

where $\nu = N/V$ is atomic density.

Identical steps lead to a complementary equation for the idler field such that we have the Maxwell-Bloch equations for our system as

$$\partial_\zeta \Omega_s(\zeta,\tau) \approx i\frac{\omega_s \nu d_{12}^2}{2v_g\epsilon_0\hbar}\overline{\rho}_{12} \tag{B34}$$

$$\partial_\zeta \Omega_s(\zeta,\tau) \approx i\frac{\omega_i \nu d_{34}^2}{2v_g\epsilon_0\hbar}\overline{\rho}_{34} \tag{B35}$$

1. Considering Noise

We must still consider an alternative possibility. In Sec. III B we discuss the inclusion of a noise term to our Hamiltonian as

$$\tilde{H}_{ab} = \tilde{H} + \frac{\hbar}{2}\left(\varepsilon_s\hat{a} + \varepsilon_s^*\hat{a}^\dagger\right) \tag{B36}$$

where we have introduced a noise term for just field $\hat{a}$. This would then result in the equation

$$\begin{aligned}
\frac{d}{dt}\hat{\varepsilon}_H(z,t) &= \frac{i}{\hbar}\left[\tilde{H}(t) + \hbar\left(\varepsilon_s\hat{a} + \varepsilon_s^*\hat{a}^\dagger\right), \hat{\varepsilon}_H(z,t)\right] \\
&= \frac{i}{\hbar}\left[\tilde{H}(t), \hat{\varepsilon}_H(z,t)\right] + \left[\hbar\left(\varepsilon_s\hat{a} + \varepsilon_s^*\hat{a}^\dagger\right), \hat{\varepsilon}_H(z,t)\right] \\
&= \frac{i}{\hbar}\left[\tilde{H}(t), \hat{\varepsilon}_H(z,t)\right] + \left[\hbar\hat{\varepsilon}_H(z,t), \hat{\varepsilon}_H(z,t)\right] \\
&= \frac{i}{\hbar}\left[\tilde{H}(t), \hat{\varepsilon}_H(z,t)\right]
\end{aligned} \tag{B37}$$

Thus, the Maxwell-Bloch equations of Eq. (B34) are suitable for a system in which such a noise term is included in the Hamiltonian.

C Four-Level Diamond Configuration Atom Code

The following is code simulating the four-level atomic system outlined in Sec. III and code specification as outlined in Sec. IV B.

Code Block 2: Python code for simulating the properties of a four-level diamond configuration atom in the presence of external driving fields tuned near resonant transition frequencies.

```
import matplotlib.pyplot as plt
import numpy as np
import random
from IPython.display import Image
from qutip import (Qobj, tensor, basis, create, destroy, qeye, mesolve,
    steadystate, expect, num)

##########################################
# CONSTANTS, BASIS STATES, AND OPERATORS #
##########################################

# Constants
hbar = 1
epsilon_0 = 1
c = 1
n = 1 # Index of refraction for Rubidium
boltzmann_const = 1
Rb_mass = 1
kappa = 1.0 # Write all parameters relative to kappa (cavity decay rate)
Gamma_12 = 0.1 * kappa
Gamma_13 = 0.1 * kappa
Gamma_24 = 0.05 * kappa
Gamma_34 = 0.05 * kappa
N_Fock = 5 # Truncate each Fock basis to N_Fock photons
d_12 = 1
d_34 = 1
omega_12 = 1 / 795
omega_13 = 1 / 780
omega_24 = 1 / 1475
omega_34 = 1 / 1530

# Atomic Basis Kets
atomicKet1 = basis(4, 0) # |1>
atomicKet2 = basis(4, 1) # |2>
atomicKet3 = basis(4, 2) # |3>
atomicKet4 = basis(4, 3) # |4>

atomic_density = lambda temp: 1 / temp * 10**(31.178 - 4040 / temp)


# Set up operators in the relevant basis
S11 = tensor(atomicKet1 * atomicKet1.dag(), qeye(N_Fock), qeye(N_Fock))
S22 = tensor(atomicKet2 * atomicKet2.dag(), qeye(N_Fock), qeye(N_Fock))
S33 = tensor(atomicKet3 * atomicKet3.dag(), qeye(N_Fock), qeye(N_Fock))
S44 = tensor(atomicKet4 * atomicKet4.dag(), qeye(N_Fock), qeye(N_Fock))

S12 = tensor(atomicKet1 * atomicKet2.dag(), qeye(N_Fock), qeye(N_Fock)) #
    Photon emission 1->2
S13 = tensor(atomicKet1 * atomicKet3.dag(), qeye(N_Fock), qeye(N_Fock)) #
    Photon emission 1->3
S24 = tensor(atomicKet2 * atomicKet4.dag(), qeye(N_Fock), qeye(N_Fock)) #
```

```
    Photon emission 2->4
S34 = tensor(atomicKet3 * atomicKet4.dag(), qeye(N_Fock), qeye(N_Fock)) #
    Photon emission 3->4

S21 = S12.dag() # Photon absorption 2->1
S31 = S13.dag() # Photon absorption 3->1
S42 = S24.dag() # Photon absorption 4->2
S43 = S34.dag() # Photon absorption 4->3

# a and b act on photon independent Fock bases as lowering operator
a = tensor(qeye(4), destroy(N_Fock), qeye(N_Fock))
b = tensor(qeye(4), qeye(N_Fock),    destroy(N_Fock))

# Creation operators
a_dag = a.dag()
b_dag = b.dag()

# Number Operator for field a
num_a_op = tensor(qeye(4), num(N_Fock),  qeye(N_Fock))
num_b_op = tensor(qeye(4), qeye(N_Fock), num(N_Fock))

# Decay operators
cavity_a_decay = np.sqrt(2*kappa) * a
cavity_b_decay = np.sqrt(2*kappa) * a
atomic_decay_12 = np.sqrt(2*Gamma_12) * S12
atomic_decay_13 = np.sqrt(2*Gamma_13) * S13
atomic_decay_24 = np.sqrt(2*Gamma_24) * S24
atomic_decay_34 = np.sqrt(2*Gamma_34) * S34
c_ops = [
    cavity_a_decay,
    cavity_b_decay,
    atomic_decay_12,
    atomic_decay_13,
    atomic_decay_24,
    atomic_decay_34
]




################
# HAMILTONIANS #
################

def form_Hamiltonian_ab(
        Omega_s,  # Signal Rabi frequency (Related to electric field strength)
        Omega_i,  # Idler Rabi frequency (Related to electric field strength)
        Omega_I,  # Pump 1 Rabi Frequency (Related to electric field strength)
        Omega_II, # Pump 2 Rabi Frequency (Related to electric field strength)
        Delta_s,  # Signal detuning
        Delta_I,  # Pump 1 detuning
        Delta_II, # Pump 2 detuning
        Delta_a,  # Cavity a detuning
        Delta_b   # Cavity b detuning
    ):

    H_atom = hbar * (
        Delta_I * S22 +
```

```
        Delta_s * S33 +
        (Delta_II + Delta_s) * S44
    )

    H_photon = hbar * (
        Delta_a * a_dag * a +
        Delta_b * b_dag * b
    )

    H_int = hbar * (
        Omega_s     * a_dag * S12 + np.conj(Omega_s)  * a * S21 +
        Omega_I     * S13         + np.conj(Omega_I)  * S31     +
        Omega_II    * S24         + np.conj(Omega_II) * S42     +
        Omega_i     * b_dag * S34 + np.conj(Omega_i)  * b * S43
    )

    epsilon_s = hbar * Omega_s / d_12
    epsilon_i = hbar * Omega_i / d_34

    H_noise = hbar * (
        epsilon_s * a + np.conj(epsilon_s) * a_dag +
        epsilon_i * b + np.conj(epsilon_i) * b_dag
    )

    H_total = H_atom + H_photon + H_int + H_noise
    return H_total




def form_Hamiltonian_a(
        Omega_s,  # Signal coupling (Related to electric field strength)
        Omega_i,  # Idler coupling (Related to electric field strength)
        Omega_I,  # Pump 1 Rabi Frequency (Related to electric field strength)
        Omega_II, # Pump 2 Rabi Frequency (Related to electric field strength)
        Delta_s,  # Signal detuning
        Delta_I,  # Pump 1 detuning
        Delta_II, # Pump 2 detuning
        Delta_a,  # Cavity a detuning
        Delta_b   # Cavity b detuning
    ):

    H_atom = hbar * (
        Delta_I * S22 +
        Delta_s * S33 +
        (Delta_II + Delta_s) * S44
    )

    H_photon = hbar * (
        Delta_a * a_dag * a +
        Delta_b * b_dag * b
    )

    H_int = hbar/2 * (
        Omega_s  * a_dag * S12 + np.conj(Omega_s)  * a * S21 +
        Omega_I  * S13         + np.conj(Omega_I)  * S31     +
        Omega_II * S24         + np.conj(Omega_II) * S42     +
        Omega_i  * S34         + np.conj(Omega_i)  * S43
```

```
    )

    epsilon_s = hbar * Omega_s / d_12

    H_noise = hbar * (
        epsilon_s * a + np.conj(epsilon_s) * a_dag
    )

    H_total = H_atom + H_photon + H_int + H_noise
    return H_total



def form_Hamiltonian_b(
        Omega_s,  # Signal coupling (Related to electric field strength)
        Omega_i,  # Idler coupling (Related to electric field strength)
        Omega_I,  # Pump 1 Rabi Frequency (Related to electric field strength)
        Omega_II, # Pump 2 Rabi Frequency (Related to electric field strength)
        Delta_s,  # Signal detuning
        Delta_I,  # Pump 1 detuning
        Delta_II, # Pump 2 detuning
        Delta_a,  # Cavity a detuning
        Delta_b   # Cavity b detuning
    ):

    H_atom = hbar * (
        Delta_I * S22 +
        Delta_s * S33 +
        (Delta_II + Delta_s) * S44
    )

    H_photon = hbar * (
        Delta_a * a_dag * a +
        Delta_b * b_dag * b
    )

    H_int = hbar/2 * (
        Omega_s  * S12         + np.conj(Omega_s)  * S21     +
        Omega_I  * S13         + np.conj(Omega_I)  * S31     +
        Omega_II * S24         + np.conj(Omega_II) * S42     +
        Omega_i  * b_dag * S34 + np.conj(Omega_i)  * b * S43
    )

    epsilon_i = hbar * Omega_i / d_34

    H_noise = hbar * (
        epsilon_i * b + np.conj(epsilon_i) * b_dag
    )

    H_total = H_atom + H_photon + H_int + H_noise
    return H_total



####################
# SOLUTION METHODS #
####################
```

```python
def Maxwell_Bloch_step(rho, Omega_s, Omega_i, Delta_s, Delta_i, temp,
    step_size):
    rho_12 = rho[1,2]
    rho_34 = rho[3,4]

    # NOTE - We are using c here instead of group velocity here
    #        So this is an approximation
    d_z_Omega_s = 1j * (omega_12 + Delta_s) * d_12**2 * atomic_density(temp) *
        rho_12 / (2 * c * epsilon_0 * hbar)
    d_z_Omega_i = 1j * (omega_34 + Delta_i) * d_34**2 * atomic_density(temp) *
        rho_34 / (2 * c * epsilon_0 * hbar)

    Omega_s_step = d_z_Omega_s * step_size
    Omega_i_step = d_z_Omega_i * step_size

    # Update rabi frequencies
    new_Omega_s = Omega_s + Omega_s_step
    new_Omega_i = Omega_i + Omega_i_step

    return new_Omega_s, new_Omega_i



def solve_for_steady_state_doppler_broadened_density_matrix(
        Omega_s,          # Signal Rabi Frequency (Related to electric field
            strength)
        Omega_i,          # Idler Rabi Frequency (Related to electric field
            strength)
        Omega_I,          # Pump 1 Rabi Frequency (Related to electric field
            strength)
        Omega_II,         # Pump 2 Rabi Frequency (Related to electric field
            strength)
        Delta_s,          # Signal detuning
        Delta_i,          # Idler detuning
        Delta_I,          # Pump 1 detuning
        Delta_II,         # Pump 2 detuning
        Delta_a,          # Cavity a detuning
        Delta_b,          # Cavity b detuning
        temp,             # Temperature of Rb gas
        num_velocity_subclasses = 20,
        Hamiltonian_Function = form_Hamiltonian_ab
    ):

    # Set up normal velocity distribution
    sigma_sq = 2 * boltzmann_const * temp / Rb_mass
    maxwell_boltzmann_dist = lambda v: 1 / np.sqrt(np.pi * sigma_sq) *
        np.exp(-v**2 / sigma_sq)

    # Range of velocities to sample over (2 standard deviations)
    min_v = -2 * np.sqrt(sigma_sq)
    max_v = 2 * np.sqrt(sigma_sq)
    v_vals = np.linspace(min_v, max_v, num_velocity_subclasses)

    rho_sum = None

    for v in v_vals:
        # Frequency dependent doppler shifts
```

```
        # We assume all beams travel along the same axis
        # Note that this is an approximation since we take n to be constant
        doppler_shift_s = n * (omega_12 + Delta_s) * v / c # Aligned
        # doppler_shift_i = -n * (omega_34 + Delta_i) * v / c # Anti-aligned

        doppler_shift_I = n * (omega_13 + Delta_I) * v / c # Aligned
        doppler_shift_II = -n * (omega_24 + Delta_I) * v / c # Anti-aligned

        doppler_shift_a = n * (omega_12 + Delta_a) * v / c # Aligned
        doppler_shift_b = -n * (omega_34 + Delta_b) * v / c # Anti-aligned

        # Solve for the steady state
        # NOTE - Hamiltonian_Function is an input argument, not an actual
            function
        H_total = Hamiltonian_Function(
            Omega_s, Omega_i, Omega_I, Omega_II,
            Delta_s * doppler_shift_s,
            Delta_I * doppler_shift_I,
            Delta_II * doppler_shift_II,
            Delta_a * doppler_shift_a,
            Delta_b * doppler_shift_b
        )
        rho = steadystate(H_total, c_ops, method='direct')

        weight = maxwell_boltzmann_dist(v)

        if rho_sum is None:
            rho_sum = weight * rho
        else:
            rho_sum += weight * rho

    # The resulting rho_sum should be approximately normalized,
    # but fix any discrepancy here
    doppler_broadened_rho = rho_sum.unit()

    return doppler_broadened_rho



def iterative_solve(
        Omega_s,            # Signal Rabi Frequency (Related to electric field
            strength)
        Omega_i,            # Idler Rabi Frequency (Related to electric field
            strength)
        Omega_I,            # Pump 1 Rabi Frequency (Related to electric field
            strength)
        Omega_II,           # Pump 2 Rabi Frequency (Related to electric field
            strength)
        Delta_s,            # Signal detuning
        Delta_i,            # Idler detuning
        Delta_I,            # Pump 1 detuning
        Delta_II,           # Pump 2 detuning
        Delta_a,            # Cavity a detuning
        Delta_b,            # Cavity b detuning
        temp,               # Temperature
        cell_length,        # Length of vacuum cell (determines number of steps)
        step_size = 0.01,   # Distance covered by each iteration
```

```
        num_velocity_subclasses = 20,
        Hamiltonian_Function = form_Hamiltonian_ab
    ):

    distance_covered = 0
    while distance_covered < cell_length:
        rho = solve_for_steady_state_doppler_broadened_density_matrix(
            Omega_s, Omega_i, Omega_I, Omega_II,
            Delta_s, Delta_i, Delta_I, Delta_II,
            Delta_a, Delta_b,
            temp,
            num_velocity_subclasses,
            Hamiltonian_Function
        )

        # Update Omega_s and Omega_i via the Maxwell-Bloch equations
        Omega_s, Omega_i = Maxwell_Bloch_step(rho, Omega_s, Omega_i, Delta_s,
            Delta_i, temp, step_size)

        # Update the distance covered to bring the loop closer to finishing
        distance_covered += step_size

    # Obtain the final density matrix with respect to the final values of
        Omega_s and Omega_i
    rho = solve_for_steady_state_doppler_broadened_density_matrix(
        Omega_s, Omega_i, Omega_I, Omega_II,
        Delta_s, Delta_i, Delta_I, Delta_II,
        Delta_a, Delta_b,
        temp,
        num_velocity_subclasses,
        Hamiltonian_Function
    )

    return rho, Omega_s, Omega_i
```